Information Frictions and the Expansion of Markets: Evidence from Mobile Phone Access in North Korea*

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Abstract

Do information frictions limit the extent of markets? Existing evidence shows that lower information costs improve market efficiency, but whether they also expand market activity remains far less clear. I examine this question using the staggered rollout of North Korea's first public mobile-phone network, which sharply reduced information costs in an economy where restrictions on mobility, media, and communication had long confined most information to people's own localities. A sequential search model illustrates how lower search costs allow producers to search longer and reject low offers, raising the expected net selling price and thereby strengthening incentives to produce for the market. Using satellite imagery, I reconstruct the installation timing of more than 1,100 cell towers and measure two dimensions of market activity: agricultural activity, proxied by a vegetation index, and exchange, captured by the physical footprints of 536 marketplaces identified through deep learning. Consistent with the model's incentive mechanism, higher levels of mobile-phone access increased agricultural activity on sloping land, where households can cultivate private plots and retain the returns. Each additional cell tower also expanded marketplace area by the equivalent of about 198 stalls, almost entirely through the emergence of new marketplaces near existing ones. These effects were larger in counties with rail stations and in those closer to cities, suggesting that mobile connectivity complements physical connectivity. These findings show that easier access to information can expand markets and stimulate local economic activity, even under severe institutional constraints.

JEL codes: D83, O12, O17, O18, O33

Keywords: Information frictions, search costs, market expansion, mobile phones, satellite imagery, North Korea.

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1 Introduction

Markets operate through decentralized exchange among producers, traders, and consumers, all of whom rely on information to decide what to produce, where to trade, and at what price. Hayek (1945) emphasized that markets coordinate these decisions by drawing on knowledge dispersed among individuals. However, acquiring and transmitting this knowledge is costly (Stigler, 1961), and information remains fragmented across space. Transportation infrastructure has long promoted market integration and economic development by lowering the cost of moving goods between previously isolated locations (Pascali, 2017; Donaldson, 2018; Asher and Novosad, 2020). More recently, mobile phones have done the same for information, enabling instantaneous communication in developing countries even where physical infrastructure remains limited (Aker and Mbiti, 2010; Hjort and Poulsen, 2019). A large body of evidence shows that, by reducing information frictions, mobile phones improve market efficiency, narrowing price dispersion (Jensen, 2007; Aker, 2010) and reallocating market share to more productive firms (Jensen and Miller, 2018).

Yet, while these findings show how information improves the efficiency of existing markets, much less is known about whether it can expand market activity itself. Better information may change not only the prices at which goods are traded but also whether and how much people produce for and trade in markets. Can improved access to information expand the scale and geographic extent of market activity, and under what conditions? These questions matter because, in many developing economies, local markets are where households earn income and obtain essential goods, so their expansion can bring benefits beyond gains in efficiency. They also have implications for policy: investments that reduce information frictions may stimulate local economic activity and support self-sustaining development driven by the decisions of better-informed individuals.

This paper provides evidence that improved access to information can expand market activity. I study this question using the staggered rollout of North Korea's first mobile-phone network available to the general public. Before the rollout, most North Koreans had limited access to information beyond their immediate localities, as the state tightly restricted domestic mobility and controlled virtually all sources of information. Telecommunications infrastructure was also severely underdeveloped, with landline subscriptions remaining at about five per 100 people for nearly two decades. The network, launched in December 2008, sharply changed this information environment. Although introduced to serve the government's political and revenue objectives, it gave ordinary citizens a new way to coordinate their own economic activities across locations in real time. This connectivity was particularly consequential for market activity, which had become central to household livelihoods

following the breakdown of the state distribution system. Among 40 recent defectors surveyed, 97.5 percent reported buying a mobile phone specifically for market activities despite high handset costs (Joung and Lee, 2023). North Korea thus offers a distinctive setting where severe information frictions coexist with heavy household reliance on markets, making it possible to study whether a substantial reduction in information costs can expand market activity even under persistent institutional constraints.

To guide the empirical analysis, I develop a simple framework that combines a production decision with a sequential search problem. Producers decide how much to produce before they sell, based on the net price they expect to receive after transportation and transaction costs. Without mobile phones, learning current prices in other markets at the time of sale is costly. Producers rely on past prices or information picked up by chance unless they travel to other markets in search of better offers, so they may end up accepting low offers. Mobile phones lower the cost of obtaining each additional price quote, allowing producers to search longer and reject low offers, which raises the prices they receive. Anticipating these higher prices when planting, producers have stronger incentives to devote labor and inputs to production for the market. Lower information costs can therefore expand market activity by shaping not only where and at what prices producers sell but also how much they produce in the first place. The framework predicts a stronger production response where producers keep more of the returns to additional output, which I test using institutional differences in production incentives across land types in North Korea.

A central challenge in studying North Korea is the lack of reliable, consistently measured data on local economic activity over time. I address it by constructing new data from satellite imagery. To measure mobile-phone access, I reconstruct the network rollout by tracing 1,127 cell towers visible in 2021 back through historical high-resolution imagery from Google Earth and Esri Wayback. Comparing the last image in which each tower is absent with the first in which it appears identifies either its construction year or, when intervening imagery is unavailable, a range of feasible years (Figure I).[1] The exact year is identified for half of the towers, and timing uncertainty is at most two years for nearly 95 percent of all towers. The baseline assumes a constant annual increase in each county's tower stock between exactly known years, and the results are robust to alternative timing assumptions. Independent evidence corroborates the reconstructed tower locations and timing and shows that mobile-phone adoption expanded alongside the network (Section 4.1).

I then measure two dimensions of market activity: agricultural activity and exchange. For agricultural activity, I use the satellite-derived Normalized Difference Vegetation Index

[1] For example, a tower absent in a 2013 image and visible in a 2014 image is dated to 2014. If the last image without the tower is instead from 2010, the tower could have been built in any year from 2011 to 2014.

(NDVI) during the growing seasons from 2002 to 2015. Under North Korea's slope-based land management policy, land with slopes below 15 degrees is reserved for collective farms, whereas land above this threshold (hereafter sloping land) is officially designated as forest land, but households have cultivated small private plots on it and keep the output. Comparing vegetation on sloping land and collective farmland within the same county and year thus contrasts agricultural activity under different production incentives. For exchange, I use the area of officially sanctioned marketplaces (*jangmadang*), the main venues for private trade in North Korea. Building on existing surveys of marketplace locations, I apply deep learning to high-resolution images across North Korea to identify previously undocumented marketplaces and measure the area of 536 marketplaces each year from 2008 to 2021. Vendors pay recurring stall fees, so they rent space only when expected returns justify the cost, and local authorities, who collect these fees, have an incentive to permit expansion. Marketplace area thus captures changes in market participation that are persistent enough to alter a marketplace's physical structure, rather than short-run fluctuations in sales.

I employ a difference-in-differences strategy that exploits variation in the timing and intensity of the rollout across North Korea's 178 second-level administrative units (cities and counties, hereafter counties). My main specification uses the cumulative number of cell towers in each county rather than a binary indicator of access, because each additional tower extends coverage and capacity to more people and places. Although more than half of the counties received their first tower in 2009, tower counts subsequently grew at different rates. Baseline regressions include county and year fixed effects and, in more demanding specifications, province-by-year fixed effects that compare counties within the same province. A causal interpretation requires that, absent the network, outcomes in counties with different rollout timing and intensity would have followed parallel trends. To test for pre-existing trends and trace how the effects evolved, I estimate event studies around the year of each county's first tower using the Callaway and Sant'Anna (2021) estimator. This estimator compares connected counties only with not-yet-connected and never-connected counties, avoiding the biases of conventional event studies under staggered adoption. Because these event studies capture the timing of initial access but not subsequent tower additions, I separately test in the robustness analysis whether earlier changes in outcomes predict these additions.

I find that higher levels of mobile-phone access increased agricultural activity on sloping land, both in absolute terms and relative to collective farmland in the same county, but had no significant effect on vegetation on collective farmland. This contrast is consistent with the framework's prediction that better information strengthens incentives to produce where households keep the returns to their output, as they do on sloping land but not on collective farms. Within sloping land, the increase is concentrated in areas with low tree

cover, where land is easier to clear and plant, with no detectable effect in densely forested areas. This pattern suggests expanded cultivation rather than a general greening. By 2015, when the average county had about six towers, the estimates imply that vegetation on sloping land rose by about 0.8 percent relative to collective farmland, roughly a quarter of a standard deviation of this difference within counties.[2] Event-study estimates show no evidence of differential pre-trends, and the difference between the two types of land begins to widen about four years after the first tower, with the increase most pronounced in areas with low tree cover. Because tower counts continued to rise after the first tower, the event study combines the effect of more towers with that of longer exposure. Separating the two shows that the vegetation response depends on the number of towers, not on the length of exposure, so the four-year delay reflects the gradual expansion of the network.

Mobile-phone access also expanded exchange. Each additional cell tower increased marketplace area by about 337 square meters, the equivalent of roughly 198 stalls or about 3.4 percent of the average county's marketplace area before the rollout.[3] Event-study estimates show no evidence of differential pre-trends, and marketplace area begins to rise about four years after the first tower, as with vegetation. Unlike vegetation, however, this expansion depended on both the number of towers and the length of exposure. An additional tower had little effect at first, but its effect grew by about 43 square meters with each year of exposure. A plausible reason is that cultivation is an individual decision that producers can adjust each season, whereas marketplace expansion requires coordination among many vendors and buyers and the approval and construction of new stalls, all of which take time.

Decomposing this growth, I find that it came almost entirely from the entry of new marketplaces, few of which replaced existing ones. The largest marketplace in each county showed no significant growth. Most new marketplaces opened near those active in 2008, where demand already existed and transaction costs were likely lower, consistent with agglomeration forces. Across counties, the effects were substantially larger in those with a train station, given the importance of rail transport in North Korea, and in those closer to cities, where most wholesale markets are located. These patterns suggest that mobile connectivity complements physical connectivity, with better information translating into more exchange where goods could move. By contrast, the effects were not significantly larger in counties closer to Pyongyang or with special economic zones, suggesting that marketplace expansion was not concentrated in areas favored by the state. In counties bordering China, which were already within range of Chinese mobile networks, the rollout had no detectable effect, plausibly because the domestic network added little new information. This null result

[2] The outcome is measured in units of $100 \times \ln$ NDVI. After removing county and year fixed effects, the difference between sloping land and collective farmland has a standard deviation of 2.8 in the same units.

[3] Based on an average stall size of 1.7 square meters (Hong et al., 2016).

also argues against cross-border trade as the main driver of marketplace expansion.

I examine the validity of the identifying assumption and the robustness of the findings in several ways. First, towers may have been built where markets would have grown even without mobile phones. Indeed, they arrived first in economically larger counties. However, counties that would later receive more towers were not already growing faster, and marketplace growth does not predict subsequent tower additions. The estimates also hold after controlling for pre-rollout trends. Matching counties on pre-rollout characteristics yields similar estimates for vegetation and smaller, less precise ones for marketplaces. Second, changes that coincided with the rollout may offer alternative explanations, but the estimates are largely robust to controlling for rainfall (for vegetation), road networks, trade with China, and exposure to sanctions. For vegetation, the response on sloping land with low tree cover is the most robust. Third, spillovers from towers in neighboring counties do not explain the results, as own-county estimates remain positive and significant after controlling for neighboring towers. Fourth, allowing for treatment-effect heterogeneity using the estimator of de Chaisemartin and D'Haultfœuille (2024) yields a smaller but still significant marketplace effect, about 40 percent as large as the baseline estimate.

Related Literature and Contributions. This paper contributes to several strands of literature. First, it extends research on information frictions and the role of communication technology in markets. Existing studies have documented that mobile phones improve market efficiency (Jensen, 2007; Aker, 2010; Aker and Fafchamps, 2015; Jensen and Miller, 2018) and that information frictions shape where producers sell and account for much of regional price dispersion (Allen, 2014). Closer to this paper, communication technologies have been shown to affect production and market participation for particular crops: internet kiosks increased soybean acreage in India (Goyal, 2010), and mobile-phone coverage encouraged banana farmers in Uganda to sell in markets (Muto and Yamano, 2009). Yet little is known about whether better information can expand market activity across an entire economy. I examine this question using the nationwide rollout of a general-purpose mobile-phone network.[4] Guided by a search framework that predicts a production response, I find that lower information costs expanded both agricultural activity and the physical extent of marketplaces, with most of the market expansion driven by the entry of new marketplaces.

Second, the paper sheds light on the conditions under which better access to information expands markets. Research on property rights shows that institutions shape production and investment incentives by determining who benefits from additional production,

[4] In Goyal (2010), the kiosks were operated by a firm that also purchased crops directly from farmers, so they provided both price information and a new buyer. By contrast, the mobile-phone network studied here was not designed to promote agriculture or market activity.

as in China's agricultural reforms (McMillan et al., 1989; Lin, 1992; Besley, 1995; Chari et al., 2021). Another strand of literature emphasizes the role of transport infrastructure in reducing trade costs and connecting producers to markets (Donaldson and Hornbeck, 2016; Donaldson, 2018; Aggarwal, 2018; Asher and Novosad, 2020). My findings link these literatures to information frictions. Mobile-phone access increased agricultural activity on privately cultivated plots but not on collective farmland. It also expanded marketplaces more in counties with a train station. Together, these findings suggest that lower information costs can create new economic opportunities, but realizing them depends on production incentives and the ability to move goods to market.

Third, the paper provides new evidence on the economic consequences of communication technology under authoritarian rule. Existing research has focused largely on its political consequences, from state surveillance and censorship (Chen and Yang, 2019; Beraja et al., 2023) to protests and trust in government (Manacorda and Tesei, 2020; Guriev et al., 2021), but much less is known about its economic effects. I find that North Korea's mobile-phone network, introduced for the state's political purposes, expanded the market activity on which households depend for their livelihoods. The paper also contributes to research on the North Korean economy, which has examined the effects of sanctions and the responses of elites and authorities (Lee, 2018; Kim et al., 2023), while evidence on ordinary households has relied largely on retrospective surveys of defectors (Kim and Song, 2008; Kim and Kim, 2019). By measuring production and exchange inside North Korea, I shift attention from the state to the general population.

Finally, the paper adds to a growing literature that uses high-resolution satellite imagery to measure economic outcomes in data-scarce settings. Existing studies typically rely on imagery from a single period and on cross-sectional variation (Jean et al., 2016; Marx et al., 2019; Michaels et al., 2021; Chi et al., 2022). Closest to this paper, Tafese et al. (2025) trace special economic zones through historical imagery to measure their expansion over time but draw outcomes from surveys. I instead use historical imagery to construct nationwide annual panels of both the treatment and the outcomes, dating cell towers by tracing them through archived images and measuring marketplaces with deep learning (Yang et al., 2026). These panels provide variation across counties and over time, making it possible to study a staggered rollout in such a setting and to trace how its effects unfolded over more than a decade.

The rest of the paper proceeds as follows. Section 2 describes the setting, and Section 3 presents the conceptual framework. Section 4 introduces the data, and Section 5 outlines the empirical strategy. Section 6 discusses the main results, followed by robustness checks in Section 7. Section 8 concludes.

2 Background

2.1 Marketization in North Korea

Market activity has become a central feature of North Korea's economy despite the country's formal centrally planned system. This marketization emerged as a survival response to the collapse of the Public Distribution System (PDS) during the "Arduous March" famine of the mid-1990s, which forced households to turn to local marketplaces, known as *Jangmadang*, to obtain food and other basic necessities (Kim, 2017).[5] Initially an informal and illegal coping mechanism, these marketplaces became increasingly widespread and difficult to suppress. The regime institutionalized them by collecting stall fees and regulating their operating hours. By 2018, more than 400 officially authorized marketplaces were estimated to operate nationwide (Noh, 2017).

A wide range of agricultural, manufactured, and imported goods are traded in these marketplaces (see Fig. B.6). Over time, market activity has expanded beyond the exchange of goods to services, transportation, labor, and real estate (Kim and Yang, 2015; Hong et al., 2016), making these marketplaces important venues for economic exchange in North Korea. Households rely on these marketplaces for both income and consumption. During the early stages of marketization, from 1996 to 2003, *Jangmadang* activity generated approximately 78% of household income and accounted for more than 90% of household consumption (Kim and Song, 2008).[6] Even hospital treatment often requires patients to obtain medicines and supplies through markets because of shortages at medical facilities (Sunhwa, 2025). State-owned enterprises and collective farms also depend on market exchange. Under the planned system, they are required to meet state production quotas, but central provisioning has failed to provide sufficient materials for production. They therefore use markets to procure these materials and sell surplus output after meeting their quotas to cover operating costs (Lee et al., 2018).[7] Market exchange has thus become essential to the functioning of North Korea's economy, sustaining both household livelihoods and production.

[5] Famine mortality is estimated at 0.6–3.5 million deaths (3–17.5% of the population) (Goodkind et al., 2011).

[6] More recent defector survey evidence indicates that households purchased approximately 90% of their clothing in these marketplaces between 2011 and 2020 (Kim et al., 2022).

[7] This reliance on markets reflects institutional reforms toward decentralization and self-financing. The July 1 Economic Management Improvement Measures of 2002 expanded managerial autonomy and introduced greater flexibility in production, input procurement, and the sale of output. Under Kim Jong-un, the Socialist Enterprise Responsibility Management System further expanded enterprise autonomy over production and market transactions, while the Field Responsibility System gave collective farms greater control over production and the use of surplus output (Lee et al., 2018).

2.2 Information Frictions and the Mobile Phone Rollout

Access to information has historically been severely restricted in North Korea. To maintain institutional stability, the regime tightly controls information flows both across locations within the country and from the outside world. It does so by restricting domestic mobility,[8] monopolizing mass media,[9] and limiting access to the internet.[10] As a result, the information available to ordinary citizens was largely confined to their immediate locality or state-controlled channels.

These institutional barriers were compounded by severe limitations in communication infrastructure. Prior to public mobile phone service, North Korea's telecommunications consisted of a fragmented network of landlines, public phones, and telegrams. Landline penetration remained extremely low at approximately five subscriptions per 100 inhabitants for two decades (Fig. B.3). This chronic underdevelopment reflected high installation costs equivalent to several years of income, lengthy administrative approvals, and severe supply-side constraints (Hwang et al., 2009).[11] For those without landlines, other means of communication were similarly limited. Public telephones were spaced 4–12 km apart, and making a single call could take half a day (Kim, 2005).[12] Workplace phone use required callers to record personal information in state ledgers, while telegrams could take up to three days for interregional delivery (Hong et al., 2001). These institutional and infrastructural constraints made it prohibitively costly for most citizens to obtain and exchange information across locations.

The launch of the *Koryolink* 3G network on December 15, 2008 marked a structural break in this environment. Developed in partnership with Egypt's Orascom Telecom, *Koryolink* was the first mobile service available to the general public. Although a 2G service had existed since 1998, access was confined to political elites and foreign diplomats and was suspended following the 2004 Ryongchon station explosion, leaving the country without mobile service for over four years before *Koryolink*.[13] The introduction of public 3G services was driven by the regime's political and revenue-generating objectives rather than efforts

[8] Interregional travel required official permits that imposed substantial time and financial costs. For example, obtaining a permit could take one to three months, although bribery could expedite the process (Han et al., 2017).

[9] The state maintains a strict monopoly over mass media, prohibiting private ownership of media outlets. Newspapers and broadcasting are confined to state-run channels such as *Rodong Sinmun* and Korean Central Television (KCTV) (NIUE, 2025).

[10] Public access to the global internet is effectively nonexistent. Instead, the regime operates a closed domestic intranet (*Kwangmyong*) (Lee et al., 2017). Access nevertheless remained extremely limited: in 2017, only 1.4% of households had intranet access at home (DPRK and UNICEF, 2018).

[11] Subscribers bore the full costs of hardware, wiring, and installation. Administrative approval for landline installation often took months with no guarantee of service. Supply-side constraints included shortages of exchange equipment and reliance on manual switching in rural areas, which could accommodate only 50–60% of peak-hour demand (Hwang et al., 2009).

[12] Public telephones were typically located at local post offices, requiring substantial travel and waiting time.

[13] Launched through a Thai joint venture and limited to the Rajin–Sonbong SEZ and a few cities, the 2G service reached 20,000 subscribers, or 0.08% of the population, by 2003. Following the 2004 explosion, which the regime alleged was an assassination attempt on Kim Jong-il using a mobile phone, all transmissions were halted and devices confiscated (Lee et al., 2017).

to promote market activity. Politically, the launch served as the symbolic centerpiece of the "Strong and Prosperous Nation" (*Kangsong Daeguk*) campaign, timed for the 2012 centennial of Kim Il-sung's birth. The regime framed the technology as a symbol of modernization and a benevolent gift from the leadership.[14] The network also generated substantial revenue for the regime. Orascom's $400 million investment was the largest FDI in North Korean history outside China and South Korea[15], while the regime's monopoly over handset sales generated an estimated $600–$900 million in revenue by 2017 (Lee et al., 2017).

Mobile penetration expanded at an unprecedented rate, increasing nearly a thousandfold from roughly 7,000 subscriptions in 2009 to more than 6 million by 2021. By then, mobile subscriptions exceeded landline subscriptions more than sixfold (Fig. B.3). This rapid diffusion reflected the greater accessibility of mobile phones relative to traditional communication channels. Unlike landlines, mobile networks required no household-level infrastructure investment, and subscribers could obtain a handset and phone number on the day of purchase (Lee et al., 2017).Mobile access was also much more widespread in rural areas than landline service, with an urban–rural gap in penetration of only 1.6-fold (80.4% vs. 50.6%), compared with a 3.4-fold gap for landlines (61.9% vs. 18.1%) (DPRK and UNICEF, 2018). Mobile telephony thus substantially lowered barriers to real-time communication across locations.

2.3 Mobile Phones and Market Activity

Although introduced to serve the regime's political and revenue interests rather than to promote market activity, households adopted mobile phones in ways that fundamentally changed how markets operated in North Korea. Among 40 defectors who left North Korea between 2018 and 2023, 97.5% reported purchasing a mobile phone specifically for market activities. Despite high handset costs,[16] all respondents reported substantial economic returns from their mobile phone use, with income increasing two to sevenfold (Joung and Lee, 2023).[17] Notably, these economic gains occurred in a setting where mobile communication relied primarily on voice calls and text messaging.[18] The effects documented in this paper

[14] This pattern mirrors the earlier 2G deployment. Kim Jong-il ordered the Pyongyang network completed by April 15, 2002, to mark Kim Il-sung's 90th birthday. Consistent with this framing, defector interviews suggest the 3G rollout instilled a sense of national pride, as residents perceived the technology as a tangible sign of modernization (Lee et al., 2017).

[15] The deal also required Orascom to complete the Ryugyong Hotel, a prominent symbol of the regime that had remained unfinished for years due to financial constraints, ahead of the 2012 centennial (Lee et al., 2017).

[16] Among the 40 respondents, 55% reported handset costs equivalent to one to three months of income, while 45% reported costs exceeding eight months of income (Joung and Lee, 2023).

[17] Among respondents, 77.5% experienced a two to threefold increase in income, while 22.5% experienced a four to sevenfold increase. A separate survey of 126 defectors yielded similar findings, with 92% identifying increased market activity and information dissemination as the most significant changes brought by mobile phones to North Korea (Noh, 2019).

[18] Mobile communication in North Korea was largely restricted to voice and SMS. Multimedia messaging was briefly available in 2011 before being suspended on security grounds, while video calling was substantially more expensive and largely limited to elites. Internet and Wi Fi were disabled at the hardware level, while applications required in person installation at state stores.

therefore primarily reflect improved information access through direct, real-time communication rather than digital services such as market analytics or agricultural advisory tools.

Mobile phones facilitated market exchange through several channels. First, real time communication improved the coordination of transactions. A market vendor who ran out of shoes, for example, could immediately call a wholesaler with available inventory and place an order rather than lose potential sales. Similarly, fish traders could learn when boats would arrive and coordinate purchases in advance, which was valuable when trading perishable goods (Joung and Lee, 2023). Second, mobile phones lowered transaction costs across locations. Previously, merchants often had to travel to purchase goods while carrying cash, incurring transportation and travel permit costs as well as risks of theft or confiscation. With mobile phones, merchants could instead arrange deliveries through private delivery services (*servi-cha*) and settle payments through local intermediaries without traveling in person (Kim, 2015).[19] Finally, mobile phones expanded the geographic scope of trading relationships. One interviewee reported that merchants who had previously relied on local suppliers could instead deal directly with a Pyongyang wholesaler connected to a cigarette factory.

A particularly important use of mobile phones was to access price information and identify profitable trading opportunities. This was especially consequential in rural areas, where households historically had limited information about prices outside their local markets. Before mobile phones, rural households often sold food cheaply to visiting urban traders because they lacked information about prevailing prices elsewhere and had limited means to transport their goods. With mobile phones, farmers could learn current prices from urban traders before selling, and some even asked to be contacted when prices reached a level at which they were willing to sell. Similarly, livestock sellers could call several potential buyers, compare their offers, and sell to the highest bidder. As a result, farmers could sell at better prices than before (Joung and Lee, 2023). Overall, by sharply reducing the cost of acquiring and transmitting market relevant information, mobile phones made it easier to identify trading opportunities, coordinate transactions, and trade across locations.

Basic utility services, such as train schedules and weather information, were available through the voice network, but agricultural advisory or market analytics services were not (Lee et al., 2017).

[19] For example, a long-distance transaction by video call cost roughly one third as much as traveling 160 km (100 miles) in person, a trip that took about 13 hours (Moon, 2019).

3 Conceptual Framework

> "The most significant fact about this system is the economy of knowledge with which it operates, or how little the individual participants need to know in order to be able to take the right action."
>
> — F. A. Hayek, *The Use of Knowledge in Society* (1945)

This section presents a simple theoretical framework for understanding how mobile-phone access can expand market activity. As discussed in Section 2.3, mobile phones facilitated market exchange through several channels, including lower search and transaction costs and improved coordination and matching among market participants. The framework focuses on one of these channels: *reducing the cost of acquiring information* about selling opportunities. Combining a production decision with a sequential search problem, the framework shows how lower search costs alter the producer's optimal search behavior and raise the expected net selling price. A higher expected net selling price, in turn, strengthens incentives for market-oriented production, thereby expanding market activity.

Production decision. Consider a farmer who chooses how much effort to devote to market-oriented production. Let h denote production effort, capturing both labor and monetary investments, such as seeds and fertilizer. Output is given by $q(h)$ and produced at cost $c(h)$, where $q'(h) > 0$, $q''(h) < 0$, $c'(h) > 0$, and $c''(h) > 0$. Thus, greater effort increases output at a diminishing rate, while marginal production cost is increasing. Because the price at which output will ultimately be sold is uncertain at the time of production, the farmer bases the production decision on the expected net selling price. Let $\mathbb{E}[p]$ denote the expected selling price the farmer will receive after search, net of transportation and transaction costs.

The producer therefore solves the profit-maximization problem:

$$\max_{h \geq 0} \left\{ \mathbb{E}[p]\, q(h) - c(h) \right\}. \tag{1}$$

The optimal production effort h^* satisfies

$$\mathbb{E}[p]\, q'(h^*) = c'(h^*). \tag{2}$$

Implicitly differentiating Eq. (2) with respect to $\mathbb{E}[p]$ yields

$$\frac{\partial h^*}{\partial \mathbb{E}[p]} = \frac{q'(h^*)}{c''(h^*) - \mathbb{E}[p]\, q''(h^*)} > 0. \tag{3}$$

Eq. (3) shows that optimal production effort increases with the expected net selling price. Thus, a higher expected net selling price strengthens the producer's incentive to devote greater effort to market-oriented production. We next examine how the producer's search for selling opportunities shapes the price he expects to receive and how mobile-phone access affects this search.

Optimal Search Behavior and Mobile-Phone Access. At harvest, a producer must decide where to sell the output. Although he knows the distribution of prices across markets, the price offered at each market is unknown before search. He therefore searches sequentially across markets to obtain price information. Because each additional search requires time and resources, however, he cannot search indefinitely. Instead, after each quotation, he chooses whether to stop and sell at the highest price observed so far or to incur an additional search cost and continue searching. This trade-off gives rise to an optimal stopping problem that determines the producer's optimal search behavior and, consequently, the expected net selling price he receives.

Formally, let p denote the net price revealed by a search. Each search draws p from a known continuous distribution with cumulative distribution function F, probability density function f, and support $[\underline{p}, \overline{p}]$. Each additional search incurs a constant search cost $s > 0$. Search opportunities are assumed to be unlimited, and recall is allowed, so previously observed prices remain available. Therefore, the producer's decision depends only on the highest price observed so far, denoted by z. Following the recursive optimal-stopping framework of DeGroot (1968), let $V(z)$ denote the maximum expected payoff given the current state z. The producer's search problem can be written recursively as

$$V(z) = \max \left\{ \underbrace{z}_{\textit{Stop and sell}}, \underbrace{-s + \mathbb{E}\left[V(\max\{z, p\})\right]}_{\textit{Continue searching}} \right\}. \tag{4}$$

The Bellman equation implies that, at each stage of the search process, the producer compares the payoff from stopping and selling at the current best price, z, with the expected payoff from conducting one additional search after incurring the search cost. The optimal solution is characterized by a stationary reservation price, $r(s)$: the producer stops searching whenever $z \geq r(s)$ and continues searching whenever $z < r(s)$ (DeGroot, 1968). The reservation price is the threshold at which the producer is indifferent between stopping and continuing to search. This indifference condition implies

$$s = \int_{r(s)}^{\overline{p}} (p - r(s))\, f(p)\, dp. \tag{5}$$

The reservation equation states that the expected gain from an additional search equals its cost. Before the introduction of mobile phones, obtaining price information often required traveling to another market or relying on slower communication channels, making information acquisition costly. In contrast, *mobile-phone access* reduces the cost of obtaining price information and therefore lowers the search cost s. Differentiating Eq. (5) with respect to s shows how lower search costs affect the producer's optimal search behavior:

$$\frac{dr(s)}{ds} = -\frac{1}{1 - F(r(s))} < 0. \tag{6}$$

Since $1 - F(r(s)) > 0$, Eq. (6) implies that lower search costs raise the producer's reservation price. Intuitively, when obtaining additional price information becomes less costly, the producer is willing to continue searching rather than accept relatively low price offers. As a result, the producer becomes more selective and searches longer before stopping.

We next examine how this change in optimal search behavior affects the expected net selling price. Under the optimal stopping rule, the producer accepts only prices at or above the reservation price. For a generic reservation price r, the expected net selling price is

$$\mathbb{E}[p] = \mathbb{E}[p \mid p \geq r] = \frac{\int_r^{\overline{p}} p f(p)\, dp}{1 - F(r)}. \tag{7}$$

Eq. (7) nests two benchmarks. At one extreme, when search is so costly that the producer accepts the first offer, $r = \underline{p}$ and $\mathbb{E}[p] = \int_{\underline{p}}^{\overline{p}} p f(p)\, dp$, the unconditional mean price. This corresponds to selling without current price information. At the other extreme, as search becomes costless ($s \to 0$), Eq. (5) implies $r(s) \to \overline{p}$, and the producer receives the highest price available across markets. Because the best of several prices is never below any single one, better information can only raise the price the producer expects to receive. Mobile-phone access lowers s and moves the producer from the first benchmark toward the second.

To see how the expected net selling price changes between these benchmarks, differentiating Eq. (7) with respect to r yields

$$\frac{\partial \mathbb{E}[p]}{\partial r} = \frac{f(r)}{1 - F(r)} \left(\mathbb{E}[p] - r\right) > 0. \tag{8}$$

Thus, a higher reservation price increases the expected net selling price. Returning to

the optimal reservation price $r(s)$, combining Eqs. (8) and (6) gives

$$\underbrace{\frac{d\,\mathbb{E}[p]}{ds}}_{\substack{\text{Search cost and}\\ \text{expected net selling price}}} = \underbrace{\frac{\partial \mathbb{E}[p]}{\partial r}}_{\substack{\text{Reservation price and}\\ \text{expected net selling price}}} \cdot \underbrace{\frac{dr(s)}{ds}}_{\substack{\text{Search cost and}\\ \text{reservation price}}} < 0. \tag{9}$$

Eq. (9) summarizes the central mechanism of the search model: lower search costs raise the expected net selling price by changing the producer's optimal search behavior. Mobile-phone access therefore raises the price the producer expects to receive by lowering information-acquisition costs. This mechanism also has a distributional implication (Appendix A.4). When search costs fall and the reservation price rises, relatively low price offers that would have been accepted under a higher search cost are instead rejected, shifting the distribution of accepted prices toward higher values (Fig. A.1). As a result, the accepted-price distribution under a lower search cost first-order stochastically dominates that under a higher search cost (Fig. A.2).

Information, Incentives, and Market Activity. As established in Eq. (3), optimal production effort increases with the expected net selling price. Combining this result with Eq. (9) yields

$$\frac{dh^*}{ds} = \frac{\partial h^*}{\partial \mathbb{E}[p]} \cdot \frac{d\,\mathbb{E}[p]}{ds} < 0. \tag{10}$$

Eq. (10) shows that lower search costs increase optimal production effort by raising the net price the producer expects to receive. The framework therefore illustrates an incentive channel through which mobile-phone access can expand market activity: lower information-acquisition costs improve expected selling opportunities and encourage greater market-oriented activity.

4 Data

Empirical research on North Korea faces two data challenges: the scarcity of reliable official data and the lack of spatially granular data measured consistently over time. To address these constraints, I rely extensively on remote-sensing data, including historical high-resolution satellite imagery from Google Earth and Esri World Imagery Wayback Basemap at about 1.2 meters per pixel.

4.1 Cell Towers

To measure the rollout of mobile-phone access, I construct a county-year panel of cell-tower stock, defined as the cumulative number of towers in each county, from 2009 to 2021. This tower stock serves as the treatment variable. I identify tower locations and reconstruct construction years using historical imagery from Google Earth and Esri World Imagery Wayback.

Locations. I obtain candidate tower locations from the AccessDPRK project, a nationwide geospatial database developed for military purposes that systematically maps facilities across North Korea using Google Earth imagery through 2021.[20] The database classifies telecommunications structures by morphology. Cell towers are generally smaller than microwave and radio towers and are typically accompanied by a control building with solar panels to cope with the unreliable electricity supply (Fig. I, Panel A). After visually verifying each structure, I confirm 1,127 of the 1,249 telecommunications structures as cell towers.

Deployment Timing. Assuming that a tower remains in place once constructed, I date each tower by comparing the last image without the completed tower to the first image in which it appears.[21] When these two images are from consecutive years, the deployment year is exactly identified (Fig. I, Panel B(i)). Otherwise, the two images define a range of feasible deployment years. When the last image before completion shows ongoing construction, I assign deployment to the following year (Panel B(ii)).[22][23]

County-Year Treatment and Timing Uncertainty. I aggregate towers to 178 second-level administrative units (ADM2). North Korea's provinces are subdivided into cities (*si*) and counties (*gun*), which are parallel rather than nested units; cities are generally more urban and counties more rural. For simplicity, I refer to both as *counties* throughout the paper (Appendix B.1).

The deployment year is exactly identified for 566 of the 1,127 towers (50.2 percent), and nearly 95 percent of towers have either an exact year or a feasible range of at most two years (Appendix Table B.1). To account for the remaining uncertainty, I construct four county-

[20] To map the country systematically, the AccessDPRK project divided North Korea into approximately 2,000 5-by-5-mile grid cells and manually reviewed each cell for identifiable facilities. Each grid cell was reviewed about four times, and major cities about ten times.

[21] The satellite-based deployment year captures the physical completion of a cell-tower site rather than the date on which it becomes operational. Orascom Telecom documents additional steps between construction and a site going "on air," including equipment installation, testing, and commissioning. Measuring deployment at the annual level may reduce sensitivity to these timing differences. See https://www.scribd.com/document/441673005/ENG-PRO-0003.

[22] At locations with frequent imagery, tower construction is typically completed within one year. In Panel B(ii), for example, the site is under construction in 2015 and the next available image, from 2018, shows the completed tower, so I assign 2016 as the deployment year.

[23] Some towers are already visible in imagery before 2009. These sites likely belong to an earlier 2G network that operated from 2002 until mobile service was suspended in 2004. Because public mobile service resumed only with the launch of Koryolink in December 2008, I assign these towers to 2009.

year measures of tower stock. The *Exact* measure retains only county-years in which tower stock is known exactly and treats the rest as missing. My baseline *Uniform* measure fills these gaps by assuming a constant annual increase in tower stock between exactly known county-years, thereby also incorporating towers with uncertain deployment timing. The *Earliest* and *Latest* measures assign each uncertain tower to the first or last year of its feasible range, respectively. For example, a tower that could have been deployed between 2014 and 2018 is assigned to 2014 under *Earliest* and to 2018 under *Latest*. These two measures therefore provide upper and lower bounds on tower stock. I use the *Exact*, *Earliest*, and *Latest* measures to test the robustness of the results to alternative timing assumptions.

Validation. The cell-tower data are consistent with external evidence on tower locations, deployment timing, and mobile-phone diffusion. First, my tower locations closely match an independent map of cellular base stations compiled by 38 North and published in *The Economist* in 2022 (The Economist, 2022). The two maps show similar spatial distributions and comparable counts: more than 1,000 sites in the 38 North map and 1,127 in my data through 2021 (Appendix Fig. B.1). Second, the deployment timing tracks independent evidence on network expansion. Depending on the timing measure, 50–57 percent of towers were in place by 2011 and 87–94 percent by 2015 (Appendix Table B.2), in line with Koryolink's reported early expansion and the timing of base-station equipment imports (Appendix Fig. B.2, Panel A).[24] Third, handset imports and ITU estimates of mobile subscriptions rose sharply after the launch of Koryolink, indicating that mobile phones diffused alongside the expansion of the network (Appendix Figs. B.2, Panel B, and B.3).

4.2 Vegetation Index

To measure agricultural activity, I use the Normalized Difference Vegetation Index (NDVI), a standard satellite measure of vegetation greenness (Tucker, 1979). NDVI data come from the MODIS Terra sensor (MOD13Q1, Collection 6.1), which provides 16-day composite images at a 250-meter resolution (Didan, 2021). I keep only pixels flagged as good quality and use the years 2002–2015.[25]

Land Types. I classify land using North Korea's slope-based land management policy. Land with a slope below 15 degrees is reserved for collective farms under the Ministry of Agriculture, whereas land at or above 15 degrees is officially designated as forest land under the Ministry of Land and Environment Protection (FAO and WFP, 2019).[26] Since the late

[24] Koryolink reported 453 base stations by the end of the third quarter of 2011. In the trade data, 45 percent of base-station equipment imports (HS 851761) arrived by 2011 and nearly all by 2015. Because North Korea does not publish official trade statistics, I use mirror data, which measure North Korea's imports by its trading partners' reported exports to North Korea.

[25] I end the sample in 2015 because the state began confiscating cultivated plots on steep slopes and expanded afforestation in the mid-2010s, which changed vegetation on sloping land for reasons unrelated to market incentives.

[26] Land below 15 degrees also includes some non-agricultural land, such as built-up areas, roads, and rivers. I refer to it as collective

1990s, however, households, including both collective-farm members and urban residents, have cultivated small private plots on this sloping land outside the state plan (Appendix Fig. B.4) (FAO and WFP, 2019).[27] Vegetation on sloping land therefore captures agricultural activity where households can keep the returns to additional output, while collective farmland serves as a comparison. I compute slope from the SRTM digital elevation model at the 250-meter resolution of the MODIS pixels, so that the land classification and the vegetation measure refer to the same area. Sloping land and collective farmland are interspersed within counties (Appendix Fig. B.5).[28]

Tree Cover. Not all sloping land can be cultivated. Additional planting is possible on slopes with few trees, but not in dense forest. I therefore divide sloping land by its tree cover in 2000, measured by the Global Forest Change data at a 30-meter resolution and averaged within each MODIS pixel (Hansen et al., 2013): low ($<$30 percent), medium (30–70 percent), and high ($\geq$70 percent). Because tree cover is measured almost a decade before the first cell towers were built, this classification cannot be affected by mobile-phone access. If the vegetation response reflects cultivation, it should be largest on sloping land with low tree cover and smallest on densely forested slopes.[29]

Measuring Agricultural Activity. For each pixel, I compute the mean NDVI in each month from April to August, which covers the main growing season, and average these monthly means. I then average across pixels within each county, land type, and year, and use 100 times the log of this average as the outcome.[30] NDVI measures greenness rather than output, but greenness during the growing season is closely related to crop growth and yields (Burke and Lobell, 2017; Asher and Novosad, 2020). My main vegetation outcome is the difference between sloping land and collective farmland within the same county and year, which removes shocks common to all land in a county, such as weather. I also report vegetation on each land type separately, which shows whether a change in the difference comes from sloping land or from collective farmland. The results are robust to alternative measures, including the April–September mean, the seasonal peak, and the difference between the peak and early-season NDVI (Asher and Novosad, 2020) (Appendix Table E.6).

farmland for simplicity.

[27] In 2012, the government banned the reclamation of slopes steeper than 15 degrees, and an afforestation campaign launched in 2014 led to a gradual decline in production on sloping land (FAO and WFP, 2019). These policies work against finding an increase in vegetation on sloping land.

[28] As a robustness check, I compute slope at the 30-meter resolution of the elevation model and assign a pixel to a land type only if more than half of its 30-meter cells fall in that type. The results are similar (Appendix Table E.6).

[29] The Global Forest Change data detect large-scale forest loss but not the gradual clearing of forest edges that is common in North Korea. North Korean forestry statistics report a loss of 1.38 million hectares of stocked forest between 1999 and 2008, whereas the Global Forest Change data record about 65,000 hectares of loss on sloping land between 2001 and 2008. Slopes classified as having high tree cover may therefore still contain cultivated plots, which would make the difference across tree-cover groups smaller.

[30] All measures use the same set of pixels: those with valid observations in both the early (April 1–May 12) and main (June 15–August 31) parts of the growing season.

4.3 Marketplace Area

I measure local market activity using the physical area of officially sanctioned marketplaces (*jangmadang*), described in Section 2.1. Using satellite imagery, I measure the annual footprints of 536 marketplaces from 2008 to 2021 and sum their areas within each county to form a county-year panel (Yang et al., 2026).

Interpreting Marketplace Area. Marketplace area measures the physical scale of market activity. Marketplaces operate at fixed and physically bounded sites under the supervision of local market management offices, and vendors pay recurring stall fees (*jangse*) to occupy designated selling space (Fig. B.7).[31] Because selling space is costly, vendors rent it only when expected returns justify the fee, so changes in marketplace area reflect vendors' entry and exit. Entry adds stalls or expands a market's perimeter, while exit can lead to the removal of market structures.[32] Marketplace area therefore captures changes in market activity that are persistent enough to alter a market's physical footprint, rather than short-run fluctuations in transaction volume or trade intensity.

Measuring Marketplace Footprints. Marketplace locations are drawn from three independent sources that identify North Korean marketplaces using satellite imagery and defector interviews (Hong et al., 2022; Cha and Collins, 2018; NGII, 2024). Merging these sources and verifying each site against historical imagery yields 482 marketplaces.[33] Marketplaces take two distinct physical forms that require different measurement approaches (Appendix Fig. B.8). Open-air markets consist of rows of stalls within a defined perimeter and are delineated using a DeepLabv3+ semantic segmentation model.[34] Enclosed markets consist of permanent buildings that are difficult to distinguish from other large-span structures and are therefore delineated manually. Applying the segmentation model to built-up areas nationwide identifies 54 additional marketplaces absent from the three original sources, bringing the total to 536 marketplaces.

[31] These official marketplaces are the primary venues for private exchange in North Korea because they give authorities, vendors, and buyers incentives to concentrate trade there. Authorities collect stall fees, vendors can trade without facing crackdowns, and buyers can inspect goods, negotiate prices, and compare a wide variety of products in one place (Kim and Ahn, 2024). By contrast, unauthorized forms of trade, including house-based markets (*maetaek*), alley markets (*golmok-sijang*), and transient "grasshopper markets" (*mettugi-sijang*), remain subject to crackdowns and are not captured by my measure.

[32] Although local authorities approve the construction and expansion of marketplaces, vendors largely finance these investments. When a marketplace is newly built or relocated, local people's committees contribute little, and prospective vendors pay the construction costs and are often mobilized as labor (Radio Free Asia, 2023). In Chongjin, for example, authorities financed a new permanent marketplace in 2004 by collecting fees from prospective vendors in advance (Daily NK, 2005). Likewise, when state control over markets loosened in the mid-2010s, marketplaces in Hyesan added stalls in previously unused space, raising stall-fee revenue for market management offices (Daily NK, 2015).

[33] The three sources contain 414, 436, and 383 marketplaces, respectively, yielding 497 unique locations after merging. Fifteen locations show no identifiable marketplace structures in historical imagery and are excluded.

[34] Model performance is evaluated by comparing predicted marketplace areas with areas manually labeled by researchers in a validation set not used in training. The F1 score for the marketplace class, which balances precision and recall, is 0.707, and the mean intersection-over-union (mIoU), which measures the spatial overlap between predicted and manually labeled areas averaged over the marketplace and non-market classes, is 0.769. Both metrics range from zero to one, with higher values indicating better performance. See Yang et al. (2026) for details on model training and evaluation.

County-Year Outcome. I sum the areas of all marketplaces within each county to obtain total marketplace area, in square meters, for each county-year. Counties without an active marketplace are assigned zero. Because imagery is not available for every marketplace in every year, however, a missing observation does not necessarily mean that a marketplace was inactive. Summing only observed areas could thus confound changes in imagery availability with changes in market activity. For example, if a marketplace is observed in 2010 and 2012 but no image is available in 2011, its absence in 2011 would mechanically lower the county total in that year and produce an artificial jump from 2011 to 2012. I therefore construct a complete annual series for each marketplace before aggregating to the county level. For years without imagery, I use three rules. First, I linearly interpolate area between two observed years. Second, when imagery is unavailable before the first observation, I carry the first observed area backward over earlier active years. Third, when imagery is unavailable after the last observation, I carry the last observed area forward over later active years.[35] I refer to this baseline measure as *Full Interpolation*.

Alternative Measures. The baseline measure can blur the timing of marketplace expansion, because interpolation spreads changes evenly across unobserved years, and can understate growth at the beginning and end of each series, because carried values are held constant. I therefore construct three alternative measures. *Interior Interpolation* applies only the first rule. *Observed Only* uses only directly observed areas, together with zeros for years in which a marketplace was not active, which avoids filling altogether but reintroduces the imagery-availability problem described above. *Complete Case* retains only county-years in which every active marketplace in the county is directly observed, which avoids both filling and this problem at the cost of a smaller sample (Section 7 and Appendix Table E.8).

Validation. Yang et al. (2026) validate marketplace area as a measure of local economic activity using four complementary approaches based on ground-level data, aggregate economic shocks, and independent satellite-based measures. First, across counties in the 2008 population census, marketplace area has a correlation of 0.897 with urban population, higher than that of total building area (0.835) or nighttime lights (0.480).[36] Second, aggregate marketplace area responds to major national economic disruptions in the expected direction, contracting after the 2016–2017 UN sanctions and the 2020 COVID-19 border closure with China. Third, county-level panel regressions with county and year fixed effects show a positive and statistically significant association between marketplace area and nighttime lights, indicating that marketplace area co-moves with an independent measure of local economic activity within counties. Finally, long-difference regressions us-

[35] Consecutive imagery observations are on average about two years apart, and more than 90 percent of gaps span three years or less.
[36] Other spatial proxies include SiScore (0.718), industrial building area (0.686), and road length (0.613). SiScore is a satellite-based composite measure of socioeconomic development (Ahn et al., 2023).

ing high-resolution building-footprint data show that marketplace expansion is associated with building growth concentrated in the immediate vicinity of market sites, and the association weakens with distance from the market. This spatial concentration suggests that marketplace area captures activity tied to the marketplaces themselves rather than broader county-level development. Together, these exercises show that marketplace area tracks economic activity across independent data sources, spatial scales, and time horizons.

5 Empirical Strategy

5.1 Identification

Baseline Specification. I estimate the effects of mobile-phone access using the two-way fixed effects (TWFE) specification

$$Y_{ct} = \beta T_{ct} + \alpha_c + \gamma_t + \varepsilon_{ct}, \tag{11}$$

where Y_{ct} is an outcome in county c and year t, and T_{ct} is the cumulative number of cell towers, constructed using the Uniform timing measure (Section 4.1). County fixed effects, α_c, absorb time-invariant differences across counties, and year fixed effects, γ_t, absorb shocks common to all counties. In alternative specifications, I replace year fixed effects with province-by-year fixed effects, which compare counties within the same province and year. Standard errors are clustered by county. For vegetation, the main outcome is the difference in $100 \times \ln(\text{NDVI})$ between sloping land and collective farmland in the same county and year (2002–2015), which removes local shocks, such as weather, that affect both land types similarly. For exchange, the outcome is total marketplace area in square meters (2008–2021).

I measure treatment as the cumulative number of towers rather than a binary indicator of access, because each additional tower extends coverage and capacity to more people and places. This choice is closer to studies of national rollouts that measure the share of the population or area covered (Manacorda and Tesei, 2020; Guriev et al., 2021) than to studies that compare places before and after first coverage (Jensen, 2007; Aker, 2010). It also suits the setting, as more than half of all counties received their first tower in 2009, while the number of towers kept growing for more than a decade.

Identifying Assumption. Cell towers were not placed at random. The identifying assumption is that counties that received towers earlier or in larger numbers would have followed the same outcome trends as other counties without the network. Because the tower count varies both in when it first rises and in how much it rises afterward, the assumption

concerns both the timing and the intensity of the rollout. I examine timing with the event study below and intensity in Section 7. Reading β as the average effect per tower further requires that counties receiving many towers were not places where mobile phones would have mattered more in any case (Callaway et al., 2024). Because towers went first to more populous counties and to those with a train station, where the marketplace effect is also larger, the TWFE estimate may overstate the average effect per tower. I therefore compare it with heterogeneity-robust estimates (de Chaisemartin and D'Haultfœuille, 2024) and with estimates from comparable counties (Section 7).

Event Study. Because counties gained access in different years, a conventional TWFE event study would use already-treated counties as comparisons and can misweight effects that vary across cohorts or over time (De Chaisemartin and d'Haultfoeuille, 2020; Sun and Abraham, 2021). I therefore use the estimator of Callaway and Sant'Anna (2021). Let G_c denote the year of a county's first tower, dated by the midpoint of its feasible range. For cohort g and year t, the estimator compares the change in outcomes since year $g-1$ between counties first treated in g and counties not yet treated in t, including those never treated,

$$ATT(g,t) = \mathbb{E}\left[Y_{ct} - Y_{c,g-1} \mid G_c = g\right] - \mathbb{E}\left[Y_{ct} - Y_{c,g-1} \mid G_c > t\right]. \tag{12}$$

I average these estimates across cohorts, weighted by cohort size, to obtain the effect $e = t - g$ years after first access, measured relative to the year before access.[37] Estimates before access show whether outcomes were already diverging, and estimates after access trace how the response evolved. Because this design uses the timing of first access but not later tower additions, it complements the TWFE estimates.

Network Scale and Duration. Equation (11) imposes the same effect for every tower, whether it is a county's first or its tenth and however long the county has been connected. I relax both restrictions by estimating

$$Y_{ct} = \underbrace{\beta_1 \mathbb{1}\{T_{ct} > 0\}}_{\text{Access}} + \underbrace{\beta_2 T_{ct}}_{\text{Network scale}} + \underbrace{\beta_3 d_{ct}}_{\text{Duration}} + \underbrace{\beta_4 (T_{ct} \times d_{ct})}_{\text{Network scale} \times \text{duration}} + \alpha_c + \gamma_t + \varepsilon_{ct}, \tag{13}$$

where d_{ct} is the number of years since the county's first tower, equal to zero in that year and before. β_1 captures any discrete change at connection beyond the linear effect of the number of towers, and an additional tower d years after access raises Y_{ct} by $\beta_2 + \beta_4 d$. β_3 absorbs changes with time since access that are common to all connected counties, so that β_4 compares counties with different network sizes but the same duration of access. The two outcomes yield different predictions for β_4. Vegetation on sloping land is closer to a flow,

[37] Pre-period estimates use a common base period ($g-1$), following Roth (2026). The default in Callaway and Sant'Anna (2021) instead compares each pre-period year with the year before it, so that pre-period and post-period estimates are not directly comparable.

because crops are replanted each season, so β_4 should be close to zero. Marketplace area is closer to a stock, because expanded space persists, so it should reflect accumulated exposure, implying $\beta_4 > 0$. Because most counties were connected in 2009, β_3 is identified from the few later-connected and never-connected counties, while β_4 is identified from whether counties with larger networks diverge over time from those with smaller ones.

5.2 Measurement Error

The construction year of about half of the towers is known within a range rather than exactly (Section 4.1). Unlike classical measurement error, which attenuates estimates toward zero, this error does not bias the TWFE estimate under two conditions. The Uniform measure spreads each uncertain tower equally across its feasible years, so if construction is equally likely in each of these years, the measured tower stock equals the expected true stock given the available imagery, and its error is uncorrelated with the measured stock. If, in addition, the timing of available imagery is unrelated to outcome shocks, OLS estimates of β remain consistent, and the error raises standard errors without attenuating the estimates (Appendix C.1). Because construction may instead be concentrated early or late within the range, I also report estimates under the Earliest and Latest measures, the two extremes of the feasible timing, and under the Exact measure, which counts towers whose construction year is known exactly. For the event study, where midpoint dating may misclassify some county-years, I report versions that use the earliest and latest feasible years and that restrict the treated sample to counties whose first year of access is known exactly (Appendix Figs. E.3 and E.4).

6 Results

6.1 Vegetation on Sloping Land

Table 1 reports estimates of Equation (11) for vegetation. Each additional cell tower increased vegetation on sloping land relative to collective farmland by 0.130 percent with county and year fixed effects and by 0.076 percent with province-by-year fixed effects (column 3). With about four towers in the average county by 2012, these estimates imply that the network raised vegetation on sloping land by roughly 0.3 to 0.5 percent relative to collective farmland by that year. Estimated separately, the response came from sloping land, where vegetation rose with tower exposure, while vegetation on collective farmland did not change significantly (columns 1 and 2). This contrast is what the conceptual framework

predicts. Households keep the returns from their private plots on sloping land, but not from collective farmland.

If the response reflects cultivation of private plots, it should be largest where land could be cleared and smallest in dense forest. Columns 4–6 split sloping land by tree cover in 2000, almost a decade before the first cell towers were built. The effect declines with tree cover, from 0.159 on land with low tree cover and 0.166 on land with medium tree cover to 0.075 on land with high tree cover, which is not statistically significant. With province-by-year fixed effects, the corresponding estimates are 0.106, 0.114, and 0.019. The difference between low and high tree cover is statistically significant (column 7). Figure III summarizes these estimates.

The increase is concentrated early in the growing season (Appendix Table E.7). Using April to mid-May alone, the estimate doubles to 0.262, while it is close to zero in the peak season from mid-June to August, when NDVI saturates over most vegetated land. The early-season increase is smaller on densely forested slopes, which argues against earlier leaf-out of existing forest. It is consistent with early-season cultivation on open slopes, although regrowth of grasses and shrubs on open land cannot be ruled out with these data.

Figure IV presents the event study around first access. Before access, the estimates are close to zero and show no trend, so counties that gained access earlier were not already on different vegetation paths. After access, vegetation on sloping land rises relative to collective farmland, with estimates that grow over time and become statistically significant five and six years after access. These late-horizon estimates are identified relative to the few never-treated counties and should be interpreted with caution. Figure V repeats the event study by tree cover. At late horizons, vegetation also rises somewhat on land with high tree cover, suggesting that a component common to all land is partly mixed in. On top of this component, however, the increase is much larger on land with less tree cover.

6.2 Marketplace Area

Table 2 reports estimates for marketplace area. Each additional cell tower increased marketplace area in a county by 337 square meters with county and year fixed effects and by 297 square meters with province-by-year fixed effects, about 2.5 percent of the average county's marketplace area in 2021. The estimates are similar across the four measures of tower timing. Applied to the 1,127 towers built by 2021, the estimate implies that the network added about 380,000 square meters of marketplace area nationwide, or roughly 220,000 stalls of 1.7 square meters (Hong et al., 2016).

Figure VI presents the event study. Marketplace area shows no differential trend before

access, so marketplaces were not already expanding faster in counties that gained access earlier. After access, the response emerges gradually. Estimates are close to zero in the first years, rise from three to four years after access, and reach about 1,700 square meters twelve years after access. The delay fits the nature of the outcome. Unlike vegetation, which responds within a season, marketplace area expands when vendors finance new stalls or new marketplaces, which may take years of coordination among vendors, buyers, and local authorities.

Because the effect of towers grows with time since access, the TWFE estimator may assign negative weights to some comparisons. In the TWFE regression with county and year fixed effects, 1,181 of the 2,208 treated county-year cells receive negative weights, which sum to -0.48 for marketplace area (-0.18 for vegetation). Heterogeneity-robust estimates are less precise, as the near-simultaneous rollout in 2009–2010 leaves few not-yet-treated comparison counties, but they remain positive in every specification (Appendix Table E.9). For marketplace area, the estimate is 146 square meters per tower, about 43 percent of the TWFE coefficient, and statistically significant. This estimate implies about 165,000 square meters nationwide, or roughly 100,000 stalls. The TWFE estimate may therefore be closer to an upper bound. For vegetation, point estimates are similar to or larger than the TWFE estimates, particularly on land with low tree cover, but are not statistically significant.

The marketplace effect is not concentrated in counties bordering China or in counties favored by the state (Section 6.4), and it does not follow the time path of trade between China and North Korea (Section 7).

6.3 Network Scale and Duration

Table 3 reports estimates of Equation (13) (Appendix C.2). Columns 1 and 4 reproduce the baseline estimates. In columns 2 and 5, the access indicator is negative and not statistically significant for either outcome (-0.298 for vegetation and -654 for marketplace area), while the coefficient on the number of towers barely changes (0.133 and 338). The effects therefore come from the size of the network rather than from connection itself.

Columns 3 and 6 allow the effect of the network to change with time since access, and the two outcomes diverge as the flow and stock distinction predicts. For vegetation, the interaction between towers and duration is zero (0.000), so the effect of an additional tower does not grow with exposure. For marketplace area, the interaction is 43 square meters and statistically significant, so each additional tower adds about 43 square meters more for every year the county has been connected. Duration alone does not raise marketplace area (-137), indicating that the growth with exposure comes through the network rather than through

the passage of time. Figure VII relaxes the linear interaction by estimating the effect of one tower separately for 0–1, 2–3, 4–6, 7–9, and 10–12 years since the first tower. The effect rises with years since access for marketplace area but not for vegetation.

6.4 How and Where Marketplaces Expanded

New versus Existing Marketplaces. Table 4 decomposes total marketplace area into marketplaces that opened after 2008 (entry), marketplaces active in 2008 that remained active (continuing), and marketplaces active in 2008 that later closed (exit).[38] The effect operates mainly through the extensive margin. New marketplaces account for nearly all of the increase, 335 of 337 square meters per tower. Continuing marketplaces grew as well, by 76 square meters, but this intensive-margin growth was offset by a decline among exiting marketplaces of 74 square meters, which is not statistically significant. If new marketplaces were relocations of existing ones, the area of exiting marketplaces would fall by a similar amount. The estimate for exiting marketplaces is small and imprecise, so relocation can explain at most about one fifth of the effect on new marketplaces, based on the point estimate.

Identifying relocations directly yields a similar conclusion (Appendix Table D.1, Panel B). I classify a new marketplace as a relocation if a 2008 marketplace within 2 kilometers was active in the year before it opened and then closed. Under a narrow definition, in which the nearby marketplace closed in the year the new one opened, 11 of the 94 new marketplaces are relocations, and they account for 35 square meters of the effect per tower, which is not statistically significant. Under a broad definition, which also counts closures within two years and contractions of more than half, 20 new marketplaces are relocations, accounting for 112 square meters. Because the broad definition also counts closures unrelated to the new marketplaces, it is an upper bound. Even under this definition, new marketplaces that were not relocations account for 223 square meters per tower, about two thirds of the effect.

Where New Marketplaces Opened. New marketplaces opened close to existing ones. Figure VIII compares the distance from each new marketplace to the nearest 2008 marketplace with the distance between 2008 marketplaces. Sixty-three percent of new marketplaces opened within 2 kilometers of a 2008 marketplace, whereas 29 percent of 2008 marketplaces had another 2008 marketplace that close. New marketplaces within 2 kilometers account for 200 of the 335 square meters per tower, while those 5 kilometers or farther away show no significant effect (Appendix Table D.1, Panel A). New marketplaces thus filled in areas where trade already took place, where demand already existed and transaction costs were

[38] Of the 536 marketplaces, 430 were active in 2008. Of these, 391 remained active through 2021 and 39 closed in at least one later year. Another 94 marketplaces first became active between 2009 and 2021. The remaining 12 were not active between 2008 and 2021 and are excluded.

likely lower, a pattern consistent with agglomeration forces (Duranton and Puga, 2004). Exchange did not concentrate in a single hub, however. The largest marketplace in each county in 2008 showed no significant growth (-47 square meters per tower), while the other marketplaces in the county grew by 383 square meters and the number of marketplaces rose (Appendix Table D.2).

Heterogeneity by County Characteristics. Table 5 examines which counties saw larger effects, interacting the number of towers with baseline county characteristics. The clearest pattern concerns rail. In counties with a train station, each additional tower increased marketplace area by about 425 square meters more than in counties without one, a difference that holds within provinces and after controlling for population. Rail remains the main mode of transporting goods in North Korea, while road density does not predict larger effects. Effects were also larger in more populous counties and in those closer to cities, where most wholesale markets are located, although these two characteristics cannot be separated. These patterns suggest that mobile connectivity complements physical connectivity, with better information translating into more exchange where goods could move. By contrast, effects were not significantly larger in counties closer to Pyongyang or with special economic zones, suggesting that marketplace expansion was not concentrated in areas favored by the state. In counties bordering China, which were already within range of Chinese mobile networks, the rollout had no detectable effect, plausibly because the domestic network added little new information.

The vegetation response, in contrast, was not concentrated in counties with better initial transport infrastructure (Appendix Table D.3). If anything, it was larger in less populous counties, although this pattern does not hold within provinces. Producers in less-connected areas thus appear to have responded to better information as well.

7 Robustness

This section examines the validity of the identifying assumption and the robustness of the findings. Section 7.1 addresses the concern that towers were built where outcomes would have grown anyway, along with alternative explanations and spillovers. Section 7.2 examines alternative measures of the treatment and outcomes, and Section 7.3 examines alternative estimators.

7.1 Threats to Identification

Selection into the Rollout. Towers did not arrive at random. Counties that gained access in 2009 were more populous and more urban than counties that gained access later, and more of them had a train station (89 versus 60 percent), industrial facilities, and ports (Appendix Table E.1). The number of towers built by 2015 and 2021 follows a similar pattern. Early and later counties did not differ, however, in road density, terrain, or pre-rollout trends in nighttime lights and vegetation. Because Koryolink charged fees for its service, it is natural that towers went first to larger markets for phones. The question is whether these counties would have seen faster growth in outcomes even without mobile phones. The next two sets of tests address this question.

Pre-Existing Trends and Reverse Causality. Appendix Table E.2 reports three sets of tests, each motivated by a different scenario. First, towers may have gone first to places that were already changing, or a temporary shock just before the first tower may have reverted afterward. Following Gonzalez (2021), I keep the placement of towers fixed and move their timing earlier, regressing outcomes on future towers in years before any tower existed. Any effect would then reflect where towers were placed rather than mobile phones. These placebo estimates are small and not significant for marketplace area and for sloping land with low tree cover, whereas future towers predict higher vegetation on collective farmland (Panel A). Second, Koryolink may have added towers where markets were growing. Holding current towers fixed, towers one to three years ahead jointly predict none of the outcomes (p-values of 0.18 to 0.88), and marketplace growth does not predict tower additions in the following three years (Panel B). Third, places already growing faster or catching up may have kept doing so. The marketplace estimate is unchanged by controlling for pre-rollout trends interacted with year fixed effects, for the pre-rollout level of marketplace area interacted with year fixed effects, or by dropping the two years before each county's first tower (Panel C). For vegetation, the response on sloping land with low tree cover survives every control. Because vegetation on collective farmland was already higher just before the first tower, the difference between sloping land and collective farmland loses significance when its pre-rollout level is controlled for (0.030), although it survives the other controls.

Comparisons among Similar Counties. Counties that received more towers may differ in ways that the fixed effects do not absorb. Appendix Table E.3 therefore compares counties with similar characteristics. The characteristics used for matching are selected by double-selection LASSO (Belloni et al., 2014) from 13 pre-rollout county characteristics, which avoids choosing them by hand. For vegetation, nearest-neighbor matching, propensity-score stratification, and entropy balancing yield estimates of 0.087 to 0.158, similar to the

baseline, even when counties are matched on their pre-rollout vegetation. For marketplace area, these estimates range from 169 to 243 square meters, about 50 to 70 percent of the baseline and less precise, close to the heterogeneity-robust estimate in Section 6.2. Comparing adjacent counties yields a marketplace estimate of 298 square meters but a vegetation estimate close to zero (0.020), plausibly because the vegetation response extends across county borders, as discussed below.

Alternative Explanations. Changes that coincided with the rollout may offer alternative explanations (Appendix Table E.4). Rainfall could raise vegetation on sloping land more than on flat land. Controlling for precipitation, including its interaction with the share of sloping land, leaves the vegetation estimate at 0.121 with year fixed effects and 0.074 with province-by-year fixed effects (Panel A). Road construction could have expanded both phone use and trade. Allowing counties to follow different paths according to road density in 2008 and road growth between 2008 and 2014 leaves the marketplace estimate similar but weakens the difference between sloping land and collective farmland, while the response on sloping land with low tree cover remains the most robust (Panel B).

Trade with China grew during the rollout and could have driven marketplace expansion. Allowing counties to follow different paths according to whether they border China, their distance to the nearest border customs office, and their exposure to the 2016–2017 UN sanctions (Kim et al., 2023) leaves the estimates similar (Panel C). The marketplace effect also did not weaken after the sanctions took effect in 2017 (Panel D). Together with the absence of an effect in border counties (Section 6.4), these results argue against cross-border trade as the main driver of marketplace expansion.

Spatial Spillovers. Towers near a county border may extend coverage into neighboring counties. If so, counties with few towers would be partly treated, and own-county estimates would understate the effects. Appendix Table E.5 adds the average number of towers in adjacent counties. For marketplace area, the own-county estimate remains between 257 and 304 square meters, and towers in adjacent counties have small and mostly insignificant effects. For vegetation, the own-county estimate remains positive and significant (0.080 with year fixed effects and 0.076 with province-by-year fixed effects), and towers in adjacent counties also raise vegetation (0.285 and 0.199). This spillover suggests that the producer response extended beyond county borders and that the total effect of the network on vegetation is likely larger than the own-county estimate (Miguel and Kremer, 2004; Butts, 2021).

Randomization Inference. Conventional standard errors may understate uncertainty if the rollout and outcomes share spatial or temporal patterns. I therefore reassign each county's entire path of towers to another county 1,000 times and re-estimate the baseline

regression (Appendix Figure E.1). When paths are reassigned within provinces, so that each placebo follows the timing of rollout in the same province, the two-sided p-values are 0.008 for marketplace area and 0.004 for vegetation (Appendix Figure E.2).

7.2 Alternative Measurement of Treatment and Outcomes

Tower Timing. The construction year of about half of the towers is known within a range rather than exactly (Section 5.2). The TWFE estimates are similar under the Earliest, Latest, and Exact measures (Tables 1 and 2). The event studies also show little evidence of differential pre-trends and the same post-access increase when first access is dated by the earliest or latest feasible year, or when the treated sample is restricted to counties whose first year of access is known exactly (Appendix Figures E.3 and E.4, Panel A).

Treatment Onset. The event studies define treatment onset as the arrival of a county's first tower. Panel B of the same figures instead defines onset as the year in which a county reached 25, 50, or 75 percent of its eventual towers. Because higher thresholds are typically reached several years after first access, the periods before onset under these definitions may already include years with mobile-phone coverage, and these estimates are less precise.

Marketplace Area. The baseline outcome uses the Full Interpolation measure, which fills years without imagery (Section 4.3). Appendix Table E.8 considers progressively less filled measures. The estimate remains positive under all four measures. The larger coefficients under Interior Interpolation and Observed Only should not, however, be read as larger economic effects. Their pre-treatment means fall from 10,158 square meters under Full Interpolation to 4,144 and 3,168 square meters, respectively, because imagery is particularly sparse in the early years of the panel, so these measures mechanically understate marketplace area when existing marketplaces are not observed. The Complete Case measure, which retains only county-years in which every active marketplace is observed, avoids this problem and yields an estimate of 277 square meters per tower, similar to the baseline but less precise because the sample falls from 2,492 to 658 observations. Holding the baseline measure fixed and restricting the sample to county-years in which at least 75 or 90 percent of marketplaces are directly observed yields estimates of 442 and 498 square meters, both significant at the 1 percent level. The event study using Interior Interpolation, the measure that relies least on filling, shows no pre-trend and the same dynamics under every definition of first access (Appendix Figure E.5).

Vegetation. Appendix Table E.6 reports alternative vegetation measures. The estimate remains positive when sloping land is restricted to slopes of 30 degrees or more, which leaves fewer counties, and when vegetation and slope are measured with Landsat imagery

at 30 meters, which is less precise because many growing-season scenes are lost to clouds. It also remains positive when the outcome is the difference in NDVI levels rather than logs.

7.3 Alternative Estimators

With a staggered rollout and effects that vary over time, results may depend on the choice of estimator. Appendix Figure E.6 re-estimates the first-access event studies with alternative estimators. For marketplace area, the estimators of Callaway and Sant'Anna (2021), Sun and Abraham (2021), Borusyak et al. (2024), and Wooldridge (2025) agree, with no pre-trend and a rise from three or four years after access, while the conventional TWFE event study yields smaller estimates. For vegetation, all estimators show significant increases five and six years after access. The estimators that use never-treated counties as the sole comparison group show negative pre-period estimates, because the never-treated counties in the vegetation sample, all in Jagang Province, do not follow the vegetation changes of the treated cohorts. For the continuous treatment, heterogeneity-robust estimates are discussed in Section 6.2.

8 Conclusion

This paper shows that lower information costs can expand market activity itself, beyond making existing markets more efficient. Because reliable statistics on local economic activity are scarce in North Korea, I construct new panel data using satellite imagery and deep learning to track cell towers, production, and exchange over more than a decade. I find that the country's first public mobile-phone network increased agricultural activity on sloping land, where households cultivate private plots and keep the returns, but not on collective farmland. This contrast is consistent with the conceptual framework's prediction that better information strengthens production incentives when producers keep the returns, affecting not only where and at what prices they sell but also how much they produce in the first place. The network also expanded exchange. The estimates imply that the towers built by 2021 increased marketplace area by the equivalent of roughly 100,000 to 220,000 stalls nationwide, depending on the estimator. This expansion came almost entirely from new marketplaces opening near established ones and was larger in counties with a train station. These patterns suggest that improved access to information complements existing market activity and physical connectivity in expanding exchange.

These findings have implications for North Korea. Like households elsewhere, North Korean households used better information to make decentralized economic decisions and

responded to incentives. Because exchange expanded more where a long-established rail network was in place, further investments in both digital and physical connectivity could bring substantial gains to ordinary North Koreans. The measures developed here also offer a way to track economic conditions in North Korea, where reliable statistics are rarely available, and could help guide humanitarian assistance or future economic cooperation.

Beyond North Korea, the findings offer two broader lessons about the economic effects of information and communication technology. First, access to information is not the same as the ability to act on it. Better information reveals new market opportunities, but people act on them when they can keep the returns from their work and get their goods to buyers. North Korea offers an extreme test of this idea. Incomes are low, markets remain constrained, and the mobile-phone network was not built to support market activity. Yet even in such a setting, a basic network that simply eased access to information expanded production and exchange where these conditions held. Similar or even larger gains are therefore plausible in other low-income and fragile economies. The same principle may extend to newer technologies, from mobile internet to AI-based tools, whose benefits will also depend on whether people can act on the information they provide. Second, some of these gains can take years to appear. New marketplaces opened gradually, perhaps because vendors and buyers needed time to coordinate. Evaluations that focus on prices or trading efficiency, or that rely on short follow-up periods, may therefore miss slower changes in market structure.

This study relies on public satellite imagery, which comes with limitations. Public archives do not provide high-resolution images of every location every year, so the construction years of some cell towers are imputed within a feasible range, and marketplace area is interpolated in years without images. Most cell towers can still be dated to within two years, and the results hold under alternative tower timings and when marketplace area is restricted to years with imagery. Vegetation indices also cannot fully separate new cultivation from the regrowth of grasses and shrubs. Future research could improve this measure by combining satellite imagery with ground-level crop surveys through machine learning. Despite these limitations, this work may encourage other researchers to draw on public imagery archives to study staggered treatments, such as cell-tower rollouts, where conflict, natural disasters, or restricted access leave data scarce.

A natural next step is to examine the network's effects on other outcomes, such as prices. Vegetation and marketplace area capture how much activity expanded, but prices can show whether markets also became more integrated. In future work, I will address this question using ten years of quarterly prices for 18 goods and services surveyed in 12 North Korean marketplaces. Whereas earlier studies typically focus on a single commodity, these data cover goods that differ in how perishable they are, such as fresh and dried fish, and

in how tradable they are, from manufactured goods to services such as haircuts. Comparing them could help broaden our understanding of which sectors gain most from better information.

Hayek (1945) argued that markets work by drawing on knowledge dispersed among individuals. The evidence from North Korea suggests that even minimal digital connectivity can help put this knowledge to use, allowing people to produce and trade more.

9 Figures

Figure I: Constructing the Cell-Tower Data

Panel A: Identifying Cell Towers



Cell Tower



Microwave Tower



Radio Tower

Panel B: Inferring Deployment Timing

(i) Cell-Tower Site in South Pyongan Province



Sep. 3, 2013



Oct. 2, 2013



Jul. 5, 2014

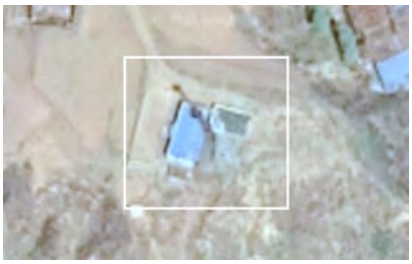
(ii) Cell-Tower Site in North Hamgyong Province



Jul. 4, 2015



Sep. 14, 2015

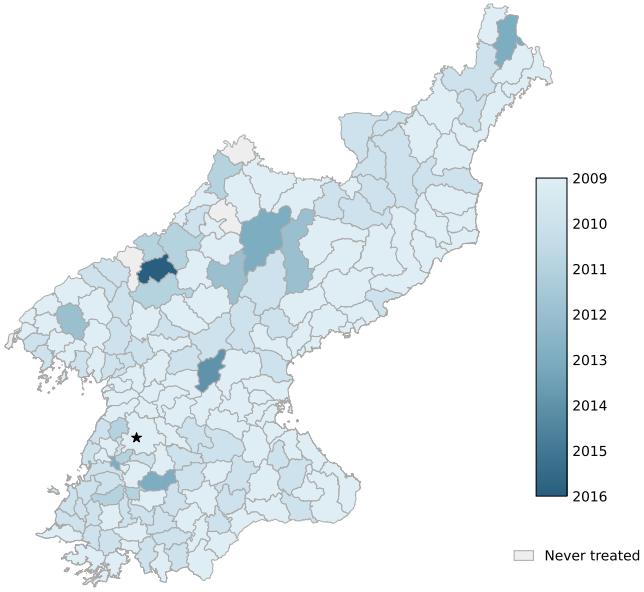


May 8, 2018

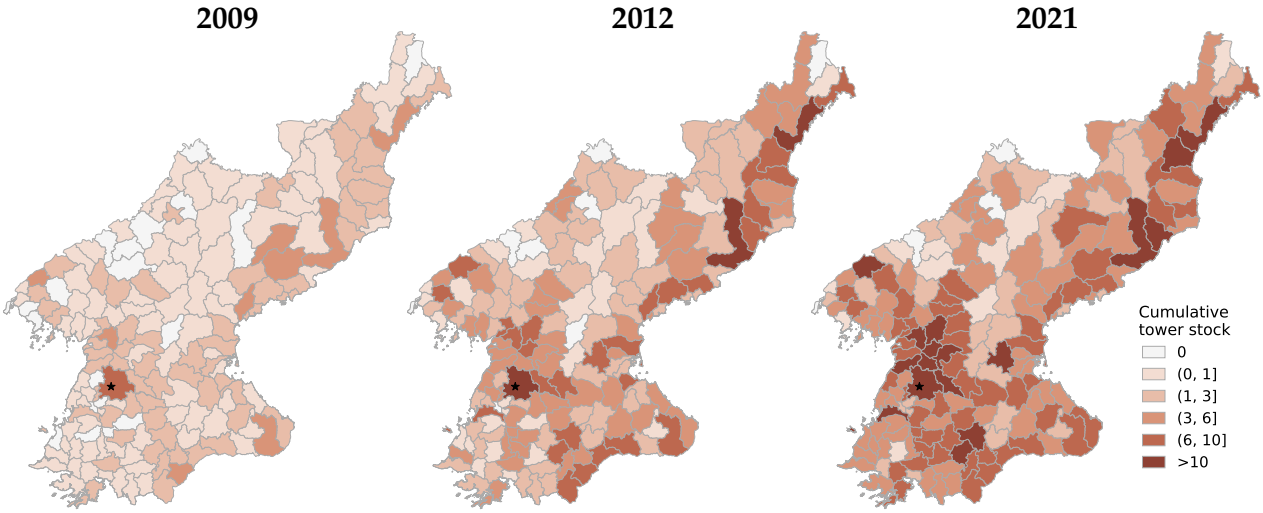
NOTES: Panel A illustrates the visual features used to distinguish cell towers from other telecommunications structures in satellite imagery. Cell-tower sites are identified by a distinctive combination of features, including the tower's long shadow, which reveals its height in overhead imagery, a small associated building, solar panels, and a clearly delineated or fenced compound. I classify a cell tower as deployed once the tower and its surrounding compound are completed.
Panel B illustrates how I infer deployment timing from historical imagery at two sites with different imagery frequency. In Panel B(i), construction is visible in 2013 and the completed site is observed in 2014, so I assign 2014 as the deployment year. In Panel B(ii), construction is visible in 2015, but the next available image is from 2018. Because construction is typically completed within about one year at sites with frequent imagery, I assign 2016 as the deployment year.

Figure II: Mobile-Phone Network Rollout in North Korea

Panel A. Timing of Initial Mobile-Phone Access

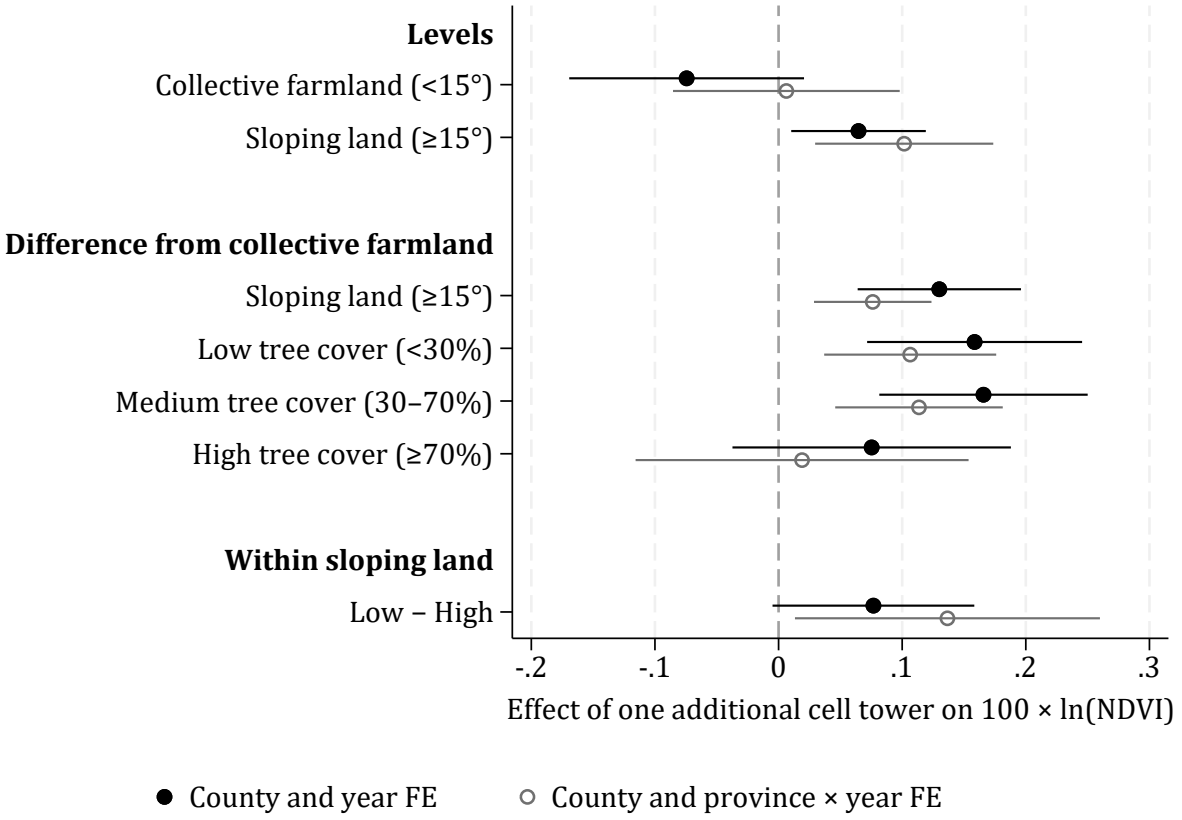


Panel B. Evolution of Network Intensity



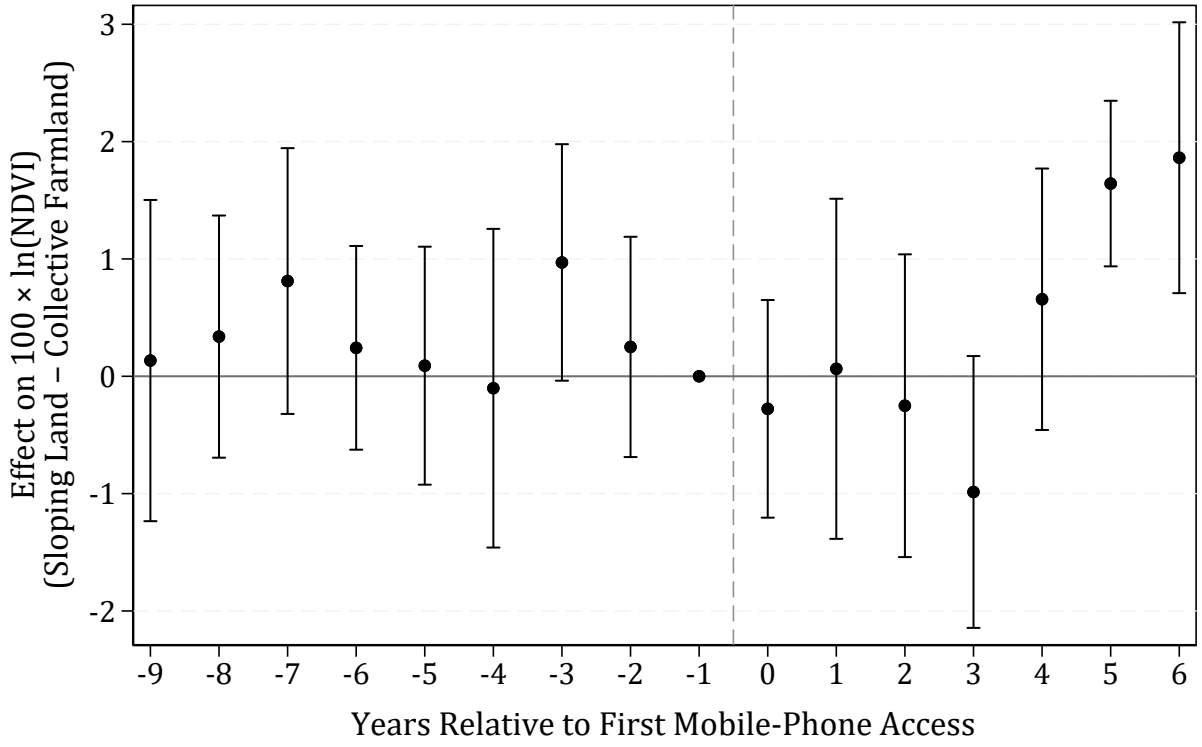
NOTES: Panel A shows the year in which each county first gained mobile-phone access, defined by the arrival of its first confirmed cell tower. Because satellite imagery is not available every year for every tower, the exact year of first access is sometimes uncertain. In these cases, I assign the first-access year to the midpoint between the earliest and latest years in which the first tower could have been deployed. Of the 178 counties in the sample, 173 receive mobile-phone access during the sample period. The five never-treated counties are Chongnam in South Pyongan Province, Sindo in North Pyongan Province, and Usi, Junggang, and Janggang in Jagang Province.
Panel B shows the expansion of the mobile-phone network across counties in 2009, 2012, and 2021, measured by the cumulative stock of confirmed cell towers in each county. The maps use the preferred Uniform measure employed in the baseline two-way fixed-effects specification, which interpolates cumulative tower stock when the exact deployment year is not observed. The resulting measure can therefore take non-integer values. The same tower-stock categories and color scale are used across all three maps. The mean cumulative tower stock across the 178 counties is 1.31 in 2009, 3.90 in 2012, and 6.33 in 2021, while the maximum is 8, 31, and 55, respectively. The black star marks Pyongyang. County boundaries correspond to the 2008 administrative boundaries used throughout the analysis.

Figure III: Mobile-Phone Access and Vegetation Greenness (NDVI) by Land Type



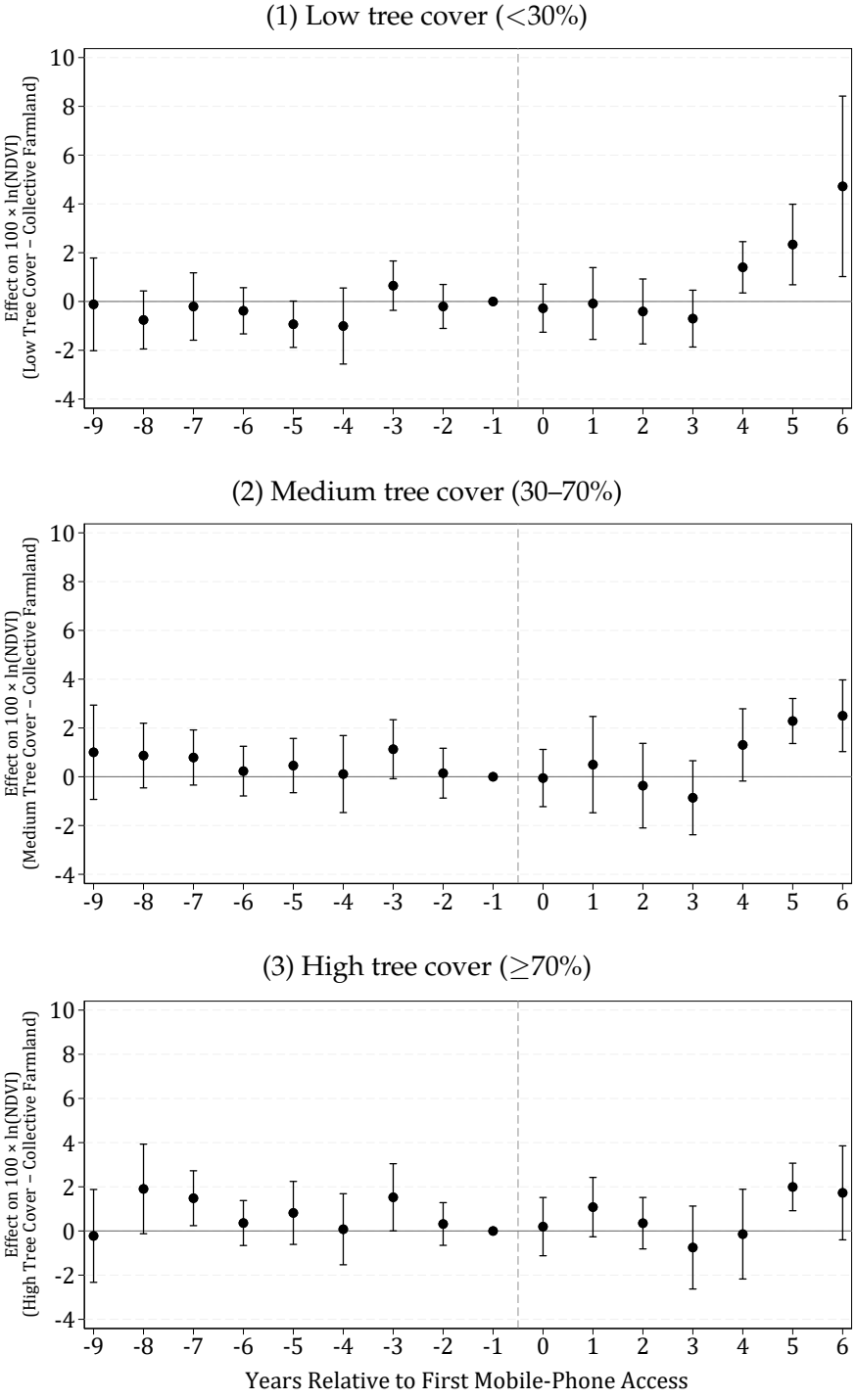
NOTES: This figure plots the coefficients on the cumulative number of cell towers (*Uniform*) from the regressions in Table 1. The dependent variable is 100 × ln(mean April–August NDVI), 2002–2015. *Levels* report collective farmland (slope below 15°) and sloping land (15° or above). *Difference from collective farmland* reports the within-county-year difference between the indicated sloping-land category and collective farmland, with sloping land divided by baseline tree cover in 2000. *Within sloping land* reports the difference between low- and high-tree-cover sloping land, estimated on the common sample of counties with high-tree-cover pixels. Filled markers include county and year fixed effects; hollow markers include county and province-by-year fixed effects. Horizontal bars show 95% confidence intervals based on standard errors clustered at the county level.

Figure IV: Dynamic Effects of First Mobile-Phone Access on Vegetation



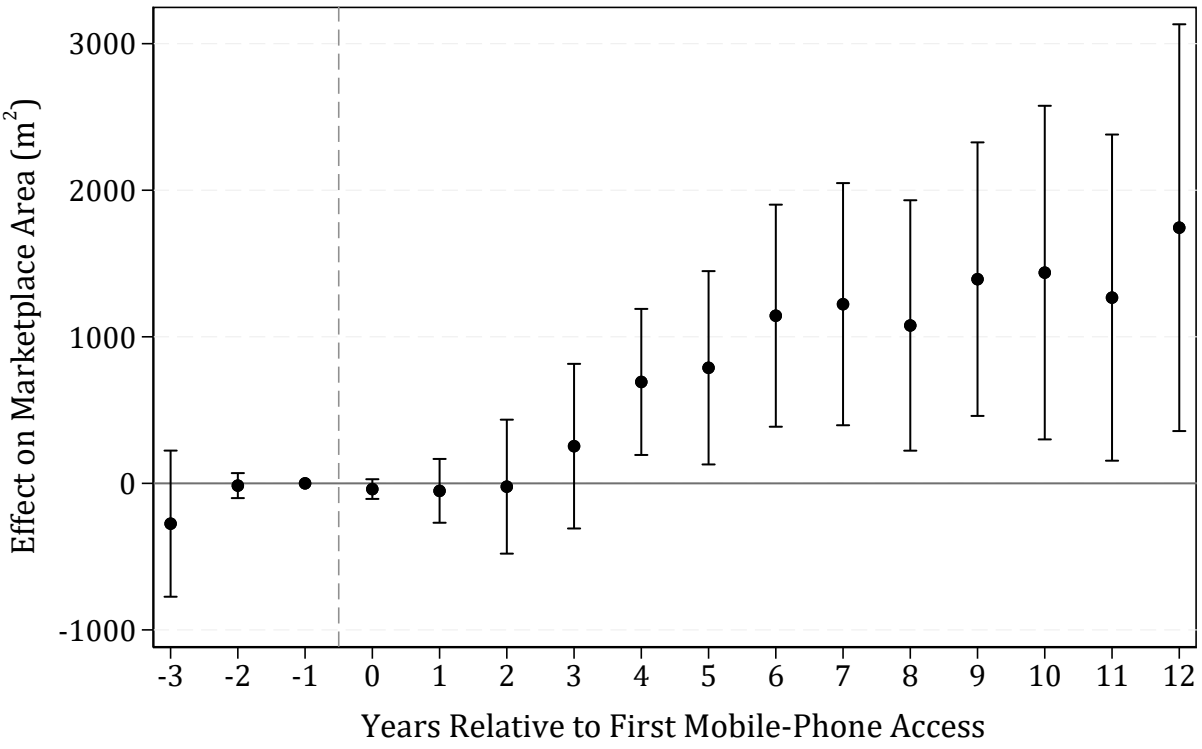
NOTES: This figure examines whether vegetation on sloping land systematically diverged from vegetation on collective farmland before first mobile-phone access and traces its subsequent evolution following access. The outcome is $100 \times \ln(\text{NDVI})$ on land with slope of at least 15 degrees minus $100 \times \ln(\text{NDVI})$ on collective farmland (slope below 15 degrees), where NDVI is the mean of April–August values (MODIS, 250m); this is the outcome in column (3) of Table 1. Treatment timing is defined as the midpoint of the earliest and latest feasible years of first cell-tower arrival. Estimates are obtained using the Callaway and Sant'Anna (2021) estimator, with not-yet-treated and never-treated counties serving as controls. All estimates are measured relative to the year before first access (event time -1), following Roth (2026). Event times are shown when at least ten treated counties contribute. The sample covers 172 counties over 2002–2015; six counties with no land above 15 degrees at 250m resolution are excluded. Estimates at event times +5 and +6 are identified relative to four never-treated counties, all located in Jagang Province, and their magnitudes should be interpreted with caution. Standard errors are clustered at the county level. The vertical dashed line separates the pre-treatment and post-treatment periods, and the horizontal solid line denotes zero effect. Vertical bars report 95 percent confidence intervals.

Figure V: Dynamic Effects on Vegetation by Tree Cover on Sloping Land



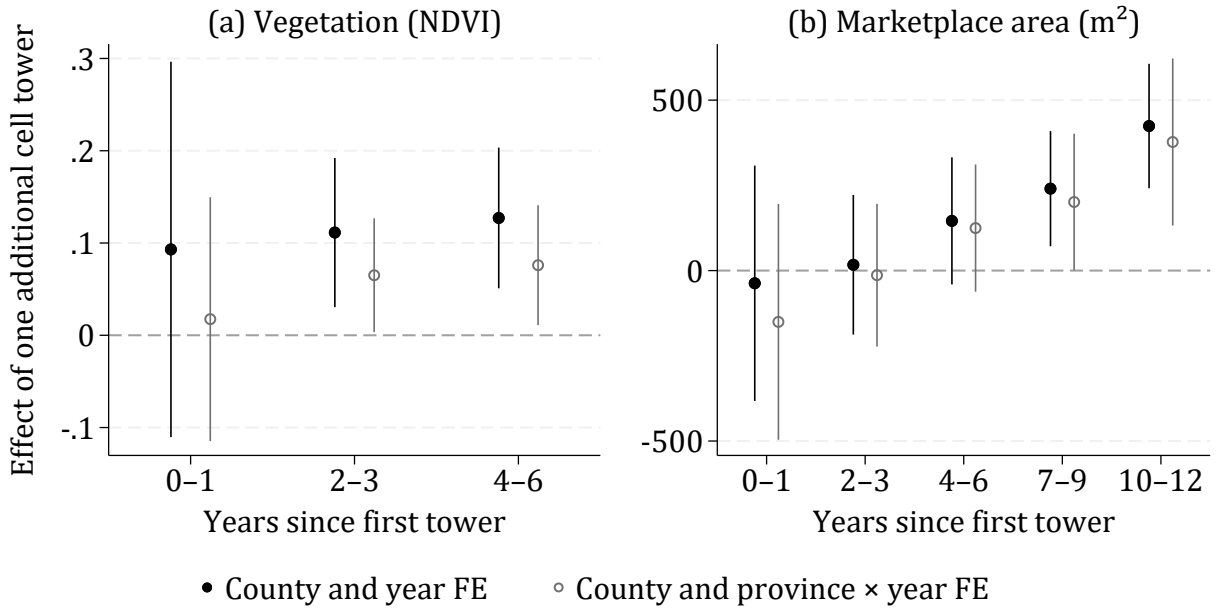
NOTES: This figure repeats the event study in Figure IV separately for sloping land (at least 15 degrees) grouped by tree cover in 2000 from Hansen et al. (2013): low tree cover (below 30 percent), medium tree cover (30–70 percent), and high tree cover (at least 70 percent). Each outcome is 100 × ln(NDVI) of the group minus 100 × ln(NDVI) of collective farmland. All panels share the same vertical axis. Estimation, treatment timing, controls, and the reference period are as in Figure IV. At longer horizons, high-tree-cover land also shows a modest increase, suggesting that part of the late-horizon estimates reflects a component common to all land, driven by the small comparison group at these horizons; on top of this common component, the increase is substantially larger on land with less tree cover. Because the 2000 tree-cover data do not capture gradual clearing after 2000, land classified as high tree cover may include some cultivated plots. Standard errors are clustered at the county level. The vertical dashed line separates the pre-treatment and post-treatment periods, and the horizontal solid line denotes zero effect. Vertical bars report 95 percent confidence intervals.

Figure VI: Dynamic Effects of First Mobile-Phone Access on Marketplace Expansion



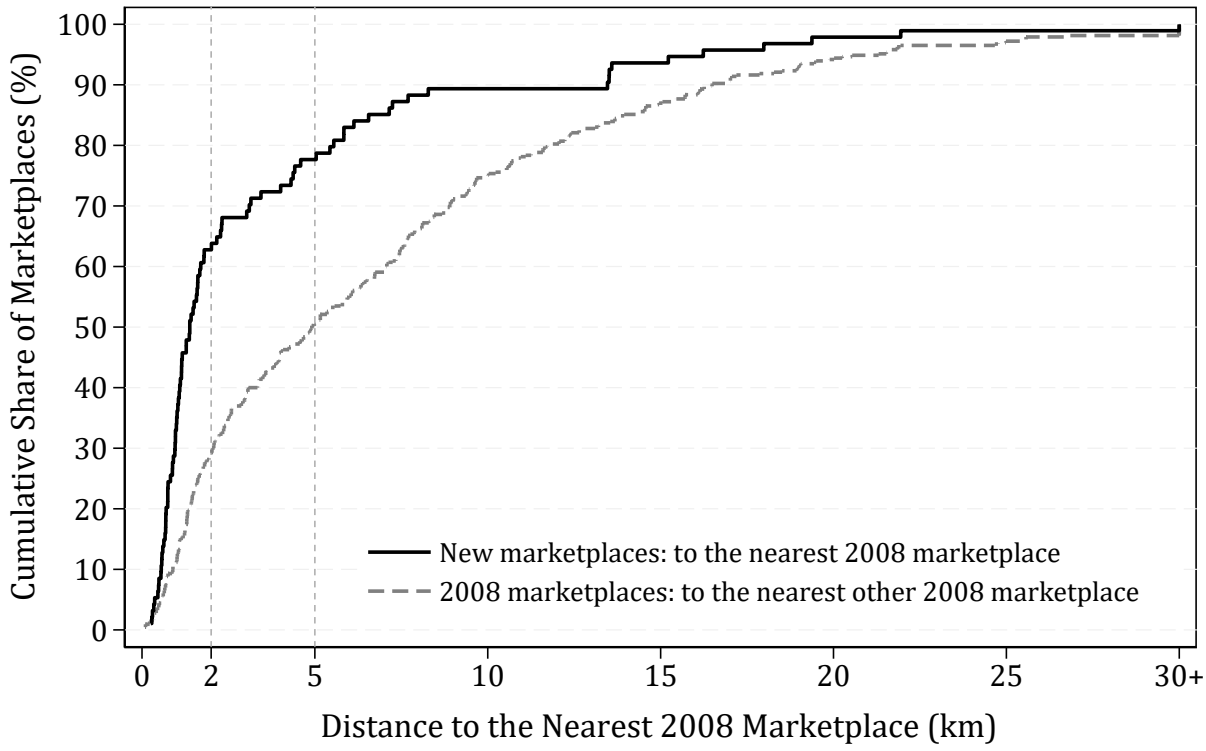
NOTES: This figure examines whether changes in marketplace activity systematically preceded first mobile-phone access and traces the subsequent evolution of marketplace activity following access. It reports event-study estimates for total marketplace area. Treatment timing is defined as the midpoint of the earliest and latest feasible years of first cell-tower arrival. Estimates are obtained using the Callaway and Sant'Anna (2021) estimator, with not-yet-treated and never-treated counties serving as controls. All estimates are measured relative to the year before first access (event time -1), following Roth (2026). Event times are shown when at least ten treated counties contribute; earlier pre-period horizons, identified from fewer than ten late-adopting counties, yield imprecise estimates that are not statistically different from zero. The sample covers all 178 counties over 2008–2021. Because marketplace data begin in 2008, counties first connected in 2009 contribute no pre-period estimates, and the pre-period estimates come from later cohorts. Standard errors are clustered at the county level. The vertical dashed line separates the pre-treatment and post-treatment periods, and the horizontal solid line denotes zero effect. Vertical bars report 95 percent confidence intervals.

Figure VII: Effect of an Additional Cell Tower by Years Since First Tower



NOTES: This figure relaxes the linear specification in columns (3) and (6) of Table 3 by estimating the effect of an additional cell tower (*Uniform*) separately for each bin of years since the first tower in the county. Each panel reports coefficients on the cumulative number of cell towers interacted with indicators for 0–1, 2–3, 4–6, 7–9, and 10–12 years since the first tower, from a regression that also includes an access indicator and indicators for these bins. Panel (a) uses the within-county-year difference in 100 × ln(mean April–August NDVI) between sloping land (15° or above) and collective farmland (below 15°), 2002–2015; because the sample ends in 2015, years since the first tower range from 0 to 6. Panel (b) uses marketplace area in square meters, 2008–2021. Filled markers include county and year fixed effects; hollow markers include county and province-by-year fixed effects. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level.

Figure VIII: Distance to the Nearest 2008 Marketplace: New versus Existing Marketplaces



NOTES: This figure examines where new marketplaces opened relative to marketplaces that already existed in 2008. The solid line plots the cumulative distribution of the distance from each of the 94 new marketplaces (inactive in 2008 and active in at least one year in 2009–2021) to the nearest of the 430 marketplaces active in 2008. As a benchmark, the dashed line plots the cumulative distribution of the distance from each of the 430 marketplaces active in 2008 to the nearest other marketplace active in 2008. Distances are straight-line (haversine) distances between marketplace coordinates. Vertical dashed lines mark 2 and 5 km. Distances above 30 km are shown at 30 km. Figure D.1 in the Appendix shows the same distances as a histogram.

10 Tables

Table 1: Mobile-Phone Access and Vegetation Greenness (NDVI) by Land Type

	Dep. Var: 100 × ln(NDVI, April–August)						
	Levels		Difference from collective farmland				Within sloping land
				Sloping land by tree cover (2000)			
	Collective farmland (<15°)	Sloping land (≥15°)	Sloping land (≥15°)	Low (<30%)	Medium (30–70%)	High (≥70%)	Low– High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. County and year fixed effects							
Uniform (baseline)	-0.074	0.065**	0.130***	0.159***	0.166***	0.075	0.077*
	(0.048)	(0.028)	(0.033)	(0.044)	(0.043)	(0.057)	(0.041)
Earliest (upper bound)	-0.089*	0.042	0.120***	0.154***	0.153***	0.072	0.075*
	(0.050)	(0.029)	(0.030)	(0.042)	(0.037)	(0.055)	(0.041)
Latest (lower bound)	-0.074	0.056*	0.122***	0.147***	0.157***	0.069	0.072*
	(0.047)	(0.028)	(0.032)	(0.041)	(0.040)	(0.055)	(0.042)
Exact (known years)	-0.059	0.076**	0.121***	0.160***	0.158***	0.042	0.116**
	(0.056)	(0.038)	(0.039)	(0.051)	(0.049)	(0.065)	(0.051)
Panel B. County and province × year fixed effects							
Uniform (baseline)	0.006	0.102***	0.076***	0.106***	0.114***	0.019	0.137**
	(0.046)	(0.036)	(0.024)	(0.035)	(0.034)	(0.068)	(0.062)
Earliest (upper bound)	0.003	0.083**	0.060***	0.097***	0.087**	-0.000	0.140**
	(0.046)	(0.038)	(0.023)	(0.034)	(0.035)	(0.070)	(0.069)
Latest (lower bound)	0.003	0.093**	0.074***	0.100***	0.112***	0.017	0.128**
	(0.046)	(0.038)	(0.024)	(0.034)	(0.035)	(0.069)	(0.064)
Exact (known years)	-0.023	0.106**	0.108***	0.137***	0.157***	-0.006	0.164
	(0.051)	(0.044)	(0.027)	(0.039)	(0.052)	(0.104)	(0.102)
Observations	2,492	2,408	2,408	2,408	2,322	1,675	1,675
N of clusters	178	172	172	172	166	121	121

NOTES: Each cell reports the coefficient on the cumulative number of cell towers from a separate regression. The dependent variable is 100 × ln(mean April–August NDVI) from MODIS MOD13Q1 at 250m resolution, 2002–2015. Columns (1) and (2) report levels for collective farmland (slope below 15°) and sloping land (15° or above). Columns (3)–(6) report the within-county-year difference between the indicated sloping-land category and collective farmland. Sloping land is divided by baseline tree cover in 2000: low (below 30%), medium (30–70%), and high (70% or above). Column (7) reports the difference between low- and high-tree-cover sloping land, estimated on the common sample of counties with high-tree-cover pixels. Sample sizes differ across columns because not every county contains pixels in each land category. Rows correspond to the four tower-stock measures described in Section 4.1: *Uniform* is the baseline measure; *Earliest* and *Latest* assign towers with uncertain deployment years to the first and last feasible years, providing upper and lower bounds on tower stock; *Exact* uses only county-years in which tower stock is known exactly. Panel A includes county and year fixed effects; Panel B includes county and province-by-year fixed effects. Observations and clusters refer to the *Uniform* specification in Panel A. Standard errors are reported in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table 2: Mobile-Phone Access and Marketplace Expansion

	Dep. Var: Marketplace Area (m^2)			
	Uniform (baseline) (1)	Earliest (upper bound) (2)	Latest (lower bound) (3)	Exact (known years) (4)
Panel A. County and year fixed effects				
Cumulative cell towers	337.215***	348.364***	324.843***	326.100***
	(93.122)	(98.900)	(101.563)	(101.472)
Effect per tower (% of pre-treatment mean)	3.4	3.5	3.7	3.3
Panel B. County and province × year fixed effects				
Cumulative cell towers	297.349***	308.235***	280.162**	289.635**
	(108.895)	(115.823)	(113.472)	(118.216)
Effect per tower (% of pre-treatment mean)	3.0	3.1	3.2	2.9
Pre-treatment mean (m^2)	9,938	9,938	8,833	9,938
Observations	2,492	2,492	2,492	1,995
N of clusters	178	178	178	178

NOTES: The dependent variable is total marketplace area in each county-year, measured in square meters, 2008–2021. The treatment variable is the cumulative number of cell towers. Columns correspond to the four tower-stock measures described in Section 4.1: *Uniform* is the baseline measure; *Earliest* and *Latest* assign towers with uncertain deployment years to the first and last feasible years, providing upper and lower bounds on tower stock; *Exact* uses only county-years in which tower stock is known exactly. Pre-treatment means are calculated separately for each measure using observations prior to the first positive tower stock implied by that measure. Effect per tower is the coefficient divided by the corresponding pre-treatment mean, in percent. Panel A includes county and year fixed effects; Panel B includes county and province-by-year fixed effects. Pre-treatment means, observations, and clusters refer to Panel A; in Panel B, column (4) drops three singleton observations. Standard errors are reported in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table 3: Network Scale and Duration of Mobile-Phone Access

	Vegetation (NDVI) Sloping land − collective farmland			Marketplace area (m^2)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. County and year fixed effects						
Access ($\mathbb{1}\{T > 0\}$)		-0.298	-0.329		-653.579	116.871
		(0.390)	(0.454)		(644.522)	(629.685)
Network scale (T)	0.130***	0.133***	0.129*	337.215***	337.804***	-81.277
	(0.033)	(0.035)	(0.072)	(93.122)	(93.014)	(128.104)
Duration (d)			0.039			-137.022*
			(0.142)			(82.599)
Network scale × duration ($T \times d$)			0.000			43.253***
			(0.016)			(13.758)
Panel B. County and province × year fixed effects						
Access ($\mathbb{1}\{T > 0\}$)		-0.443	-0.377		-772.384	-38.668
		(0.271)	(0.318)		(675.733)	(713.045)
Network scale (T)	0.076***	0.080***	0.095*	297.349***	295.248***	-102.500
	(0.024)	(0.025)	(0.050)	(108.895)	(108.079)	(134.627)
Duration (d)			-0.106			-118.332
			(0.107)			(110.288)
Network scale × duration ($T \times d$)			-0.002			41.960**
			(0.013)			(16.426)
Observations	2,408	2,408	2,408	2,492	2,492	2,492
N of clusters	172	172	172	178	178	178

NOTES: This table reports estimates of Equation (13). The dependent variable in columns (1)–(3) is the within-county-year difference in 100 × ln(mean April–August NDVI) between sloping land (15° or above) and collective farmland (below 15°), 2002–2015. The dependent variable in columns (4)–(6) is total marketplace area in square meters, 2008–2021. T is the cumulative number of cell towers (*Uniform*); access is an indicator for at least one tower; duration d is the number of years since the first cell tower in the county, equal to zero in that year and in years before access. Columns (1) and (4) reproduce the baseline estimates in Tables 1 and 2. In columns (3) and (6), the coefficient on network scale is the effect of an additional tower in the year of first access; the effect d years after access is $\beta_2 + \beta_4 d$. Panel A includes county and year fixed effects; Panel B includes county and province-by-year fixed effects. Observations and clusters refer to Panel A. Standard errors are reported in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table 4: Decomposing the Effect on Marketplace Area: New, Continuing, and Exiting Marketplaces

	Dep. Var: Marketplace Area (m^2)			
			Existing in 2008	
	Total	New	Continuing	Exiting
	(1)	(2)	(3)	(4)
Panel A. County and year fixed effects				
Cumulative cell towers	337.215***	335.038***	75.863**	-73.686
	(93.122)	(72.879)	(31.942)	(61.934)
Share of total effect (%)	100	99	22	-22
Panel B. County and province × year fixed effects				
Cumulative cell towers	297.349***	257.753***	98.055**	-58.459
	(108.895)	(92.797)	(42.930)	(68.874)
Share of total effect (%)	100	87	33	-20
Mean area per county, 2021 (m^2)	13,644	2,431	11,112	101
Mean area per marketplace, 2021 (m^2)	4,966	4,755	5,058	2,575
Marketplaces, 2008–2021	524	94	391	39
Observations	2,492	2,492	2,492	2,492
N of clusters	178	178	178	178

NOTES: The dependent variable is total marketplace area in square meters of the indicated group of marketplaces in each county-year, 2008–2021. A marketplace is active in a year if its measured area is positive. *New*: inactive in 2008 and active in at least one year in 2009–2021. *Continuing*: active in every year from 2008 to 2021. *Exiting*: active in 2008 and inactive in at least one later year (32 of the 39 remained closed in 2021). Twelve marketplaces never active during 2008–2021 are excluded. Columns (2)–(4) partition all marketplaces, so their coefficients sum to column (1); share of total effect is each coefficient divided by the coefficient in column (1) of the same panel. If new marketplaces were relocations of existing ones, the area of exiting marketplaces would fall by a similar amount; the point estimate in column (4) implies that relocation can account for at most about one-fifth of the effect in column (2). The treatment variable is the cumulative number of cell towers (*Uniform*). Mean area per county is the 2021 mean of the dependent variable across counties; mean area per marketplace is the 2021 mean across marketplaces active in that year (for exiting marketplaces, the 7 that reopened). Panel A includes county and year fixed effects; Panel B includes county and province-by-year fixed effects. Means, marketplaces, observations, and clusters refer to Panel A. Standard errors are reported in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table 5: Heterogeneous Effects on Marketplace Area by Baseline County Characteristics

	Dep. Var: Marketplace Area (m^2)		
	(1)	(2)	(3)
Cumulative cell towers ×			
Panel A. Local demand			
ln Population, 2008	82.744***	84.258***	
	(24.656)	(26.308)	
ln Marketplace area, 2008	7.969	5.153	-160.442**
	(36.010)	(36.533)	(62.549)
Panel B. Transport and market access			
Rail station	424.521***	431.720***	331.974*
	(114.960)	(163.251)	(191.566)
Distance to the nearest city (10 km)	-44.759**	-44.177**	-25.929
	(18.804)	(19.920)	(48.595)
Road density, 2008	9.478	7.005	-108.207
	(46.636)	(50.483)	(96.287)
Panel C. Proximity to the regime			
Distance to Pyongyang (10 km)	-5.241	-5.731	-2.945
	(5.123)	(5.903)	(6.071)
Special economic zone	8.811	31.550	-555.826
	(195.308)	(179.525)	(422.646)
Panel D. Proximity to China			
China border county	-530.506*	-574.290*	-468.257
	(287.606)	(308.180)	(287.136)
Distance to the nearest border crossing (10 km)	-12.021	-8.073	-9.173
	(10.239)	(11.317)	(9.807)
Mean area per county, 2021 (m^2)	13,644	13,644	13,644
Observations	2,492	2,492	2,492
N of clusters	178	178	178
County FE	Yes	Yes	Yes
Year FE	Yes		Yes
Province × year FE		Yes	
Cell towers × ln population, 2008			Yes
ln population, 2008 × year FE			Yes

NOTES: Each cell reports the coefficient on cumulative cell towers interacted with the indicated county characteristic, from a separate regression that also includes cumulative cell towers and the characteristic interacted with year fixed effects (2008–2021). Characteristics are measured before the network rollout. Fixed effects and controls are indicated at the bottom of the table. Road density is road length per km^2 in 2008. Most special economic zones were designated in 2013–2014. China border county equals one for the 26 counties sharing a border with China; distance to the nearest border crossing is measured to the nearest customs post. Negative coefficients on distances indicate larger effects closer by. Standard errors clustered at the county level are in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

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Appendices

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A Conceptual Framework

A.1. Production decision

The producer solves

$$\max_{h \geq 0} \{\mathbb{E}[p]q(h) - c(h)\}. \tag{A.1}$$

The first-order condition is

$$\mathbb{E}[p]q'(h^*) - c'(h^*) = 0. \tag{A.2}$$

Differentiating equation (A.2) with respect to $\mathbb{E}[p]$ gives

$$q'(h^*) + \mathbb{E}[p]q''(h^*)\frac{\partial h^*}{\partial \mathbb{E}[p]} - c''(h^*)\frac{\partial h^*}{\partial \mathbb{E}[p]} = 0. \tag{A.3}$$

Collecting terms,

$$\left[c''(h^*) - \mathbb{E}[p]q''(h^*)\right]\frac{\partial h^*}{\partial \mathbb{E}[p]} = q'(h^*). \tag{A.4}$$

Therefore,

$$\frac{\partial h^*}{\partial \mathbb{E}[p]} = \frac{q'(h^*)}{c''(h^*) - \mathbb{E}[p]q''(h^*)}. \tag{A.5}$$

Since $q'(h^*) > 0$, $q''(h^*) < 0$, and $c''(h^*) > 0$, it follows that

$$\frac{\partial h^*}{\partial \mathbb{E}[p]} > 0. \tag{A.6}$$

A.2. Sequential Search Problem

Let z denote the highest net-price quotation observed so far. The Bellman equation is

$$V(z) = \max\left\{z,\ -s + \mathbb{E}\left[V(\max\{z, p\})\right]\right\}. \tag{A.7}$$

Let $r(s)$ denote the stationary reservation price. At $z = r(s)$, the producer is indifferent between stopping and continuing, so

$$r(s) = -s + \mathbb{E}\left[V(\max\{r(s), p\})\right]. \tag{A.8}$$

For every realization of p, $\max\{r(s), p\} \geq r(s)$, so the updated state after an additional search lies in the stopping region. If $p < r(s)$, recall allows the producer to retain the current best quotation, so the updated state remains $r(s)$ and $V(r(s)) = r(s)$. If $p \geq r(s)$, the updated state is p, and $V(p) = p$.

Thus, $V(\max\{r(s), p\}) = \max\{r(s), p\}$, and equation (A.8) becomes

$$r(s) = -s + \mathbb{E}\left[\max\{r(s), p\}\right]. \tag{A.9}$$

The expectation can be divided according to whether the newly observed quotation lies below or above the reservation price:

$$\begin{aligned}\mathbb{E}\left[\max\{r(s), p\}\right] &= \int_{\underline{p}}^{r(s)} r(s) f(p)\, dp + \int_{r(s)}^{\overline{p}} p f(p)\, dp \\ &= r(s)F(r(s)) + \int_{r(s)}^{\overline{p}} p f(p)\, dp.\end{aligned} \tag{A.10}$$

Substituting equation (A.10) into equation (A.9) gives

$$r(s) = -s + r(s)F(r(s)) + \int_{r(s)}^{\overline{p}} pf(p)\,dp. \tag{A.11}$$

Rearranging,

$$\begin{aligned} s &= \int_{r(s)}^{\overline{p}} pf(p)\,dp - r(s)[1 - F(r(s))] \\ &= \int_{r(s)}^{\overline{p}} pf(p)\,dp - \int_{r(s)}^{\overline{p}} r(s)f(p)\,dp \\ &= \int_{r(s)}^{\overline{p}} [p - r(s)]f(p)\,dp. \end{aligned} \tag{A.12}$$

To derive the comparative static with respect to the search cost, rewrite equation (A.12) as

$$s = \int_{r(s)}^{\overline{p}} pf(p)\,dp - r(s)\int_{r(s)}^{\overline{p}} f(p)\,dp. \tag{A.13}$$

Differentiating equation (A.13) with respect to s gives

$$1 = \frac{d}{ds}\left[\int_{r(s)}^{\overline{p}} pf(p)\,dp\right] - \frac{d}{ds}\left[r(s)\int_{r(s)}^{\overline{p}} f(p)\,dp\right]. \tag{A.14}$$

Applying Leibniz's rule to the first term,

$$\frac{d}{ds}\left[\int_{r(s)}^{\overline{p}} pf(p)\,dp\right] = -r(s)f(r(s))\frac{dr(s)}{ds}. \tag{A.15}$$

For the second term, applying the product rule gives

$$\frac{d}{ds}\left[r(s)\int_{r(s)}^{\overline{p}} f(p)\,dp\right] = \frac{dr(s)}{ds}\int_{r(s)}^{\overline{p}} f(p)\,dp + r(s)\frac{d}{ds}\left[\int_{r(s)}^{\overline{p}} f(p)\,dp\right]. \tag{A.16}$$

Applying Leibniz's rule to the last term gives

$$\frac{d}{ds}\left[\int_{r(s)}^{\overline{p}} f(p)\,dp\right] = -f(r(s))\frac{dr(s)}{ds}. \tag{A.17}$$

Therefore,

$$\frac{d}{ds}\left[r(s)\int_{r(s)}^{\overline{p}} f(p)\,dp\right] = \frac{dr(s)}{ds}\int_{r(s)}^{\overline{p}} f(p)\,dp - r(s)f(r(s))\frac{dr(s)}{ds}. \tag{A.18}$$

Substituting equations (A.15) and (A.18) into equation (A.14) gives

$$1 = -r(s)f(r(s))\frac{dr(s)}{ds} - \left[\frac{dr(s)}{ds}\int_{r(s)}^{\overline{p}} f(p)\,dp - r(s)f(r(s))\frac{dr(s)}{ds}\right]. \tag{A.19}$$

Expanding the second term,

$$1 = -r(s)f(r(s))\frac{dr(s)}{ds} - \frac{dr(s)}{ds}\int_{r(s)}^{\overline{p}} f(p)\,dp + r(s)f(r(s))\frac{dr(s)}{ds}. \tag{A.20}$$

The first and third terms cancel, leaving

$$1 = -\frac{dr(s)}{ds}\int_{r(s)}^{\overline{p}} f(p)\,dp. \tag{A.21}$$

Since F is the cumulative distribution function and the density integrates to one,

$$\begin{aligned}\int_{r(s)}^{\overline{p}} f(p)\,dp &= \int_{\underline{p}}^{\overline{p}} f(p)\,dp - \int_{\underline{p}}^{r(s)} f(p)\,dp \\ &= 1 - F(r(s)).\end{aligned} \tag{A.22}$$

and $0 < F(r(s)) < 1$, it follows that $1 - F(r(s)) > 0$. Therefore,

$$\frac{dr(s)}{ds} = -\frac{1}{1 - F(r(s))} < 0. \tag{A.23}$$

A.3. Expected net selling price

Under the optimal reservation rule, the expected net selling price is

$$\begin{aligned}\mathbb{E}[p] = \mathbb{E}\left[p \mid p \geq r(s)\right] &= \int_{r(s)}^{\overline{p}} p f(p \mid p \geq r(s))\,dp \\ &= \int_{r(s)}^{\overline{p}} p \frac{f(p)}{1 - F(r(s))}\,dp \\ &= \frac{\int_{r(s)}^{\overline{p}} p f(p)\,dp}{1 - F(r(s))}.\end{aligned} \tag{A.24}$$

For notational convenience, let $r = r(s)$. Differentiating equation (A.24) with respect to r, by Leibniz's rule,

$$\frac{d}{dr}\left[\int_{r}^{\overline{p}} p f(p)\,dp\right] = -r f(r), \tag{A.25}$$

and, since $F'(r) = f(r)$,

$$\frac{d}{dr}\left[1 - F(r)\right] = -f(r). \tag{A.26}$$

Applying the quotient rule then gives

$$\begin{aligned}\frac{\partial \mathbb{E}[p]}{\partial r} &= \frac{-r f(r)[1 - F(r)] + f(r)\int_{r}^{\overline{p}} p f(p)\,dp}{\left[1 - F(r)\right]^2} \\ &= \frac{f(r)}{\left[1 - F(r)\right]^2}\left[\int_{r}^{\overline{p}} p f(p)\,dp - r[1 - F(r)]\right].\end{aligned} \tag{A.27}$$

Since

$$\begin{aligned}\int_{r}^{\overline{p}} p f(p)\,dp - r[1 - F(r)] &= \int_{r}^{\overline{p}} p f(p)\,dp - \int_{r}^{\overline{p}} r f(p)\,dp \\ &= \int_{r}^{\overline{p}} (p - r) f(p)\,dp,\end{aligned} \tag{A.28}$$

evaluating at $r = r(s)$ and using equation (A.12) yields

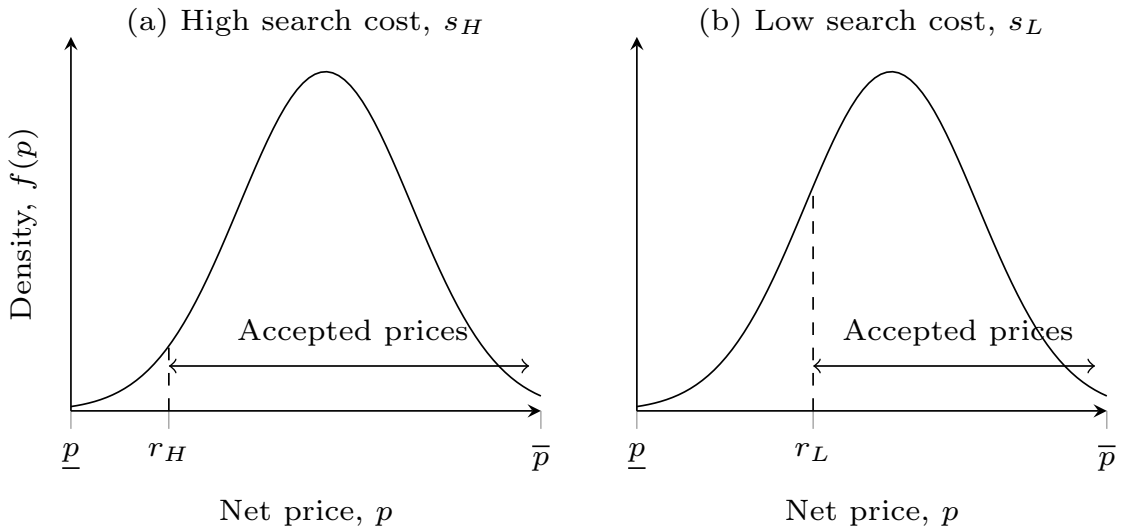
$$\begin{aligned}\frac{\partial \mathbb{E}[p]}{\partial r} &= \frac{f(r(s))}{[1 - F(r(s))]^2} \int_{r(s)}^{\overline{p}} [p - r(s)] f(p)\, dp \\ &= \frac{s f(r(s))}{[1 - F(r(s))]^2} > 0. \end{aligned} \tag{A.29}$$

A.4. Distributional Implications of Lower Search Costs

To illustrate the distributional implications of mobile-phone access, consider a higher search cost s_H and a lower search cost s_L, with $s_L < s_H$. The lower search cost represents the reduction in information-acquisition costs enabled by mobile phones. Let $r_H = r(s_H)$ and $r_L = r(s_L)$ denote the corresponding reservation prices. As shown in equation (A.23), a lower search cost raises the reservation price, so that $r_L > r_H$.

Figure A.1 illustrates how lower search costs raise the reservation price. Both panels show the same underlying net-price density $f(p)$. In Panel (a), under the higher search cost s_H, the producer accepts any quotation at or above r_H. In Panel (b), when the search cost falls to s_L, the producer accepts only quotations at or above r_L. Thus, quotations between r_H and r_L that would have been accepted under the higher search cost are rejected when the search cost is lower. Lower search costs do not change the underlying distribution of prices; they change which prices the producer is willing to accept.

Figure A.1: Search Costs and Reservation Prices

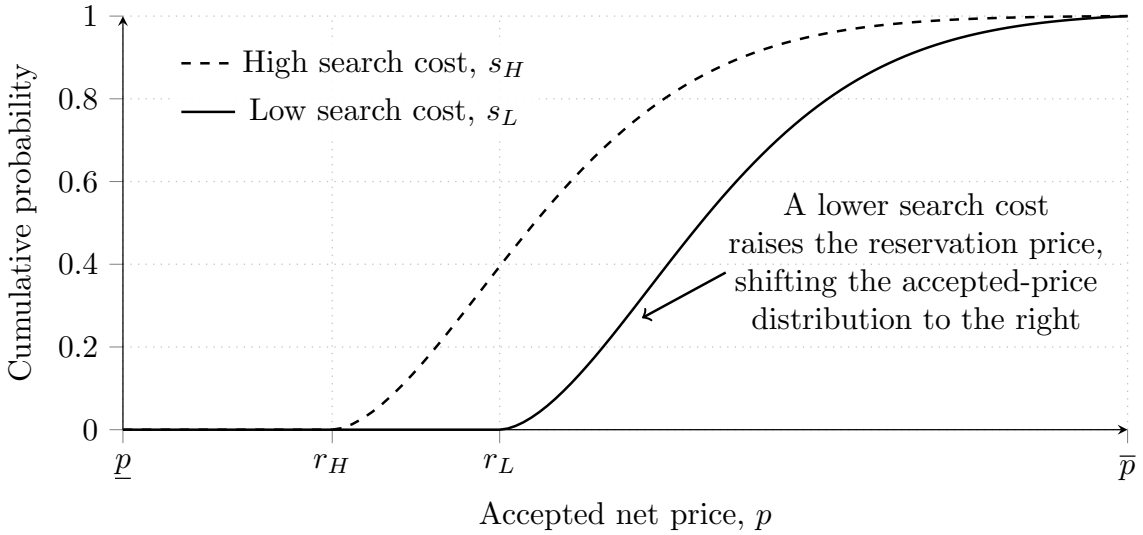


NOTES: Both panels show the same underlying net-price density $f(p)$. Panel (a) corresponds to the higher search cost s_H, with reservation price r_H, and Panel (b) to the lower search cost s_L, with reservation price $r_L > r_H$. The arrows indicate the range of prices accepted under each search cost. The distribution and parameter values are chosen for illustrative purposes.

The resulting change in the acceptance rule also changes the distribution of prices ultimately accepted by the producer. Figure A.2 shows the corresponding cumulative distributions of accepted net prices. At any given net price, the cumulative probability is weakly lower under s_L than under

s_H, implying a greater probability of accepting a price above that level under the lower search cost. Thus, the accepted-price distribution under the lower search cost first-order stochastically dominates that under the higher search cost.

Figure A.2: First-order Stochastic Dominance of Accepted Prices



NOTES: The figure shows the cumulative distributions of accepted net prices under the higher search cost s_H and the lower search cost s_L. The dashed curve corresponds to s_H, and the solid curve corresponds to s_L. Because prices below the reservation price are not accepted, each cumulative distribution is zero below its corresponding reservation price. At each net-price level, the cumulative probability under s_L is weakly lower than under s_H, illustrating first-order stochastic dominance of the accepted-price distribution under the lower search cost. The distributions and parameter values are chosen for illustrative purposes.

Consistent with equation (A.29), which shows that the expected net selling price increases with the reservation price, the lower search cost therefore yields a higher expected net selling price. Let a denote an arbitrary net-price level. The overall relationship can be summarized as

$$
\begin{aligned}
s_L < s_H \Rightarrow \quad & \underbrace{r_L > r_H}_{\substack{\text{Higher reservation price under} \\ \text{lower search cost (Figure A.1)}}} \\
\Rightarrow \quad & \underbrace{\Pr(p \le a \mid p \ge r_L) \le \Pr(p \le a \mid p \ge r_H) \quad \forall a}_{\substack{\text{First-order stochastic dominance of} \\ \text{the accepted-price distribution (Figure A.2)}}} \\
\Rightarrow \quad & \underbrace{\mathbb{E}[p \mid p \ge r_L] > \mathbb{E}[p \mid p \ge r_H]}_{\text{Higher expected net selling price}}. \qquad \text{(A.30)}
\end{aligned}
$$

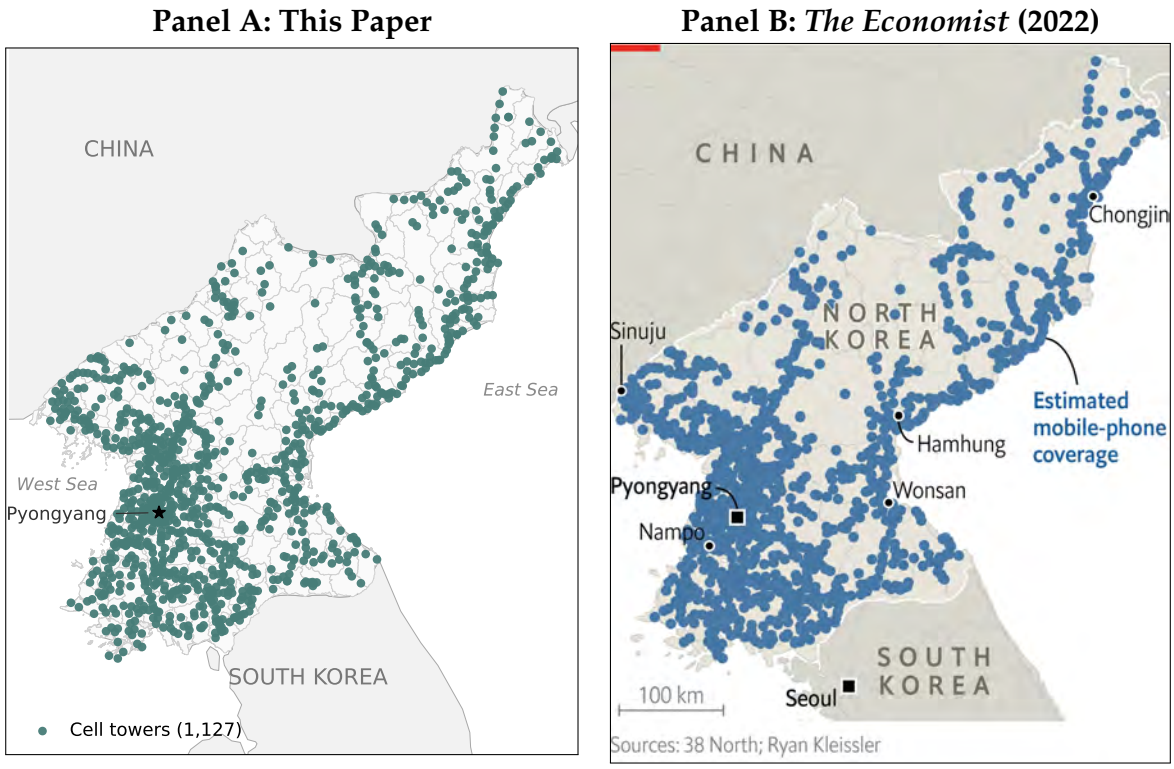
B Data

B.1. County Boundary Data

Several publicly available shapefiles delineate North Korea's county boundaries, but they differ in how units are defined and how many units they contain. I use the shapefiles from Kim et al. (2023), who digitized the official county and city maps from the 2008 North Korean census (DPRK, 2009) into 178 second-level administrative units. These boundaries predate the cell-tower rollout, which keeps county definitions fixed before treatment, and they match the geographic units of the census population data. All spatial datasets are converted to a common coordinate system (WGS 1984, EPSG:4326) to ensure spatial alignment across data sources.

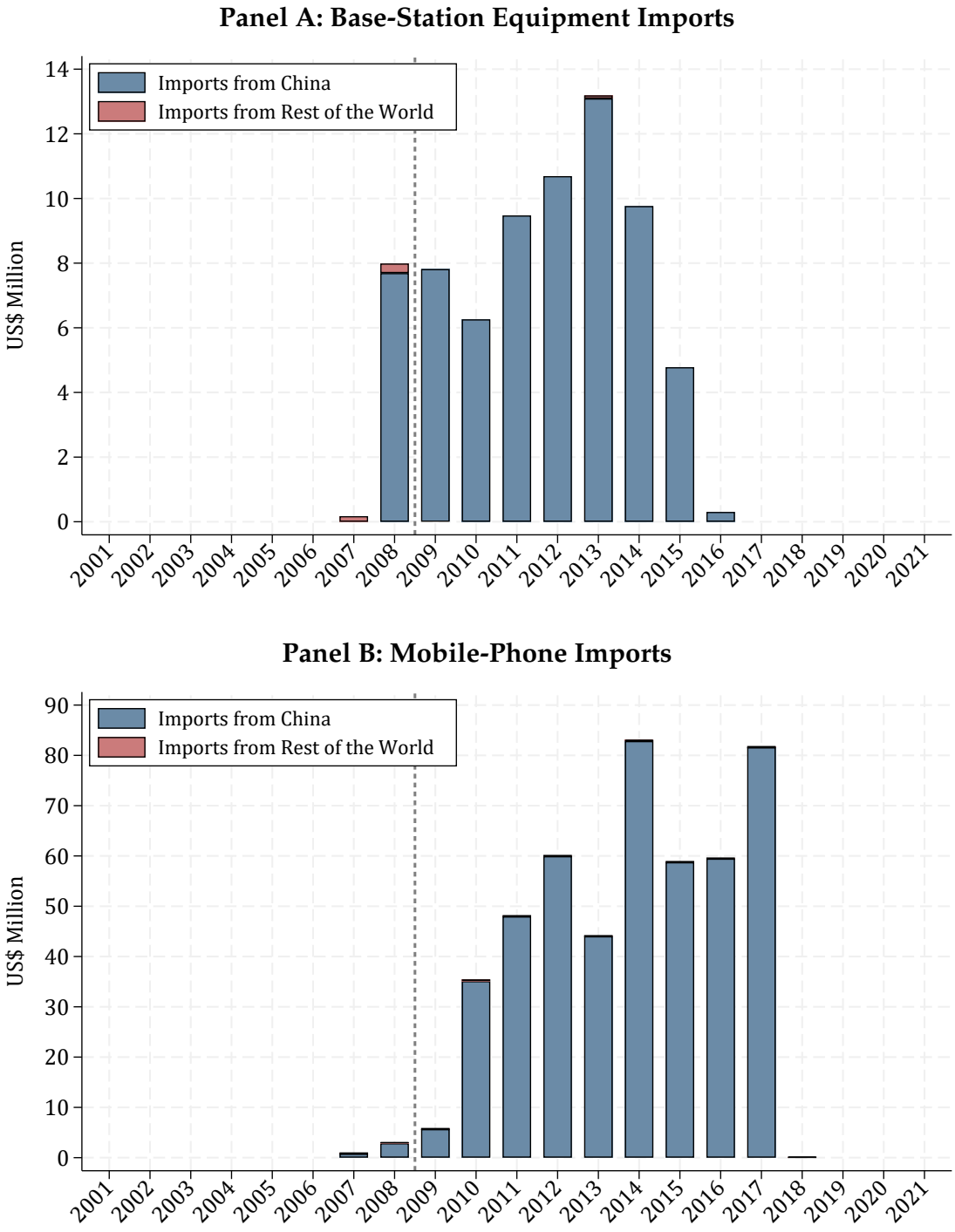
B.2. Cell-Tower Data

Figure B.1: Cell-Tower Locations: Comparison with an Independent Map



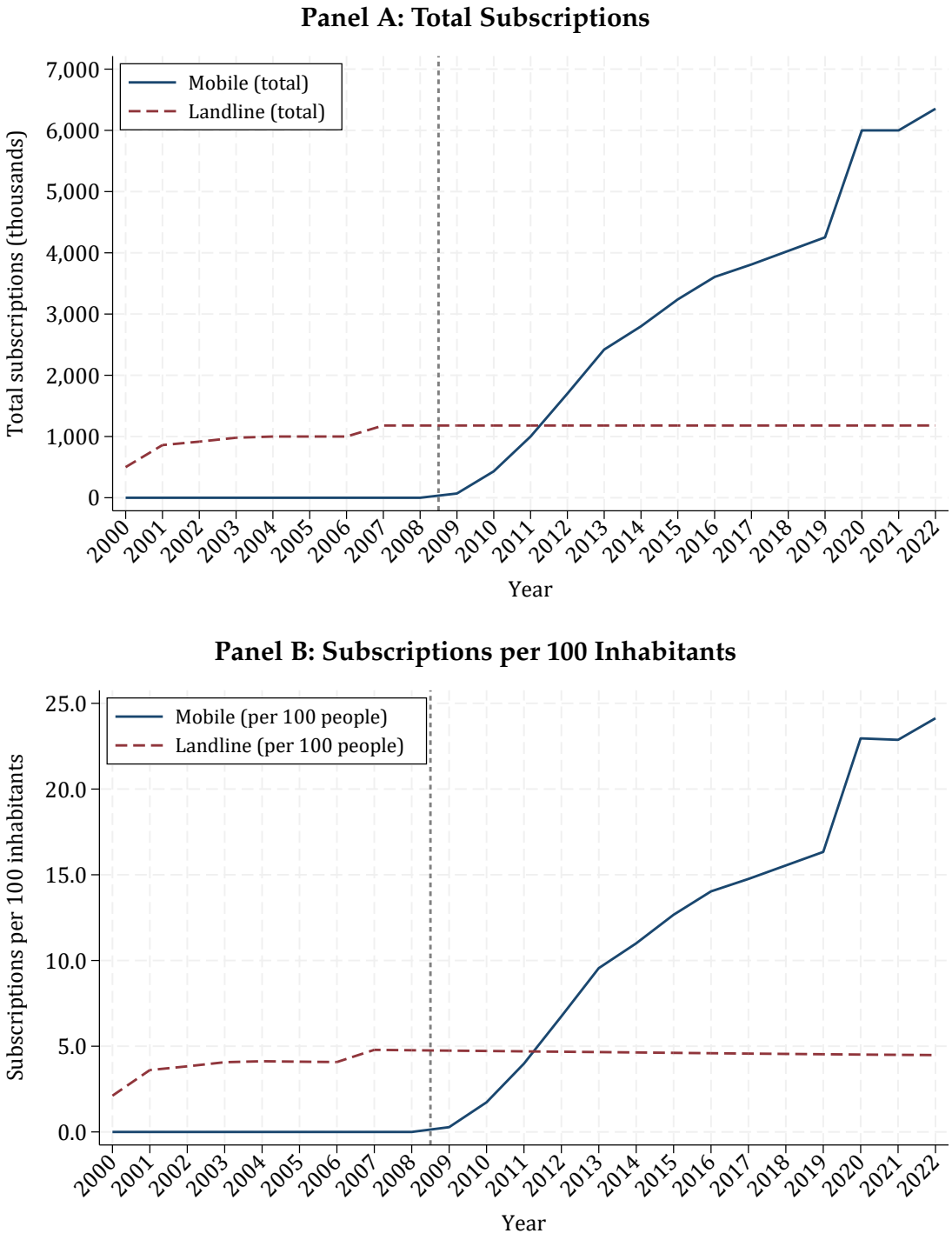
SOURCES: Google Earth; *The Economist* (The Economist, 2022).
NOTES: Panel A shows the locations of the 1,127 cell towers confirmed in this paper using historical satellite imagery from Google Earth and Esri World Imagery Wayback. The black star marks Pyongyang. Panel B is taken from *The Economist* (November 2022) and shows a map of North Korea's cellular network based on the 38 North Digital Atlas. Drawing on satellite imagery, photographs, and videos, 38 North identified more than 1,000 cellular base stations and used their locations to estimate mobile-phone coverage, but it does not report the date of the most recent imagery used.

Figure B.2: Telecommunications Equipment Imports into North Korea



NOTES: This figure reports telecommunications equipment imports into North Korea. Because North Korea does not publish official trade statistics, imports are measured using mirror data, that is, exports to North Korea reported by its trading partners. Panel A reports imports of base-station equipment (HS 851761), and Panel B reports imports of mobile phones (HS 851712). The total height of each bar represents imports from all trading partners, decomposed into imports from China and the rest of the world. The dashed vertical line marks the launch of Koryolink in December 2008. Imports fall to near zero after 2017 following UN Security Council Resolution 2397 (December 2017), which prohibited exports of electrical equipment to North Korea; border closures during the COVID-19 pandemic further restricted trade from 2020.

Figure B.3: Mobile and Landline Subscriptions in North Korea



NOTES: This figure reports mobile and landline subscriptions in North Korea from 2000 to 2022, based on estimates from the World Telecommunication/ICT Indicators Database of the International Telecommunication Union (ITU). Panel A reports total subscriptions (in thousands), and Panel B reports subscriptions per 100 inhabitants. Solid lines represent mobile subscriptions, and dashed lines represent landline subscriptions. The dotted vertical line marks the launch of Koryolink in December 2008.

Table B.1: Distribution of Uncertainty in Cell-Tower Deployment Years

Year Gap	Frequency	Percent (%)	Cumulative (%)
0 (Exact)	566	50	50
1	381	34	84
2	121	11	95
3	36	3	98
4	12	1	99
5	9	1	100
7	1	0	100
8	1	0	100
Total	1,127	100	100

NOTES: This table reports the distribution of deployment-year uncertainty across all 1,127 confirmed cell towers. The year gap is the width of the feasible deployment-year interval, and a gap of zero indicates that the deployment year is exactly identified. The table reports the number of towers, their share of all confirmed towers, and the cumulative share for each year gap.

Table B.2: Distribution of Cell Towers by Deployment Year under Alternative Timing Assignments

	Earliest-Year Assignment			Latest-Year Assignment		
Installation Year	**Frequency**	**Percent (%)**	**Cumulative (%)**	**Frequency**	**Percent (%)**	**Cumulative (%)**
2009	464	41	41	194	17	17
2010	51	5	46	199	18	35
2011	133	12	57	165	15	50
2012	106	9	67	74	7	56
2013	64	6	73	161	14	70
2014	151	13	86	90	8	78
2015	94	8	94	101	9	87
2016	30	3	97	76	7	94
2017	24	2	99	48	4	98
2018	2	0	99	9	1	99
2019	6	1	100	7	1	100
2020	1	0	100	2	0	100
2021	1	0	100	1	0	100
Total	1,127	100	100	1,127	100	100

NOTES: This table reports the distribution of all 1,127 confirmed cell towers by deployment year under alternative timing assignments. Towers with exactly identified deployment years are assigned to their observed year under both assignments. Towers with uncertain timing are assigned to the first year of their feasible range under the earliest-year assignment and to the last year under the latest-year assignment. Both assignments therefore contain the same 1,127 towers.

Table B.3: Distribution of Uncertainty in County First-Tower Timing

Year Gap	Frequency	Percent (%)	Cumulative (%)
0 (Exact)	110	64	64
1	47	27	91
2	11	6	97
3	2	1	98
4	1	1	99
5	1	1	99
8	1	1	100
Treated Total	173	100	100
Never Treated	5	.	.
All Counties	178	.	.

NOTES: This table reports the distribution of uncertainty in the timing of first cell-tower arrival across counties. For each county, the year gap is the difference between the latest and earliest possible years of its first tower arrival, obtained by assigning all towers with uncertain timing to the last or first year of their feasible ranges, respectively. For example, a county whose first tower could have arrived in any year from 2010 to 2012 has a year gap of two. A gap of zero indicates that the first-arrival year is exactly identified. Percentages are computed over the 173 treated counties that receive at least one confirmed tower. The five never-treated counties, which receive no confirmed tower during the sample period, are reported separately.

Table B.4: Distribution of Counties by First Cell-Tower Arrival Year under Alternative Timing Assignments

	Earliest-Year Assignment			Midpoint Assignment			Latest-Year Assignment		
First-Tower Arrival Year	Frequency	Percent (%)	Cumulative (%)	Frequency	Percent (%)	Cumulative (%)	Frequency	Percent (%)	Cumulative (%)
2009	157	91	91	97	56	56	97	56	56
2010	4	2	93	59	34	90	49	28	84
2011	6	3	97	8	5	95	16	9	94
2012	2	1	98	3	2	97	2	1	95
2013	2	1	99	4	2	99	5	3	98
2014	1	1	99	1	1	99	2	1	99
2015	1	1	100	0	0	99	0	0	99
2016	0	0	100	1	1	100	0	0	99
2017	0	0	100	0	0	100	2	1	100
Treated Total	173	100	100	173	100	100	173	100	100
Never Treated	5	.	.	5	.	.	5	.	.
All Counties	178	.	.	178	.	.	178	.	.

NOTES: This table reports the distribution of counties by the year of first cell-tower arrival under alternative timing assignments. The earliest-year and latest-year assignments use the earliest and latest feasible first-arrival years defined in Table B.3, and the midpoint assignment uses the rounded midpoint of the two. Counties with exactly identified first-arrival years receive the same year under all three assignments. The first-arrival year marks when a county first gains cell-tower access and does not reflect the number of towers deployed in that year. Each column includes all 178 counties, with the five never-treated counties reported separately.

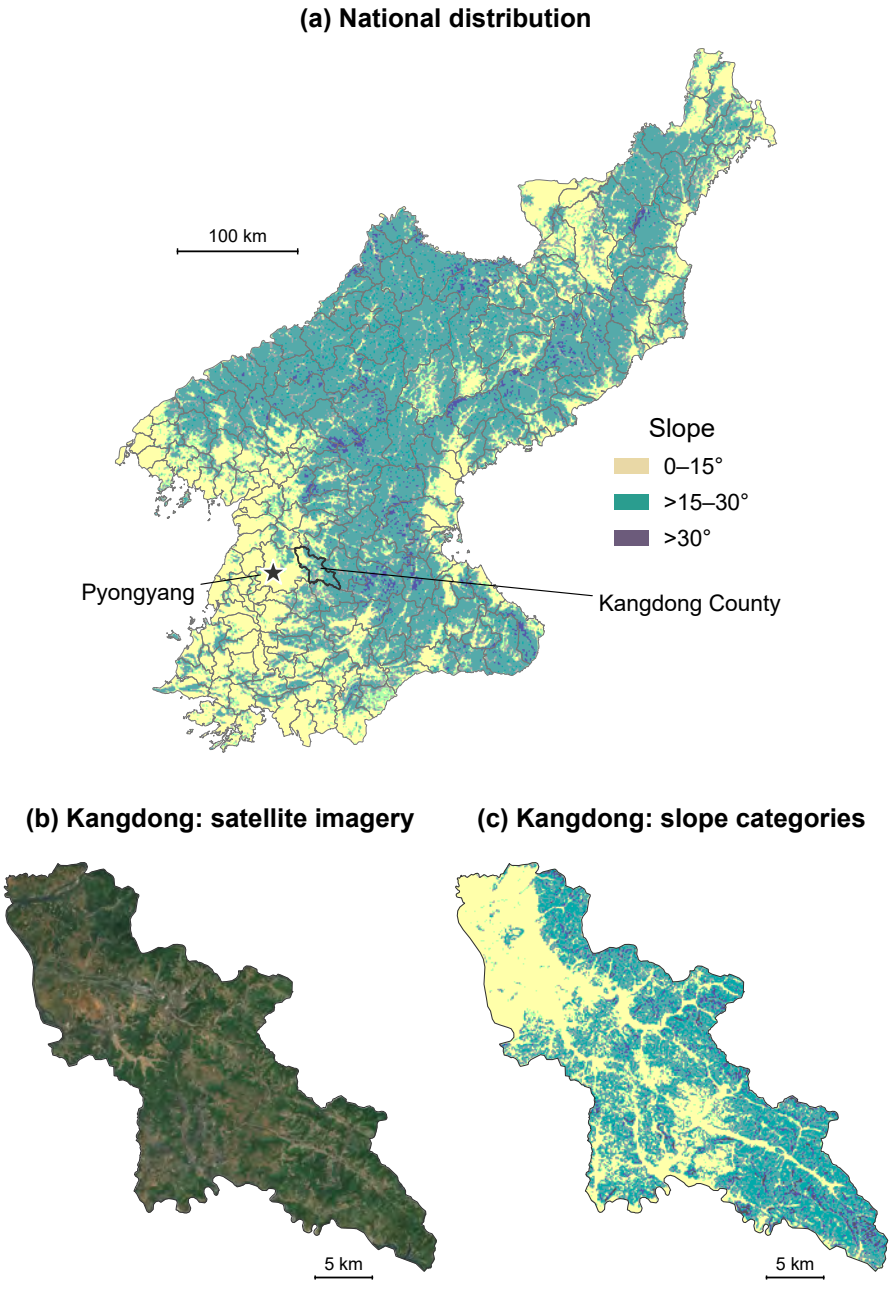
B.3. Vegetation Index

Figure B.4: Examples of Cultivation on Sloping Land in North Korea



SOURCE: (Lee, 2023) (top) and (Moon, 2025) (bottom).
NOTES: These photographs illustrate cultivation on sloping land in North Korea. The top photograph shows small cultivated plots, while the bottom shows a woman tending corn in Sakju County, North Pyongan Province, in June 2015.

Figure B.5: Spatial Distribution of Sloping Land in North Korea



SOURCE: SRTM elevation data; Sentinel-2 imagery.
NOTES: Panel (a) shows the spatial distribution of the three slope categories used in this study: 0–15°, > 15–30°, and > 30°. Thin gray lines indicate administrative boundaries, the star marks Pyongyang, and the highlighted boundary identifies Kangdong County. Panels (b) and (c) show a natural-color composite of Sentinel-2 imagery from 2020 and the corresponding slope categories for Kangdong County, respectively. These panels illustrate how the three slope categories are distributed within the same county.

B.4. Marketplace Area

Figure B.6: Examples of Goods Traded in North Korean Marketplaces



SOURCE: First row from *Korea Report*, March 14, 2022 (https://www.koreareport.co.kr/news/articleView.html?idxno=10515), and *Seoul Economic Daily*, February 7, 2024 (https://www.sedaily.com/article/13838842), both in Korean. Remaining rows from *Daily NK*, May 4, 2017 (https://www.dailynk.com/english/exclusive-video-of-north-korean-ma/).
NOTES: These photographs illustrate the variety of goods traded in North Korean marketplaces. The first row shows agricultural products, including vegetables and fruits. The second row shows imported products with English labels (circled in yellow). The remaining rows show household appliances, winter clothing, and used books.

Figure B.7: Unauthorized Street Trade and Official Marketplaces



(a) Street market (not included)



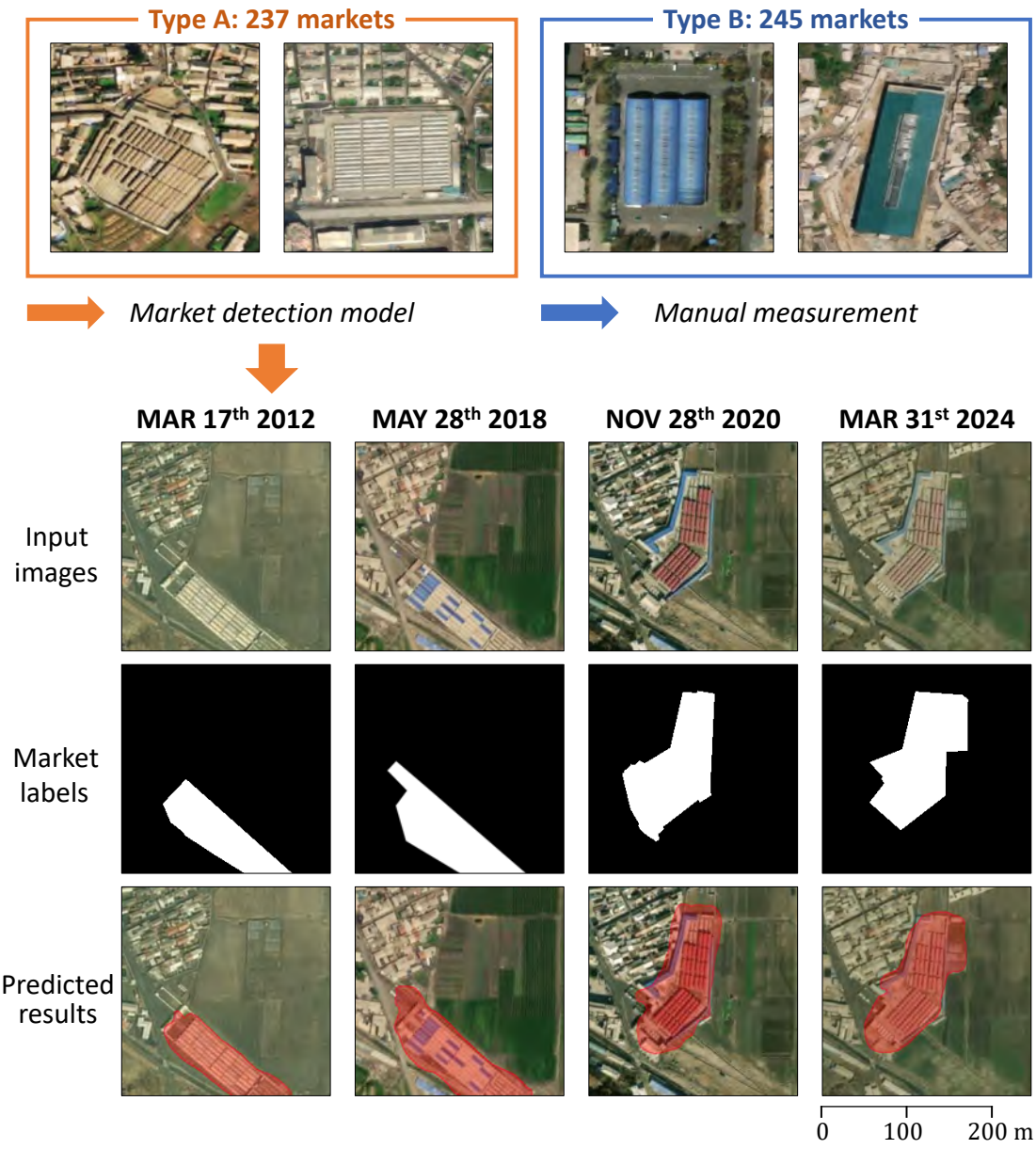
(b) Marketplace entrance



(c) Rows of designated stalls

SOURCE: Panels (b) and (c) from (Jung, 2013) and (Ahn, 2018).
NOTES: Panel (a) shows a roadside street market, an unauthorized form of trade that is not captured by the marketplace-area measure. Panels (b) and (c) show officially sanctioned marketplaces (*Jangmadang*), which operate at fixed and bounded sites. Panel (b) shows the gated entrance of a marketplace, and panel (c) shows rows of designated stalls where vendors pay recurring fees to occupy selling space.

Figure B.8: Marketplace Types and Measurement Framework



SOURCE: Yang et al. (2026).

NOTES: This figure illustrates the marketplace classification and measurement framework used to construct the marketplace-area panel. Marketplaces are classified into two morphological types. Type A markets consist of rows of open-air stalls within a defined perimeter and are delineated using a deep learning segmentation model (237 markets), whereas Type B markets consist of permanent enclosed structures and are delineated manually from satellite imagery (245 markets). The bottom panel illustrates the segmentation procedure for a Type A market at four points in time (March 2012, May 2018, November 2020, and March 2024). For each date, the top row shows the input satellite imagery, the middle row shows the manually annotated market labels used for training, and the bottom row shows the model's predicted segmentation overlaid on the imagery.

Figure B.9: Examples of Segmentation Results: Marketplace Expansion



SOURCE: Yang et al. (2026).

NOTES: This figure presents examples of marketplace expansion. Shaded areas show predicted marketplace footprints, and numbers in parentheses report the predicted area for each year. Images from 2023–2024 fall outside the 2008–2021 analysis period and are shown for illustration only. (a) Yaksu Market in Kaechon City (125.894E, 39.714N). (b) Namsinuiju Market in Sinuiju City (124.429E, 40.065N). (c) Paechon Market in Paechon County (126.308E, 37.990N). (d) Ichon Market in Ichon County (126.894E, 38.492N). (e) Sakju Market in Sakju County (125.049E, 40.386N).

Figure B.10: Examples of Segmentation Results: Marketplace Contraction



SOURCE: Yang et al. (2026).
NOTES: This figure presents examples of marketplace contraction. Shaded areas show predicted marketplace footprints, and numbers in parentheses report the predicted area for each year. Images from 2022–2024 fall outside the 2008–2021 analysis period and are shown for illustration only. (a) Changdo Market in Changdo County (127.713E, 38.642N). (b) Hoeyang Market in Hoeyang County (127.597E, 38.707N). (c) Woneup Market in Onchon County (125.179E, 38.849N). (d) Haksan Market in Unsan County (126.021E, 39.431N). (e) Munhwa Market in Tanchon City (128.812E, 40.921N).

C Empirical Strategy

C.1. Measurement Error in the Treatment

This appendix shows that the Uniform tower measure yields a consistent estimate of β in Equation (11), even though the construction year of many towers is not observed exactly.

Let C_{ct} denote the true cumulative number of towers in county c and year t, and suppose the outcome follows

$$Y_{ct} = \beta C_{ct} + \alpha_c + \gamma_t + \varepsilon_{ct}, \tag{C.1}$$

where α_c and γ_t are county and year fixed effects. Because imagery is not available every year, C_{ct} is not observed for some county-years. Let Ω_c denote the imagery available for county c, that is, the dates of the available images and whether each tower appears in them. Assume that the imagery available for a county is unrelated to outcome shocks in any year,

$$\mathbb{E}[\varepsilon_{cs} \mid \Omega_c] = 0 \quad \text{for all } s. \tag{C.2}$$

Define the measured tower stock as the expected true stock given the available imagery,

$$T_{ct} = \mathbb{E}[C_{ct} \mid \Omega_c], \tag{C.3}$$

and the measurement error as $u_{ct} = C_{ct} - T_{ct}$. By construction,

$$\mathbb{E}[u_{cs} \mid \Omega_c] = 0 \quad \text{for all } s. \tag{C.4}$$

Substituting $C_{ct} = T_{ct} + u_{ct}$ into equation (C.1) gives

$$Y_{ct} = \beta T_{ct} + \alpha_c + \gamma_t + \nu_{ct}, \qquad \nu_{ct} = \varepsilon_{ct} + \beta u_{ct}. \tag{C.5}$$

From equations (C.2) and (C.4),

$$\mathbb{E}[\nu_{cs} \mid \Omega_c] = 0 \quad \text{for all } s. \tag{C.6}$$

Because T_{ct} is a function of Ω_c in every year, the within-county deviations of T_{ct} are uncorrelated with those of ν_{ct}, and the fixed effects estimator of β in equation (C.5) is consistent. The error u_{ct} enters the error term and generally increases the standard error of $\hat{\beta}$, but it does not attenuate $\hat{\beta}$. This contrasts with classical measurement error, in which the observed regressor equals the true value plus independent noise, so that the noise is correlated with the observed regressor and biases $\hat{\beta}$ toward zero. Because u_{ct} is serially correlated within a county, standard errors are clustered by county.

The Uniform Measure. For a tower whose construction year is known exactly, its contribution to C_{ct} is observed. For a tower that is absent in one image and present in a later image, the construction year lies in a range of feasible years. If construction is equally likely in each feasible year, the tower's expected contribution to C_{ct} rises by an equal amount in each year of the range. For example, a tower absent in a 2010 image and present in a 2013 image has feasible construction years 2011, 2012, and 2013, and contributes $1/3$, $2/3$, and 1 to the expected tower stock in these years. Summing these contributions across towers in a county gives $\mathbb{E}[C_{ct} \mid \Omega_c]$, which is the *Uniform* measure.

The Earliest and Latest Measures. If construction is instead concentrated early or late within each range, $\mathbb{E}[C_{ct} \mid \Omega_c]$ differs from the Uniform measure. Whatever the distribution of construction years within each range, however, the expected tower stock lies between the *Latest* measure, which assigns every uncertain tower to the last feasible year, and the *Earliest* measure, which assigns it to the first. Estimates under these two measures therefore show how sensitive the results are to the assumed timing within each range.

C.2. Network Scale and Duration

This appendix describes how the coefficients in Equation (13) are interpreted and estimated. The effect of one additional tower on Y_{ct} is

$$\frac{\partial Y_{ct}}{\partial T_{ct}} = \beta_2 + \beta_4 d_{ct}, \qquad T_{ct} > 0. \tag{C.7}$$

In the year of first access ($d_{ct} = 0$), the first tower raises Y_{ct} by $\beta_1 + \beta_2$, while each additional tower raises it by β_2. If connection itself mattered, β_1 would be large, and if the size of the network were what mattered, β_1 would be close to zero. Equation (C.7) shows that β_4 measures how much the effect of one tower changes with each year of exposure. Once the interaction is included, β_2 is the effect of one tower in the year of first access rather than the average effect across all years.

I estimate Equation (13) in three steps. The first restricts $\beta_1 = \beta_3 = \beta_4 = 0$,

$$Y_{ct} = \beta_2 T_{ct} + \alpha_c + \gamma_t + \varepsilon_{ct}, \tag{C.8}$$

which reproduces Equation (11) and asks whether outcomes rise with the number of towers. The second adds the access indicator ($\beta_3 = \beta_4 = 0$),

$$Y_{ct} = \beta_1 \mathbb{1}\{T_{ct} > 0\} + \beta_2 T_{ct} + \alpha_c + \gamma_t + \varepsilon_{ct}, \tag{C.9}$$

which asks whether the effect comes from connection itself or from the size of the network. The third estimates the full specification in Equation (13), which asks whether the effect of the network grows with time since access, beyond changes common to all connected counties. As in the baseline, I report estimates with county and year fixed effects and with county and province-by-year fixed effects, and standard errors are clustered by county.

D Results

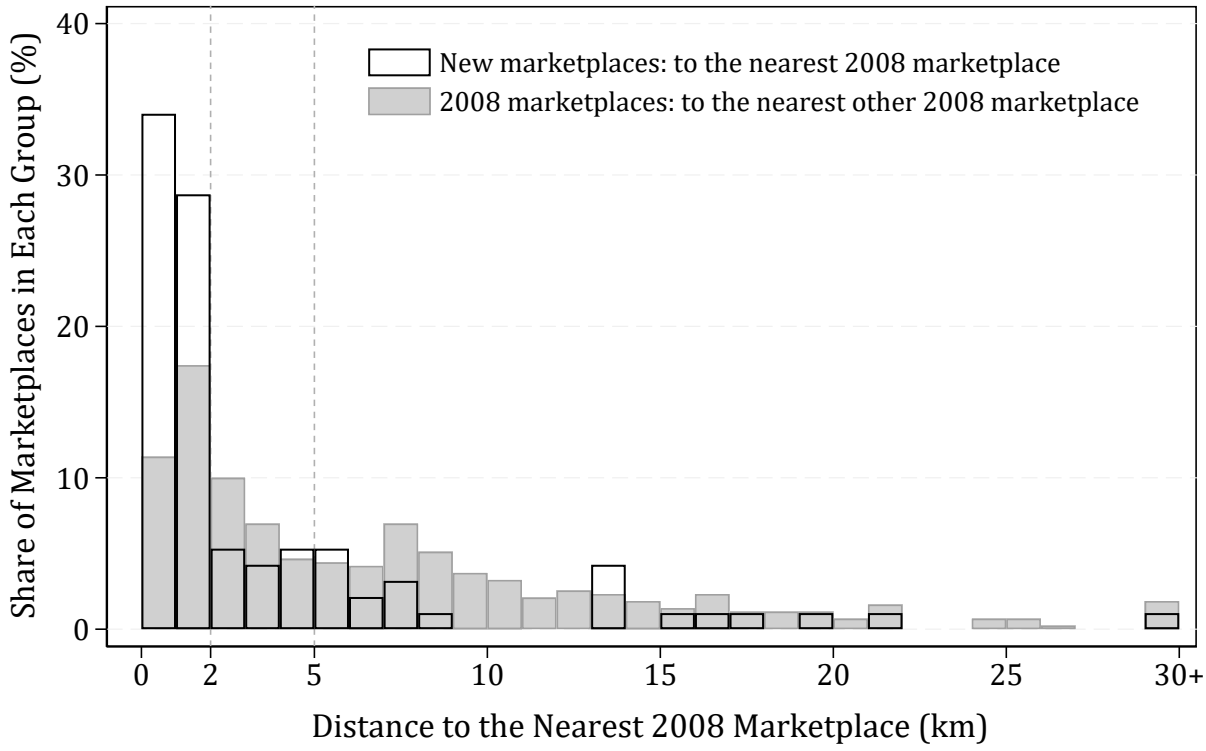
Table D.1: New Marketplaces by Location and Relocation

	Dep. Var: Area of New Marketplaces (m^2)			
Panel A. New marketplaces by distance to the nearest 2008 marketplace				
	All new	$<$ 2 km	2–5 km	$\geq$ 5 km
	(1)	(2)	(3)	(4)
Cumulative cell towers				
County and year FE	335.038***	199.775***	90.256**	45.007
	(72.879)	(48.991)	(42.157)	(48.276)
County and province $\times$ year FE	257.753***	114.612*	84.901**	58.240
	(92.797)	(63.701)	(38.933)	(53.664)
Mean area per county, 2021 (m^2)	2,431	1,521	391	519
New marketplaces	94	59	14	21

Panel B. New marketplaces by relocation from a 2008 marketplace within 2 km		Narrow definition		Broad definition	
	All new	Relocated	Not relocated	Relocated	Not relocated
	(1)	(2)	(3)	(4)	(5)
Cumulative cell towers					
County and year FE	335.038***	34.791	300.247***	112.331*	222.707***
	(72.879)	(46.715)	(71.279)	(60.171)	(69.116)
County and province $\times$ year FE	257.753***	-43.209	300.962***	48.174	209.580**
	(92.797)	(61.500)	(87.200)	(72.543)	(83.187)
Mean area per county, 2021 (m^2)	2,431	371	2,060	645	1,786
New marketplaces	94	11	83	20	74
Observations	2,492	2,492	2,492	2,492	2,492
N of clusters	178	178	178	178	178

NOTES: The sample is the 94 new marketplaces in column (2) of Table 4: marketplaces inactive in 2008 and active in at least one year in 2009–2021. The dependent variable is the total area in square meters of the indicated new marketplaces in each county-year, 2008–2021. Within each panel, columns (2) onward partition the new marketplaces, so their coefficients sum to column (1). Panel A groups new marketplaces by the straight-line distance from their coordinates to those of the nearest marketplace active in 2008. Panel B classifies a new marketplace as a relocation if a marketplace active in 2008 and located within 2 km was active in the year before the new marketplace opened and then closed. Narrow: it closed in the year the new marketplace opened. Broad: it closed by two years after the opening, or its area fell by more than half over that period; the broad definition is an upper bound, since it also counts closures and contractions unrelated to the new marketplace. Relocations are identified relative to marketplaces active in 2008; only 3 of the 94 new marketplaces closed after opening, so relocations among new marketplaces are negligible. The treatment variable is the cumulative number of cell towers (*Uniform*). Mean area per county is the 2021 mean of the dependent variable across counties. Observations and clusters refer to the county and year fixed-effects specification. Standard errors are reported in parentheses. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure D.1: Histogram of Distance to the Nearest 2008 Marketplace: New versus Existing Marketplaces



NOTES: This figure shows the distances in Figure VIII in 1 km bins. Outlined bars show the distance from each of the 94 new marketplaces (inactive in 2008 and active in at least one year in 2009–2021) to the nearest of the 430 marketplaces active in 2008. Shaded bars show the distance from each of the 430 marketplaces active in 2008 to the nearest other marketplace active in 2008. Each bar reports the share of marketplaces in the group whose distance falls in the bin, so the bars sum to 100 percent within each group. Distances are straight-line (haversine) distances between marketplace coordinates. Vertical dashed lines mark 2 and 5 km. Distances above 30 km are included in the last bin.

Table D.2: Effects on Marketplace Area: Largest versus Other Marketplaces

	Marketplace Area (m^2)			
	Total	Largest marketplace in 2008	Other marketplaces	Number of marketplaces
	(1)	(2)	(3)	(4)
Panel A. County and year fixed effects				
Cumulative cell towers	336.049***	-46.528	382.577***	0.063***
	(95.371)	(63.973)	(74.466)	(0.016)
Panel B. County and province × year fixed effects				
Cumulative cell towers	288.520**	-15.744	304.263***	0.059***
	(111.997)	(76.869)	(91.642)	(0.017)
Mean per county, 2021	14,055.3	6,244.5	7,810.8	2.8
Observations	2,380	2,380	2,380	2,380
N of clusters	170	170	170	170

NOTES: The sample includes the 170 counties with at least one marketplace active in 2008, so that each county has a largest marketplace in 2008; the period is 2008–2021. Column (2) is the area of the marketplace that was the largest in each county in 2008, tracked in every year. Column (3) is the area of all other marketplaces in the county, including those that opened after 2008, so columns (2) and (3) sum to column (1). Column (4) is the number of active marketplaces. Within counties, marketplaces that were larger in 2008 also did not respond more to tower exposure (not reported). Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table D.3: Heterogeneous Effects on Vegetation by Baseline County Characteristics

	Dep. Var: 100 × ln(NDVI), Sloping Land – Collective Farmland		
	(1)	(2)	(3)
Cumulative cell towers ×			
Panel A. Local demand			
ln Population, 2008	-0.034**	-0.004	
	(0.015)	(0.013)	
ln Marketplace area, 2008	-0.042**	-0.008	-0.133**
	(0.019)	(0.013)	(0.067)
Panel B. Transport and market access			
Rail station	-0.080	0.116	-0.048
	(0.120)	(0.090)	(0.130)
Distance to the nearest city (10 km)	0.004	-0.009	-0.009
	(0.008)	(0.006)	(0.014)
Road density, 2008	-0.015	-0.004	0.020
	(0.012)	(0.010)	(0.039)
Panel C. Proximity to the regime			
Distance to Pyongyang (10 km)	0.000	-0.001	-0.001
	(0.001)	(0.001)	(0.001)
Special economic zone	-0.124*	-0.075	-0.131
	(0.064)	(0.050)	(0.135)
Panel D. Proximity to China			
China border county	-0.092	-0.060	-0.176*
	(0.080)	(0.082)	(0.096)
Distance to the nearest border crossing (10 km)	-0.002	-0.004	0.001
	(0.003)	(0.003)	(0.003)
Mean of dep. var., 2002–2008	17.66	17.66	17.66
Observations	2,408	2,408	2,408
N of clusters	172	172	172
County FE	Yes	Yes	Yes
Year FE	Yes		Yes
Province × year FE		Yes	
Cell towers × ln population, 2008			Yes
ln population, 2008 × year FE			Yes

NOTES: This table repeats Table 5 with vegetation as the outcome. The dependent variable is the difference between 100 × ln(mean April–August NDVI) on sloping land (15° or above) and on collective farmland (below 15°), for 2002–2015. Each cell reports the coefficient on cumulative cell towers interacted with the indicated county characteristic, from a separate regression that also includes cumulative cell towers and the characteristic interacted with year fixed effects. Characteristics are measured in 2008, before the network rollout. The sample includes 172 counties with NDVI observations on both land types. Fixed effects and controls are indicated at the bottom of the table. Variable definitions follow Table 5. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

E Robustness

Table E.1: Pre-Rollout Characteristics by Timing of Mobile-Phone Access

	Unconditional comparison				Conditional on all characteristics		
	Access 2009 (SD) (1)	Later access (SD) (2)	(1)–(2) (SE) (3)	KS test p-value (4)	Access 2009 (0/1) (SE) (5)	Towers 2015 (SE) (6)	Towers 2021 (SE) (7)
Local demand							
ln population, 2008	11.70	11.17	0.53***	0.00	0.26***	4.05***	4.46***
	(0.70)	(0.58)	(0.10)		(0.08)	(1.28)	(1.47)
Urban population share, 2008	0.52	0.41	0.12***	0.00	-0.00	0.09	0.18
	(0.24)	(0.19)	(0.03)		(0.06)	(0.44)	(0.48)
City (*si*), 2008 (indicator)	0.23	0.05	0.18***		-0.05	-1.50*	-1.71*
	(0.42)	(0.22)	(0.05)		(0.05)	(0.86)	(1.00)
Industry							
ln(1 + number of industrial facilities), 2008	5.27	4.48	0.78***	0.05	0.05	0.13	0.14
	(1.91)	(1.94)	(0.29)		(0.04)	(0.25)	(0.26)
ln(1 + number of mines), 2008	1.65	1.35	0.30***	0.11	-0.03	0.80***	0.91***
	(0.72)	(0.64)	(0.10)		(0.04)	(0.30)	(0.33)
Infrastructure							
Railway station, 2008 (indicator)	0.89	0.60	0.28***		0.10**	-0.58*	-0.68*
	(0.32)	(0.49)	(0.06)		(0.04)	(0.34)	(0.37)
Road density (km/km^2), 2008	2.09	2.05	0.04	0.45	-0.03	0.19	-0.04
	(1.26)	(1.46)	(0.21)		(0.09)	(0.65)	(0.69)
Port, 2008 (indicator)	0.07	0.01	0.06**		-0.03	1.20**	1.19*
	(0.26)	(0.11)	(0.03)		(0.04)	(0.59)	(0.63)
Road growth, 2008–14 (%)†	20.65	15.73	4.92	0.94			
	(70.26)	(46.25)	(8.79)				
Land and climate							
Share of land $\geq$15°	0.22	0.21	0.01	0.45	0.10	-0.73	-0.79
	(0.17)	(0.19)	(0.03)		(0.07)	(0.51)	(0.54)
Share $\geq$15°, tree cover $\geq$70%, 2000	0.03	0.04	-0.01	0.81	-0.03	0.37	0.34
	(0.05)	(0.07)	(0.01)		(0.05)	(0.50)	(0.56)
Precipitation (100 mm), 2002–08 mean	7.24	7.10	0.15	0.50	0.07	0.17	0.16
	(1.62)	(1.32)	(0.22)		(0.04)	(0.30)	(0.31)
Baseline outcomes (vegetation: 100 × ln NDVI)							
ln(1 + marketplace area), 2008	8.73	7.99	0.74**	0.00	-0.01	-0.03	-0.11
	(1.91)	(2.26)	(0.32)		(0.04)	(0.19)	(0.23)
Vegetation, sloping land, 2002–08 mean	-32.04	-32.29	0.24	0.99	-0.03	-0.11	-0.04
	(5.12)	(5.79)	(0.84)		(0.05)	(0.36)	(0.38)
Vegetation, collective farmland, 2002–08 mean	-50.67	-50.98	0.31	0.34	0.05	1.22	1.13
	(12.60)	(15.03)	(2.10)		(0.12)	(0.75)	(0.76)
Vegetation gap (sloping–farmland), 2002–08 mean‡	17.95	17.31	0.64	0.05			
	(9.57)	(11.33)	(1.62)				
Pre-rollout trends (annual change)							
Nighttime lights, 1992–2008	0.04	0.05	-0.01	0.21	0.03	0.14	0.15
	(1.42)	(0.94)	(0.18)		(0.05)	(0.33)	(0.32)
Vegetation, sloping land, 2002–08	-0.22	-0.28	0.07	0.05	0.01	0.25	0.34
	(0.55)	(0.46)	(0.08)		(0.05)	(0.34)	(0.38)
Vegetation, collective farmland, 2002–08	-0.40	-0.67	0.27**	0.05	0.09	0.31	0.21
	(0.66)	(0.78)	(0.11)		(0.06)	(0.36)	(0.39)
Levels jointly zero: F (p)					5.85 (0.000)	8.26 (0.000)	7.00 (0.000)
Trends jointly zero: F (p)					1.63 (0.185)	1.04 (0.378)	0.95 (0.420)
R^2					0.264	0.611	0.608
Counties	97	81			172	172	172

NOTES: Columns (1)–(4) compare the 97 counties that gained mobile-phone access in 2009 (midpoint timing) with the 81 counties that gained access later or never; for indicators, means are shares of counties. Column (3) reports the difference between columns (1) and (2), with robust standard errors in parentheses, and column (4) reports the p-value of a Kolmogorov–Smirnov (KS) test of equal distributions (not reported for indicators). Column (5) is a linear probability model in which the outcome equals one if the county received its first tower in 2009 (midpoint timing) and zero otherwise, capturing the timing of access. Columns (6)–(7) are OLS regressions of the cumulative number of towers, capturing the intensity of access, measured in 2015 and 2021, the last years of the vegetation and marketplace samples. Columns (5)–(7) regress each outcome on all characteristics at once (except those marked † and ‡), using the 172 counties with all characteristics observed. Each characteristic is standardized to have mean zero and standard deviation one, so a coefficient is the change in the outcome associated with a one-standard-deviation increase in that characteristic, holding the others fixed. Robust standard errors are in parentheses. "Levels jointly zero" tests that the coefficients on all characteristics other than the pre-rollout trends are jointly zero, and "Trends jointly zero" tests that the coefficients on the three pre-rollout trends are jointly zero. The share of land with slope of 15° or above is time-invariant. Vegetation is 100 × ln mean April–August NDVI on sloping land (slope of 15° or above) and on collective farmland (below 15°); since NDVI lies below one, its log is negative. Pre-rollout trends are slopes from county-specific regressions on year; the nighttime-light trend uses 100 × ln(1 + DMSP nighttime lights) and carries limited information because most counties are unlit. †Measured over the rollout period; see Table E.4. ‡The vegetation gap is vegetation on sloping land minus that on collective farmland; it is excluded from columns (5)–(7) because both of its components are already included. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.2: Timing Tests and Pre-Rollout Differences

		Vegetation (100 × ln NDVI)			
	Marketplace area (m^2)	Sloping – farmland	Sloping land	Collective farmland	Sloping land, low tree cover
	(1)	(2)	(3)	(4)	(5)
Panel A. Were outcomes already changing before towers arrived?					
Did outcomes differ before the first tower arrived?					
Placebo: Cell towers$_{t+1}$	268.336	-0.391**	0.347*	0.799***	0.230
	(258.804)	(0.183)	(0.192)	(0.288)	(0.194)
Placebo: Cell towers$_{t+2}$	110.801	-0.206*	0.146	0.425**	0.059
	(114.345)	(0.112)	(0.097)	(0.179)	(0.107)
Placebo: Cell towers$_{t+3}$	49.998	-0.060	0.103	0.200**	0.037
	(65.672)	(0.052)	(0.062)	(0.089)	(0.078)
Did outcomes differ before additional towers arrived?					
Cell towers$_{t+1,\ t+2,\ t+3}$ controlling for Cell towers$_t$ (joint p-value)	0.183	0.558	0.417	0.311	0.880
Panel B. Did marketplace growth attract towers?					
Marketplace growth$_t$ → new cell towers$_{t+1}$	-0.009				
	(0.009)				
Marketplace growth$_t$ → new cell towers$_{t+2}$	-0.015*				
	(0.008)				
Marketplace growth$_t$ → new cell towers$_{t+3}$	-0.001				
	(0.006)				
Panel C. Do the effects remain after controlling for pre-rollout differences?					
Baseline	337.215***	0.130***	0.065**	-0.074	0.097***
	(93.122)	(0.033)	(0.028)	(0.048)	(0.028)
Pre-rollout trends × year	359.569***	0.117***	0.043	-0.074*	0.077**
	(85.585)	(0.028)	(0.029)	(0.040)	(0.031)
Pre-rollout level of outcome × year	364.236***	0.030	0.078***	-0.013	0.103***
	(97.154)	(0.027)	(0.026)	(0.052)	(0.029)
Dropping two years before first tower	386.547***	0.130***	0.072**	-0.056	0.095***
	(106.206)	(0.037)	(0.029)	(0.050)	(0.031)

NOTES: Column (1) uses marketplace area (m^2), 2008–2021. Columns (2)–(5) use 100 × ln mean April–August NDVI, 2002–2015: sloping land (slope of 15° or above) minus collective farmland (below 15°), each land type separately, and sloping land with tree cover below 30% in 2000. Cell towers are the cumulative number of towers in a county, and a subscript $t+k$ denotes k years after the outcome year t. All regressions include county and year fixed effects, with standard errors clustered by county in parentheses. In Panels A and B, insignificant estimates support the design; in Panel C, estimates that remain significant do. Panel A: the placebo rows regress the outcome on cell towers in $t+k$ alone, using only years before each county's first tower, so that no county has a tower in the sample; this keeps where towers were placed but moves their timing earlier. For marketplace area, this leaves 127 observations in 21 counties because the data begin in 2008. The joint test regresses the outcome on cell towers in t and in $t+1$ to $t+3$, using all years, and tests that the coefficients on towers in $t+1$ to $t+3$ are zero; with towers in $t+1$ to $t+2$ or $t+1$ to $t+4$, the p-values are similar for vegetation, and 0.269 and 0.072 for marketplace area. Panel B regresses the number of new cell towers in $t+k$ on the change in marketplace area from $t-1$ to t (per 1,000 m^2). Panel C: each row changes the baseline specification separately. The second row adds the three pre-rollout trends in Table E.1 (the slopes of ln nighttime lights over 1992–2008 and of vegetation on sloping land and on collective farmland over 2002–08), interacted with year fixed effects. The third row adds the pre-rollout level of the outcome (log marketplace area in 2008; the 2002–08 mean of each vegetation outcome), interacted with year fixed effects. The last row drops the two years before each county's first tower. In Table E.1, the pre-rollout trends do not jointly predict the rollout, but with province fixed effects they jointly predict the number of towers in 2015 and 2021 ($p = 0.032$ and 0.019). With province-by-year fixed effects, the four Panel C estimates are 297, 307, 323, and 328 for marketplace area; 0.076, 0.070, 0.034, and 0.062 for sloping land minus collective farmland; 0.102, 0.078, 0.090, and 0.124 for sloping land; 0.006, 0.007, 0.013, and 0.054 for collective farmland; and 0.135, 0.122, 0.140, and 0.152 for sloping land with low tree cover. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.3: Estimates within Comparable Counties

		Vegetation (100 × ln NDVI)			
	Marketplace area (m^2)	Sloping – farmland	Sloping land	Collective farmland	Sloping land, low tree cover
	(1)	(2)	(3)	(4)	(5)
Baseline (county and year FE)	337.215***	0.130***	0.065**	-0.074	0.097***
	(93.122)	(0.033)	(0.028)	(0.048)	(0.028)
Panel A. Do the effects remain among counties of similar location, size, and terrain?					
Same province (province × year FE)	297.349***	0.076***	0.102***	0.006	0.135**
	(108.895)	(0.024)	(0.036)	(0.046)	(0.054)
Adjacent county pairs (pair × year FE)	298.402***	0.020	0.074***	0.051**	0.051
	(65.758)	(0.014)	(0.023)	(0.020)	(0.032)
Similar population (tercile × year FE)	281.067**	0.054	0.071**	-0.001	0.082**
	(108.606)	(0.033)	(0.032)	(0.049)	(0.032)
Similar terrain (tercile × year FE)	302.594***	0.079***	0.074**	-0.026	0.096***
	(101.155)	(0.030)	(0.029)	(0.047)	(0.030)
Similar population and terrain (both)	266.253**	0.042	0.073**	0.002	0.078**
	(114.207)	(0.031)	(0.032)	(0.050)	(0.033)
Panel B. Do the effects remain among counties with similar characteristics?					
Nearest-neighbor matching (3 matches)	243.466**	0.097***	0.078**	-0.085**	0.086***
	(108.356)	(0.027)	(0.033)	(0.037)	(0.028)
Propensity-score strata (quintile × year FE)	238.615**	0.160***	0.047	-0.146**	0.083***
	(107.726)	(0.042)	(0.036)	(0.068)	(0.031)
Entropy balancing	226.662	0.145***	0.040	-0.131	0.064
	(139.775)	(0.039)	(0.040)	(0.082)	(0.040)

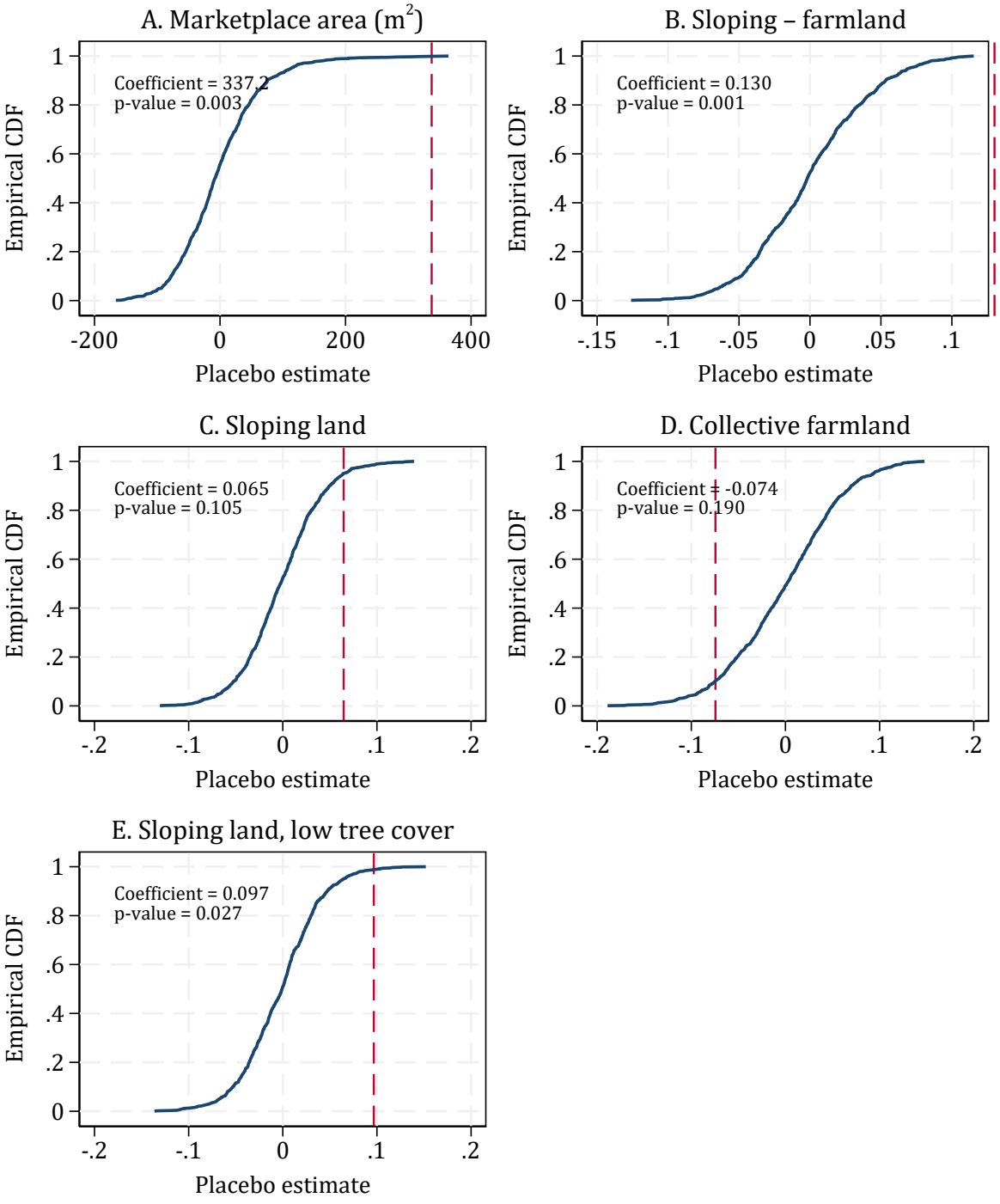
NOTES: Each cell reports the coefficient on the cumulative number of cell towers from a separate regression. Panel A compares counties of similar location, size, and terrain. "Adjacent county pairs" stacks every pair of counties that share a border (409–447 pairs, depending on the outcome), so a county appears once for each neighbor, and also includes pair × county fixed effects; the effect is identified from differences in towers between neighboring counties in the same year. Population terciles are defined by ln population in 2008, and terrain terciles by the share of land with slope of 15° or above. Panel B compares counties with similar pre-rollout characteristics, chosen separately for each outcome. The characteristics always include ln population in 2008 and the outcome's own pre-rollout level (ln(1 + marketplace area) in 2008; mean vegetation in 2002–08), and add those that double-selection LASSO (Belloni, Chernozhukov, and Hansen 2014) selects, from 13 county characteristics measured in 2008 or earlier, as predictors of the number of towers at the end of the sample, of access in 2009, or of the outcome's own pre-rollout level and trend. For every outcome, LASSO selects a railway station, the share of sloping land with high tree cover, and road density; it also selects city status for marketplace area, precipitation for the two sloping-land outcomes, and the share of flat land for collective farmland and for sloping land minus collective farmland. Nearest-neighbor matching pairs each county with the three counties closest in Mahalanobis distance (indicators matched exactly; 345–377 pairs) and includes pair × county and pair × year fixed effects. The propensity score is the predicted probability of access in 2009 given the same characteristics. Entropy balancing (Hainmueller 2012) reweights counties without access in 2009 so that their mean characteristics equal those of counties with access in 2009, with county and year fixed effects. Coefficients in columns (1)–(5) under alternative choices are as follows. Terrain quintiles instead of terciles: 312, 0.072, 0.082, −0.015, 0.101 (all $p < 0.05$ except column 4). Population quintiles instead of terciles: 230 ($p = 0.04$), 0.044, 0.055, −0.017, 0.050 ($p = 0.12$–0.74 in columns 2–5). ln population in 2008 × year instead of population terciles: 137, −0.003, 0.044, 0.008, 0.053 (none significant). One match per county: 169, 0.060, 0.076, −0.026, 0.051. Three matches within a 10% caliper: 162, 0.093, 0.117, −0.112, 0.081. Counties with a propensity score between 0.1 and 0.9 only: 125, 0.210, 0.005, −0.214, 0.071. Standard errors, in parentheses, are clustered by county (by county and pair for pair designs). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.4: Alternative Explanations: Weather, Roads, and Trade with China

		Vegetation (100 × ln NDVI)			
	Marketplace area (m^2)	Sloping – farmland	Sloping land	Collective farmland	Sloping land, low tree cover
	(1)	(2)	(3)	(4)	(5)
Baseline (county and year FE)	337.215***	0.130***	0.065**	-0.074	0.097***
	(93.122)	(0.033)	(0.028)	(0.048)	(0.028)
Panel A. Do the effects remain after controlling for weather?					
Apr–Aug precipitation		0.121***	0.078***	-0.051	0.105***
		(0.029)	(0.028)	(0.041)	(0.030)
May–Jun and other-month precipitation		0.108***	0.088***	-0.028	0.114***
		(0.027)	(0.026)	(0.036)	(0.029)
+ precipitation × sloping-land share		0.108***	0.100***	-0.021	0.128***
		(0.028)	(0.026)	(0.037)	(0.030)
Panel B. Do the effects remain across counties with different road networks?					
Road growth 2008–14 × year FE	336.050***	0.129***	0.067**	-0.071	0.099***
	(93.643)	(0.033)	(0.027)	(0.047)	(0.028)
Road density 2008 × year FE	320.966***	0.035	0.065**	-0.005	0.077**
	(99.505)	(0.033)	(0.031)	(0.053)	(0.036)
Panel C. Do the effects remain in counties with different exposure to China?					
China-border county × year FE	346.539***	0.102***	0.068**	-0.034	0.101***
	(90.705)	(0.029)	(0.028)	(0.042)	(0.030)
Distance to border customs × year FE	340.339***	0.111***	0.071***	-0.043	0.099***
	(93.876)	(0.027)	(0.026)	(0.034)	(0.028)
Export sanction exposure × year FE	330.361***	0.109***	0.065**	-0.056	0.092***
	(94.088)	(0.032)	(0.028)	(0.047)	(0.029)
Input sanction exposure × year FE	330.036***	0.141***	0.063**	-0.085*	0.094***
	(95.212)	(0.036)	(0.029)	(0.051)	(0.029)
Panel D. Did the marketplace effect weaken after the sanctions?					
Cell towers × post-2017	206.840***				
	(67.710)				

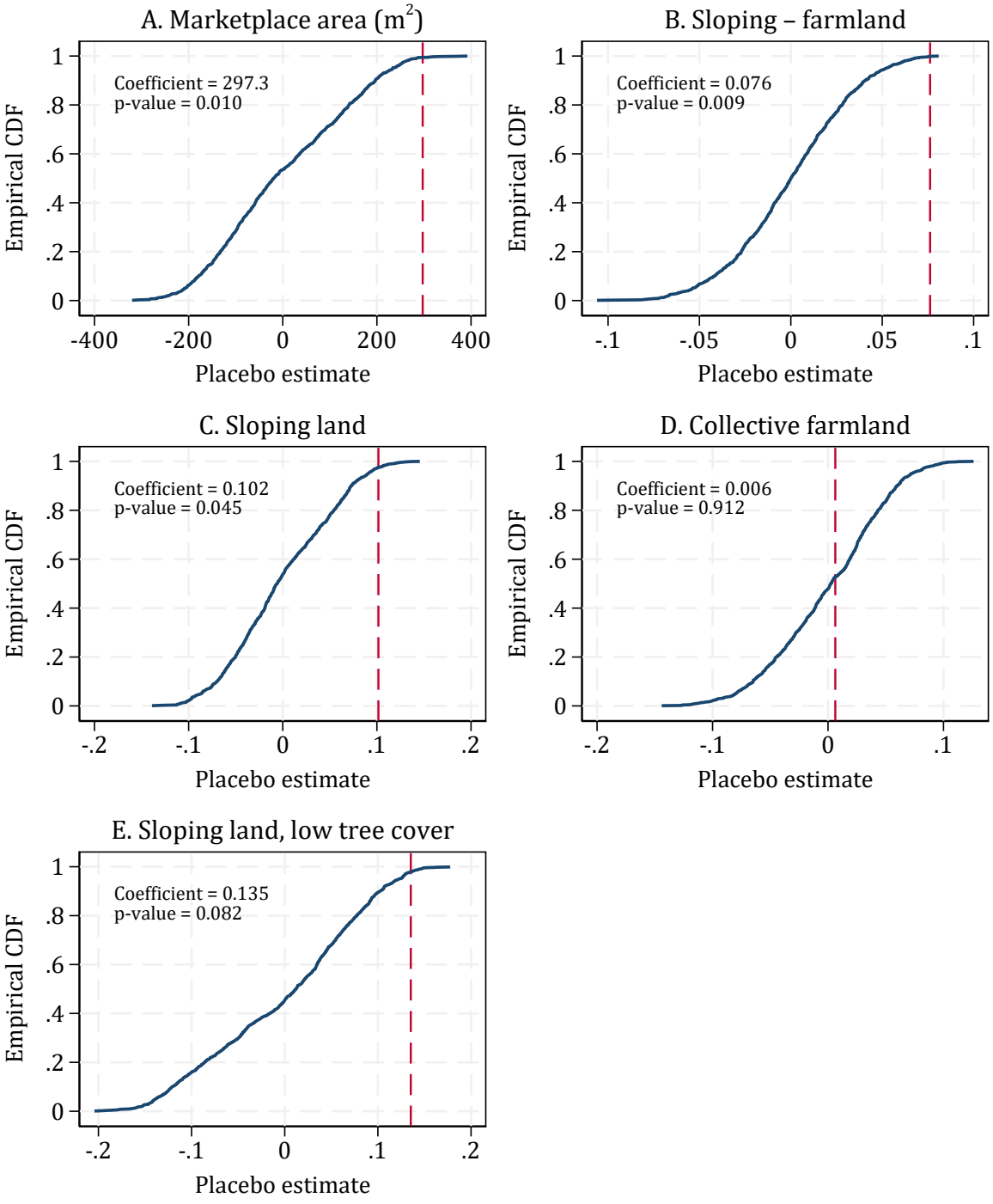
NOTES: Each cell in Panels A–C reports the coefficient on the cumulative number of cell towers from a separate regression with county and year fixed effects; each row adds the listed controls to the baseline. Panel A controls for precipitation (CHIRPS, 100 mm): total April–August precipitation and its square; May–June precipitation and precipitation in the other months (April, July, and August), each with its square; and, in the last row, each of these interacted with the share of land with slope of 15° or above, so that sloping counties may respond differently to the same rainfall. Precipitation is available through 2015, the end of the vegetation sample, so column (1) is not reported. Panel B allows counties to follow different yearly paths according to road growth between 2008 and 2014 and road density in 2008, the two years with road maps. Panel C does the same according to whether a county borders China (26 counties), its distance to the nearest border customs office, and its exposure to the 2016–2017 UN export and intermediate-input sanctions, measured as the share of the county's manufacturing exports (imported intermediate inputs) subject to sanctions given the industry mix of its firms, taken from the replication data of Kim et al. (2023); two counties without these measures are dropped in those rows. Continuous variables are standardized except sanction exposure. Panel D, for marketplace area only (the vegetation sample ends in 2015), reports the interaction from a regression of marketplace area on cell towers and cell towers × post-2017, the year in which the UN sanctions took effect. Because the effect of towers grows with time since access, this interaction shows that the effect did not weaken after the sanctions but does not isolate their effect. Standard errors, clustered by county, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure E.1: Permutation Tests: Tower Paths Reassigned across All Counties



NOTES: Each panel shows the empirical cumulative distribution of placebo estimates of the coefficient on cumulative cell towers from 1,000 permutations. In each permutation, each county's entire path of cell towers over time is reassigned to a randomly chosen county, and the regression with county and year fixed effects is re-estimated; the same reassignment is used in all five panels. Outcomes are as in Table E.2: marketplace area (2008–2021) and vegetation, measured as 100 × ln mean April–August NDVI (2002–2015). The dashed line marks the coefficient from the actual assignment. The two-sided p-value is the share of permutations, including the actual assignment, whose estimate is at least as large in absolute value as the actual coefficient.

Figure E.2: Permutation Tests: Tower Paths Reassigned within Provincess



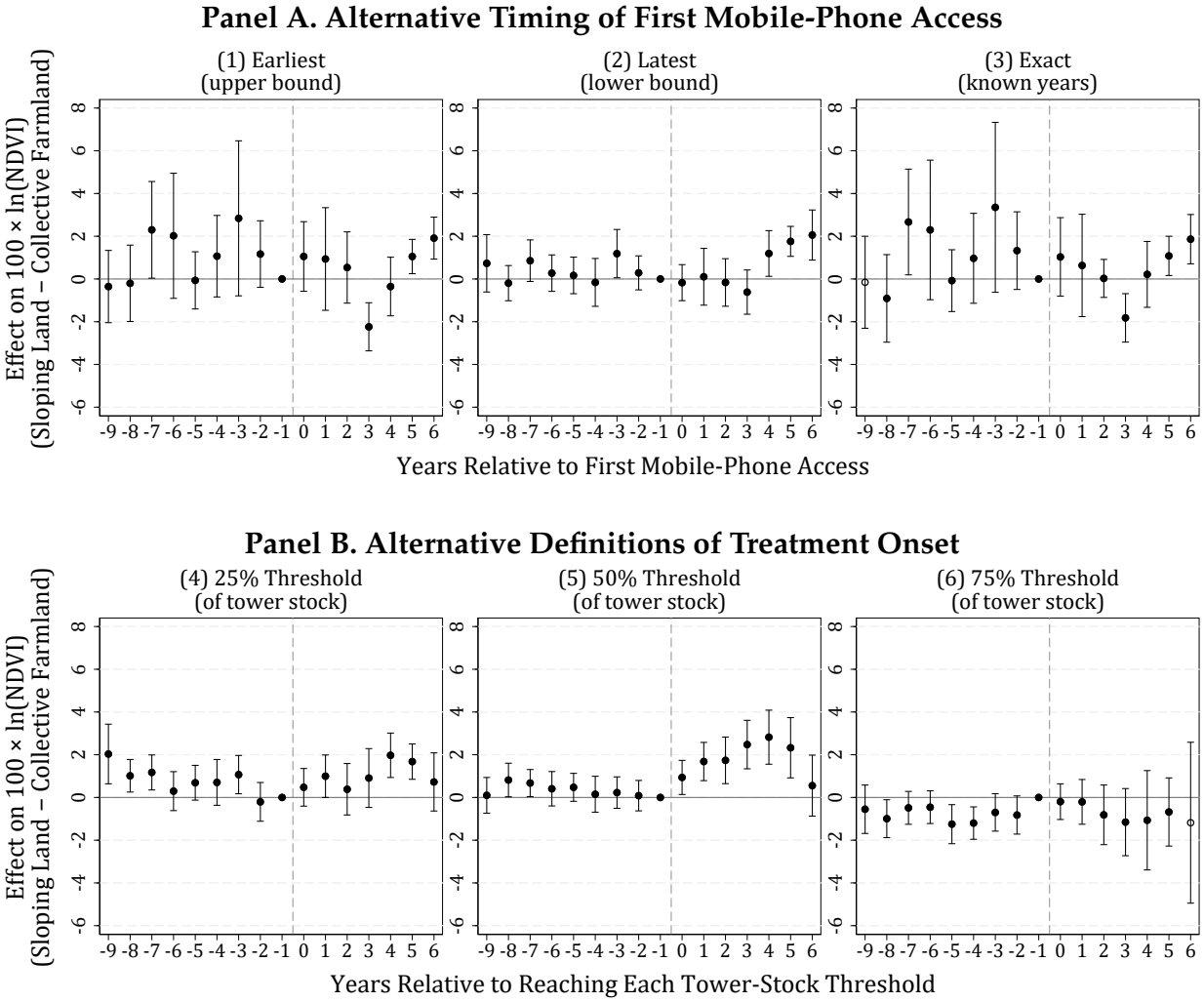
NOTES: As in Figure E.1, except that tower paths are reassigned only among counties in the same province and each regression includes county and province × year fixed effects.

Table E.5: Spillovers to Neighboring Counties

		Vegetation (100 × ln NDVI)			
	Marketplace area (m^2)	Sloping – farmland	Sloping land	Collective farmland	Sloping land, low tree cover
	(1)	(2)	(3)	(4)	(5)
Baseline (county and year FE)	337.215***	0.130***	0.065**	-0.074	0.097***
	(93.122)	(0.033)	(0.028)	(0.048)	(0.028)
Panel A. Own and neighboring counties' towers, county and year FE					
Cell towers	304.267***	0.080***	0.067**	-0.024	0.080**
	(101.062)	(0.027)	(0.032)	(0.046)	(0.034)
Cell towers in neighboring counties (mean)	166.850*	0.285***	-0.016	-0.293***	0.097
	(95.248)	(0.071)	(0.080)	(0.092)	(0.097)
Panel B. Own and neighboring counties' towers, county and province × year FE					
Cell towers	293.706***	0.076***	0.101***	0.007	0.136**
	(110.338)	(0.023)	(0.038)	(0.045)	(0.055)
Cell towers in neighboring counties (mean)	158.698	0.199**	0.088	-0.104	0.154*
	(113.077)	(0.077)	(0.072)	(0.075)	(0.091)
Panel C. Placebo: did neighbors' future towers predict vegetation before a county's first tower?					
Cell towers in neighboring counties$_{t+1}$		-0.608**	0.034	0.953***	-0.280
		(0.235)	(0.342)	(0.310)	(0.489)
Cell towers in neighboring counties$_{t+2}$		-0.343**	0.007	0.476**	-0.253
		(0.163)	(0.187)	(0.219)	(0.242)
Cell towers in neighboring counties$_{t+3}$		-0.131	-0.043	0.129	-0.173
		(0.097)	(0.120)	(0.126)	(0.166)

NOTES: Each panel reports coefficients from a separate regression for each outcome. Panels A and B regress the outcome on the county's cumulative number of cell towers and the mean cumulative number of cell towers in neighboring counties, defined as counties sharing a border, with county and year fixed effects (Panel A) or county and province × year fixed effects (Panel B). Panel C regresses vegetation on the mean number of cell towers in neighboring counties one, two, or three years ahead, using only years before the county's own first tower, with county and year fixed effects; marketplace area is not reported because the marketplace sample begins in 2008, when few county-years precede the first tower. If towers raised outcomes in neighboring counties, counties with few towers would be partly treated, and the own-county estimates would understate the effects. Weighting neighbors by shared border length leaves the own-county coefficients in columns (1)–(5) at 306, 0.089, 0.069, −0.035, and 0.081 with county and year fixed effects and at 295, 0.076, 0.102, 0.006, and 0.135 with county and province × year fixed effects. Defining treatment as regional towers, the county's own towers plus the sum of its neighbors' towers, gives 64.2, 0.059, −0.002, −0.063, and 0.022 per regional tower with county and year fixed effects (54.9, 0.036, 0.017, −0.027, and 0.034 with county and province × year fixed effects). Standard errors, clustered by county, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure E.3: Alternative Measures of Mobile-Phone Treatment: Vegetation



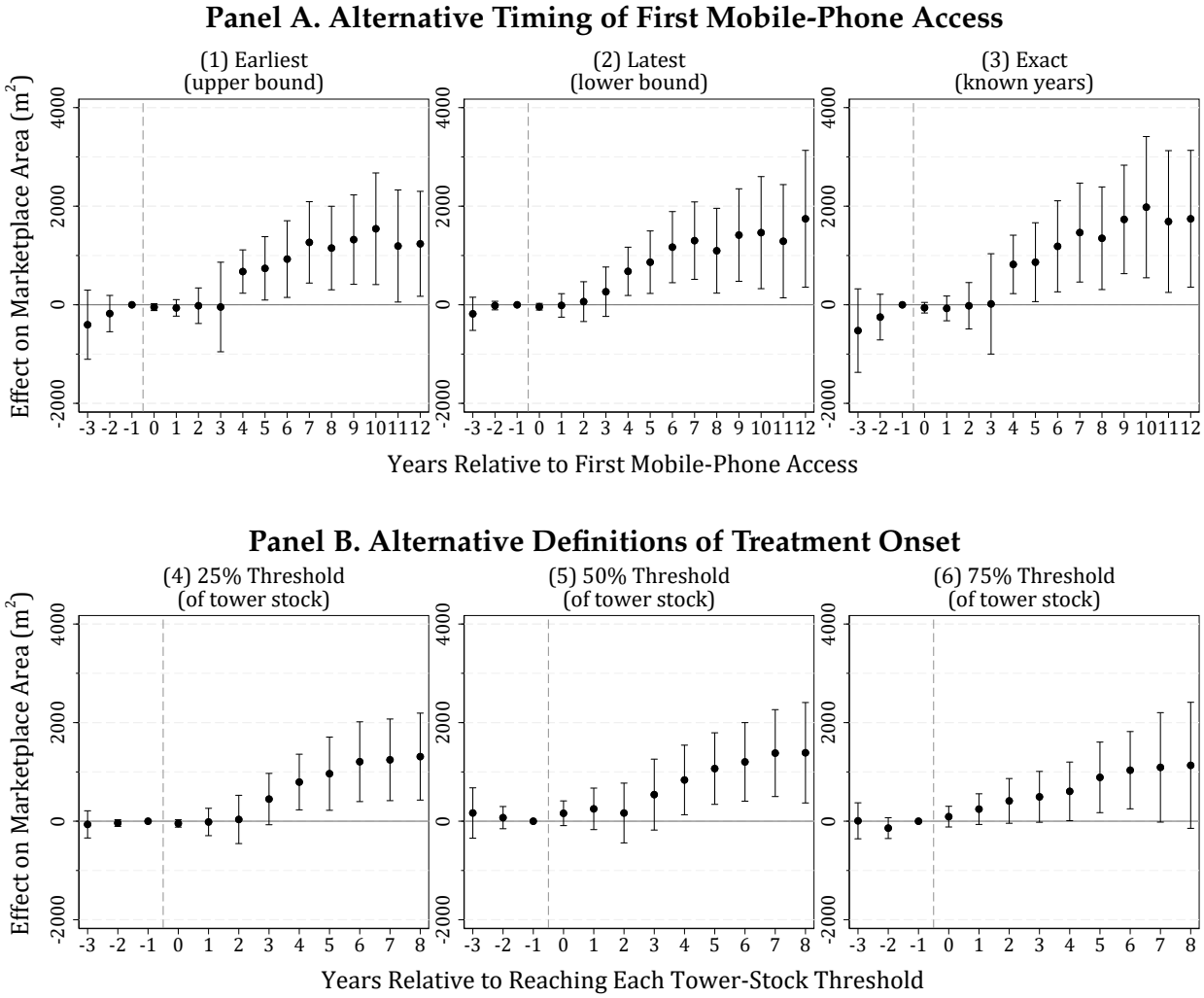
NOTES: This figure examines the robustness of the vegetation event-study estimates reported in Figure IV to alternative definitions of treatment timing and treatment onset. The dependent variable is 100 × ln(NDVI) on sloping land (at least 15 degrees) minus 100 × ln(NDVI) on collective farmland (below 15 degrees). Estimates are obtained using the Callaway and Sant'Anna (2021) estimator, with not-yet-treated and never-treated counties serving as controls. As in Figure IV, all estimates are measured relative to the year before treatment onset (event time −1), following Roth (2026).

Panel A examines whether the estimated dynamics depend on the midpoint convention used to date first mobile-phone access. As described in Section 4.1, let g_c^E and g_c^L denote the earliest and latest feasible years of first cell-tower arrival in county c. The baseline timing is defined as $g_c^M = \text{round}\left[(g_c^E + g_c^L)/2\right]$. When the midpoint falls between two calendar years, it is assigned to the later year. The Earliest and Latest specifications instead assign first access to g_c^E and g_c^L, respectively. The Exact specification restricts the treated sample to counties for which $g_c^E = g_c^L$, so that the year of first access is known exactly.

Panel B examines whether the estimated dynamics depend on defining treatment onset as the arrival of the first cell tower. Holding the Uniform measure of the tower rollout fixed, let s_{ct} denote the cumulative share of county c's eventual cell-tower stock reached by year t. For each threshold $q \in \{0.25, 0.50, 0.75\}$, treatment is defined to begin in $g_c^q = \min\{t : s_{ct} \geq q\}$. The pre-treatment periods in Panel B are defined relative to each rollout threshold and therefore need not precede first mobile-phone access. For higher thresholds, which are typically reached several years after first access, the pre-treatment periods may already include years with mobile-phone coverage. Counties whose treatment onset occurs after 2015 are treated as not yet treated during the sample period.

All panels report event times from nine years before through six years after treatment onset, the same window as Figure IV. Hollow markers denote event times supported by fewer than ten treated counties. As in Figure IV, estimates at the longest horizons rely on a small comparison group and should be interpreted with caution. The vertical dashed line separates periods before and after the relevant treatment onset, the horizontal solid line denotes zero effect, and vertical bars report 95 percent confidence intervals.

Figure E.4: Alternative Timing and Onset of Mobile-Phone Access: Marketplace Area



NOTES: This figure examines the robustness of the first-access event-study estimates reported in Figure VI to alternative definitions of treatment timing and treatment onset. Estimates are obtained using the Callaway and Sant'Anna (2021) estimator, with not-yet-treated and never-treated counties serving as controls. As in Figure VI, all estimates are measured relative to the year before treatment onset (event time -1), following Roth (2026).

Panel A examines whether the estimated dynamics depend on the midpoint convention used to date first mobile-phone access. As described in Section 4.1, let g_c^E and g_c^L denote the earliest and latest feasible years of first cell-tower arrival in county c. The baseline timing is defined as $g_c^M = \text{round}\left[(g_c^E + g_c^L)/2\right]$. When the midpoint falls between two calendar years, it is assigned to the later year. The Earliest and Latest specifications instead assign first access to g_c^E and g_c^L, respectively. The Exact specification restricts the treated sample to counties for which $g_c^E = g_c^L$, so that the year of first access is known exactly.

Panel B examines whether the estimated dynamics depend on defining treatment onset as the arrival of the first cell tower. Holding the Uniform measure of the tower rollout fixed, let s_{ct} denote the cumulative share of county c's eventual cell-tower stock reached by year t. For each threshold $q \in \{0.25, 0.50, 0.75\}$, treatment is defined to begin in $g_c^q = \min\{t : s_{ct} \geq q\}$. The pre-treatment periods in Panel B are defined relative to each rollout threshold and therefore need not precede first mobile-phone access. For higher thresholds, which are typically reached several years after first access, the pre-treatment periods may already include years with mobile-phone coverage. Panel A reports event times from three years before through twelve years after first access, the same window as Figure VI. Panel B reports event times from three years before through eight years after each rollout threshold; because the panel ends in 2021, longer horizons are identified from the few cohorts that reached each threshold in the first years of the rollout. Because marketplace data begin in 2008, counties first treated in 2009 contribute no pre-treatment estimates. The vertical dashed line separates periods before and after the relevant treatment onset, the horizontal solid line denotes zero effect, and vertical bars report 95 percent confidence intervals.

Table E.6: Alternative Vegetation Measures

	Dep. Var: NDVI, Sloping Land − Collective Farmland			
	100 × ln(NDVI)			100 × NDVI
	Baseline	Slope≥30°	Landsat 30m	Level difference
	(1)	(2)	(3)	(4)
Cumulative cell towers				
County and year fixed effects	0.130***	0.162*	0.155**	0.082***
	(0.033)	(0.084)	(0.071)	(0.019)
County and province × year fixed effects	0.076***	0.148*	0.002	0.058***
	(0.024)	(0.086)	(0.061)	(0.014)
Pre-treatment mean	17.50	11.59	29.69	11.26
Observations	2,408	1,544	2,408	2,408
N of clusters	172	111	172	172

NOTES: The dependent variable is NDVI on sloping land minus NDVI on collective farmland in each county-year, 2002–2015, measured as 100 × ln(NDVI) in columns (1)–(3) and as 100× NDVI in column (4). All columns use the preferred Uniform measure of cumulative cell-tower exposure. Each cell reports the coefficient on the cumulative number of cell towers, estimated with county and year fixed effects (first row) or county and province-by-year fixed effects (second row).
Column (1) reproduces the baseline outcome in column (3) of Table 1, 100 × ln(NDVI) on land with slope of at least 15 degrees minus 100 × ln(NDVI) on collective farmland (slope below 15 degrees), where NDVI is the mean of April–August MODIS values at 250m resolution. Column (2) restricts sloping land to slopes of 30 degrees or more, the threshold used in forest-management reports. Because slope is evaluated at 250m resolution, few pixels exceed 30 degrees, so this sample covers fewer counties, and a 15–30 degree group is nearly identical to the baseline. Column (3) uses Landsat imagery at 30m resolution, with slope also evaluated at 30m, for the same counties as column (1). MODIS composites select the best daily observation within each 16-day period, whereas Landsat revisits each location every 8–16 days, so many growing-season scenes are lost to clouds. Landsat composites combine TM, ETM+ (with scan-line gaps after 2003), and OLI imagery harmonized to ETM+ (Roy et al., 2016). Sensor changes affect all counties in a given year and are differenced out within each county-year, but the small number of cloud-free scenes makes these estimates imprecise. Column (4) uses the difference in NDVI levels, multiplied by 100, instead of the difference in logs, so its magnitude is not directly comparable with the other columns.
Pre-treatment means are calculated using observations prior to the first positive tower stock. Pre-treatment means, observations, and the number of clusters refer to the specification with county and year fixed effects. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table E.7: Timing within the Growing Season

	Dep. Var: 100 × ln(NDVI)						
	Levels		Difference from collective farmland				Within sloping land
				Sloping land by tree cover (2000)			
	Collective farmland (<15°)	Sloping land (≥15°)	Sloping land (≥15°)	Low (<30%)	Medium (30–70%)	High (≥70%)	Low- High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. County and year fixed effects							
April–August (baseline)	-0.074	0.065**	0.130***	0.159***	0.166***	0.075	0.077*
	(0.048)	(0.028)	(0.033)	(0.044)	(0.043)	(0.057)	(0.041)
April–September	-0.045	0.055**	0.096***	0.124***	0.123***	0.054	0.066*
	(0.039)	(0.023)	(0.028)	(0.039)	(0.035)	(0.040)	(0.034)
May–September	-0.038	0.004	0.042*	0.071**	0.059**	0.037	0.038
	(0.037)	(0.022)	(0.024)	(0.033)	(0.029)	(0.036)	(0.048)
Early season (April 1–May 12)	-0.061	0.157**	0.262***	0.299***	0.313***	0.183	0.125
	(0.082)	(0.074)	(0.073)	(0.083)	(0.096)	(0.181)	(0.142)
Peak season (June 15–August 31)	0.020	0.001	-0.010	0.027	0.001	-0.007	0.043
	(0.029)	(0.017)	(0.022)	(0.030)	(0.027)	(0.028)	(0.038)
Panel B. County and province × year fixed effects							
April–August (baseline)	0.006	0.102***	0.076***	0.106***	0.114***	0.019	0.137**
	(0.046)	(0.036)	(0.024)	(0.035)	(0.034)	(0.068)	(0.062)
April–September	0.035	0.097***	0.050**	0.079**	0.073***	-0.006	0.135**
	(0.040)	(0.030)	(0.021)	(0.032)	(0.025)	(0.054)	(0.052)
May–September	0.057	0.092***	0.024	0.044	0.019	-0.016	0.107***
	(0.035)	(0.022)	(0.020)	(0.030)	(0.020)	(0.030)	(0.037)
Early season (April 1–May 12)	-0.007	0.079	0.137*	0.203**	0.225***	0.132	0.200
	(0.102)	(0.086)	(0.075)	(0.088)	(0.076)	(0.174)	(0.160)
Peak season (June 15–August 31)	0.064**	0.066***	0.009	0.030	-0.007	-0.023	0.068**
	(0.029)	(0.024)	(0.014)	(0.021)	(0.019)	(0.028)	(0.029)
Observations	2,492	2,408	2,408	2,408	2,322	1,675	1,675
N of clusters	178	172	172	172	166	121	121

NOTES: This table repeats the specification in Table 1 for alternative windows of the growing season. Columns are as in Table 1; each row uses NDVI averaged over the indicated window, applied to both land types in every difference. The first row reproduces the Uniform row of Table 1. The early season covers the MODIS 16-day composites starting between April 1 and May 12; the peak season covers those starting between June 15 and August 31. All columns use the preferred Uniform measure of cumulative cell-tower exposure. Panel A includes county and year fixed effects; Panel B includes county and province-by-year fixed effects. Column (7) uses the common sample of counties with high-tree-cover pixels.

The increase in vegetation on sloping land relative to collective farmland is concentrated in the early growing season, when vegetation indices best distinguish land with growing plants from bare or sparsely covered land, and on land with low or moderate tree cover. It is not visible in the peak season, when NDVI saturates over most vegetated land. Observations and the number of clusters refer to the April–August specification with county and year fixed effects. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table E.8: Alternative Measures of Marketplace Area

	Dep. Var: Marketplace Area (m^2)			
Panel A. Alternative marketplace-area measures				
	Full Interpolation (1)	Interior Interpolation (2)	Observed Only (3)	Complete Case (4)
Cumulative cell towers				
County and year fixed effects	337.215***	2,037.250***	2,431.550***	296.893*
	(93.122)	(495.173)	(830.098)	(177.377)
County and province × year fixed effects	297.349***	1,627.693***	2,075.517**	257.929
	(108.895)	(555.976)	(857.580)	(231.522)
Pre-treatment mean (m^2)	9,938	4,011	3,115	2,870
Observations	2,492	2,100	1,141	658
N of clusters	178	178	175	157
Panel B. Samples restricted by imagery coverage				
	Full Sample (5)	≥75% Observed (6)	≥90% Observed (7)	100% Observed (8)
Cumulative cell towers				
County and year fixed effects	337.215***	468.340***	531.469***	296.893*
	(93.122)	(154.680)	(189.824)	(177.377)
County and province × year fixed effects	297.349***	403.258	445.041	257.929
	(108.895)	(250.532)	(289.706)	(231.522)
Pre-treatment mean (m^2)	9,938	3,314	2,870	2,870
Observations	2,492	731	668	658
N of clusters	178	166	159	157

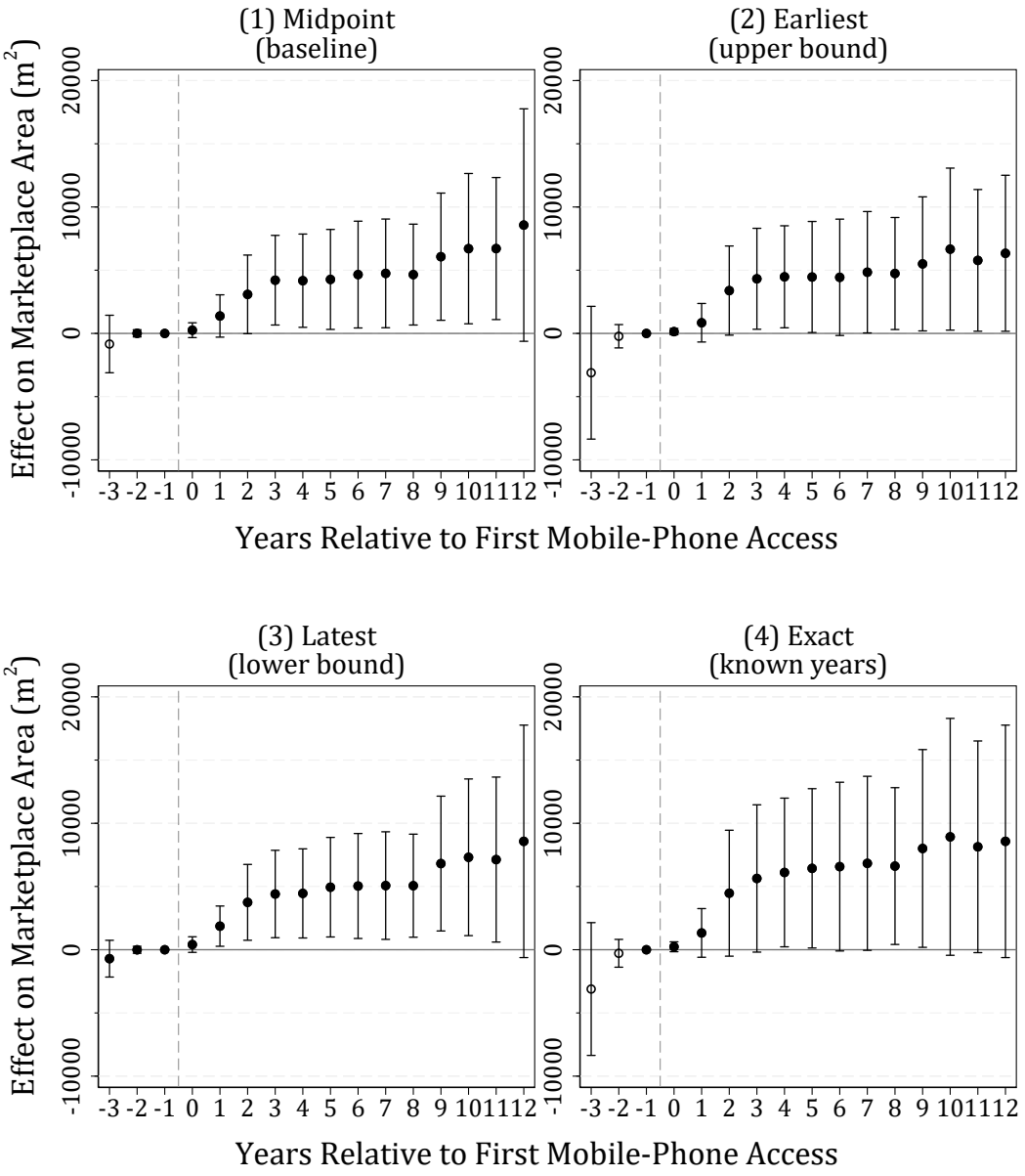
NOTES: The dependent variable is total marketplace area in each county-year, measured in square meters. All columns use the preferred Uniform measure of cumulative cell-tower exposure. Each cell reports the coefficient on the cumulative number of cell towers, estimated with county and year fixed effects (first row) or county and province-by-year fixed effects (second row).

Panel A varies the construction of marketplace area. Column (1) uses Full Interpolation, which linearly interpolates marketplace area between observed images and fills values before the first and after the last observed image. Column (2) uses Interior Interpolation, which linearly interpolates only between observed images and does not fill values before the first or after the last observed image. Column (3) uses Observed Only, which relies only on directly observed marketplace areas and structural zeros, without interpolation or extrapolation. Column (4) uses Complete Case, which restricts the Observed Only measure to county-years in which all relevant marketplaces are observed. Each specification is estimated using the observations available under the corresponding measure, so the samples, and hence the magnitudes, are not directly comparable across columns.

Panel B holds the Full Interpolation measure fixed and restricts the sample by imagery coverage. Column (5) uses the full sample. Columns (6) and (7) restrict the sample to county-years in which at least 75 and 90 percent of marketplaces, respectively, are directly observed in satellite imagery. Column (8) restricts the sample to county-years in which all relevant marketplaces are directly observed; this sample coincides with the Complete Case sample in column (4). Counties with no marketplaces are retained as structural-zero cases in the coverage-restricted samples. The coverage-restricted samples retain few county-years from the initial rollout period (2008–2010) and few observations within each province-year cell, so estimates with province-by-year fixed effects are imprecise; the point estimates remain similar in magnitude to those with year fixed effects.

Pre-treatment means are calculated using observations prior to the first positive tower stock. Pre-treatment means, observations, and the number of clusters refer to the specification with county and year fixed effects. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure E.5: Event Studies with Interior-Interpolated Marketplace Area

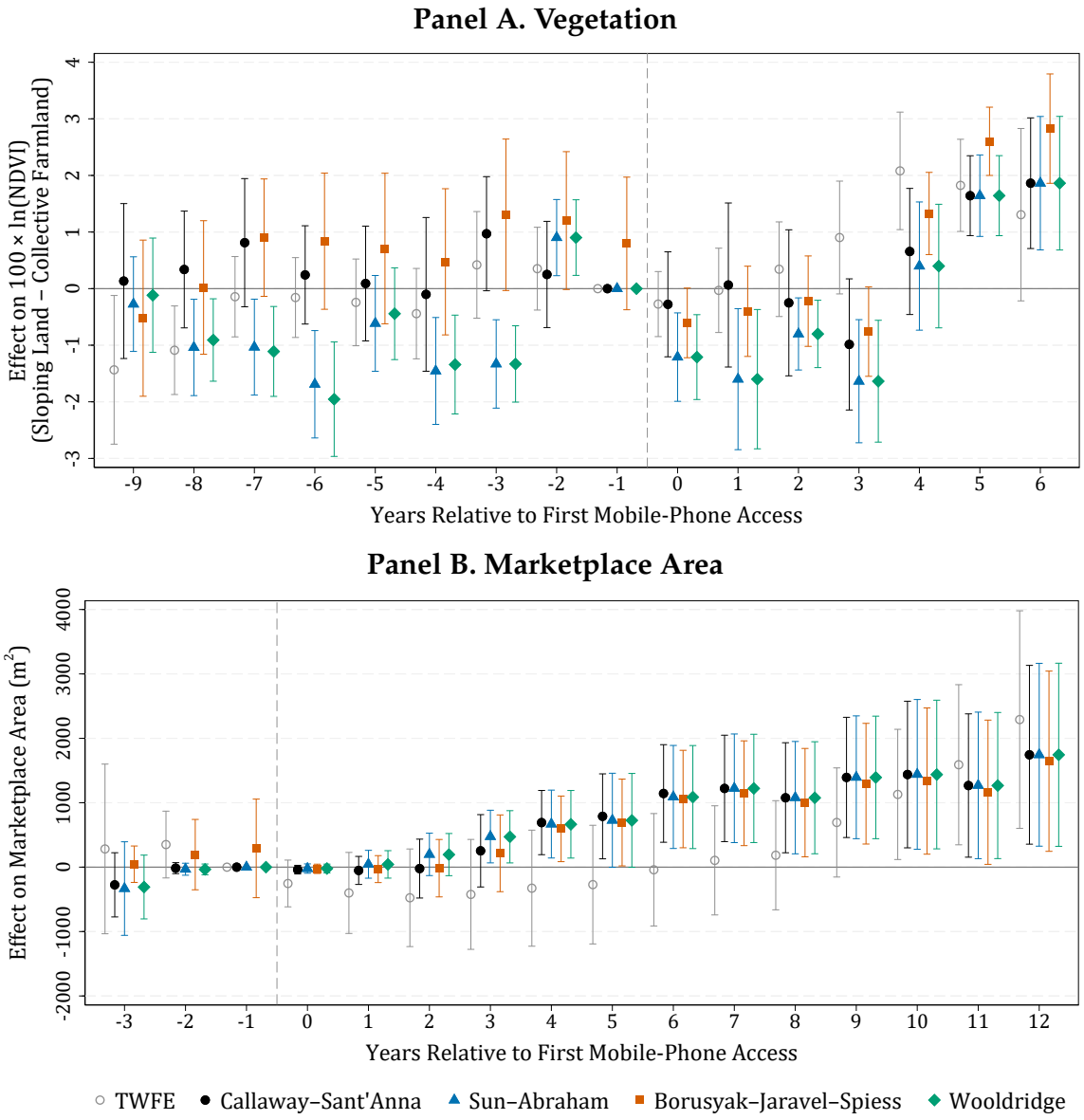


NOTES: This figure re-estimates the first-access event study in Figure VI using the Interior Interpolation measure of marketplace area, which linearly interpolates marketplace area only between observed satellite images and does not fill values before the first or after the last observed image in each county. Table E.8 compares the alternative marketplace-area measures; this figure asks whether, under the measure that relies least on interpolation, there is no evidence of pre-trends and the post-treatment dynamics are the same regardless of how the timing of first mobile-phone access is defined. Panel (1) uses the baseline midpoint timing, $g_c^M = \text{round}\left[(g_c^E + g_c^L)/2\right]$. Panels (2) and (3) assign first access to the earliest (g_c^E, upper bound) and latest (g_c^L, lower bound) feasible year, respectively, and panel (4) restricts the treated sample to counties for which $g_c^E = g_c^L$, as in Figure E.4.

Estimates are obtained using the Callaway and Sant'Anna (2021) estimator, with not-yet-treated and never-treated counties serving as controls, and are measured relative to the year before first access (event time -1), following Roth (2026). Because the Interior Interpolation measure is missing in years before a county's first or after its last observed image, each estimate uses only treated counties observed both in the year before first access and in the relevant post- or pre-treatment year, so fewer counties contribute than in Figure VI. For the same reason, the Observed Only and Complete Case measures, which exist only in years with satellite imagery, cannot support an event study around first access and are reported only in Table E.8.

All panels report event times from three years before through twelve years after first access, the same window as Figure VI, and share a common vertical axis. Hollow markers denote event times supported by fewer than ten treated counties. The vertical dashed line separates periods before and after first access, the horizontal solid line denotes zero effect, and vertical bars report 95 percent confidence intervals.

Figure E.6: First-Access Event Studies with Alternative Estimators



NOTES: This figure re-estimates the first-access event studies in Figures IV and VI with alternative estimators. In Panel A, the dependent variable is 100 × ln(NDVI) on sloping land (at least 15 degrees) minus 100 × ln(NDVI) on collective farmland (below 15 degrees), over 2002–2015. In Panel B, the dependent variable is total marketplace area in square meters over 2008–2021. Treatment is first mobile-phone access, dated by the baseline midpoint timing. The estimators are a two-way fixed effects (TWFE) event-study regression with county and year fixed effects; Callaway and Sant'Anna (2021), with not-yet-treated and never-treated counties as controls (the baseline in the main figures); Sun and Abraham (2021), with never-treated counties as controls; the imputation estimator of Borusyak et al. (2024); and the extended TWFE estimator of Wooldridge (2025), with never-treated counties as controls. Except for Borusyak et al. (2024), estimates are measured relative to the year before first access (event time −1). The pre-period coefficients of Borusyak et al. (2024) come from its pre-trend test, which uses untreated observations, and are not normalized to zero at event time −1. In the TWFE and Sun and Abraham (2021) specifications, event times before the start of the window are binned at its first event time. The estimator of de Chaisemartin and D'Haultfœuille (2024) is numerically identical to Callaway and Sant'Anna (2021) in this setting (binary, staggered treatment without covariates) and is not shown.

The Sun and Abraham (2021) and Wooldridge (2025) estimators use never-treated counties as the sole controls. In the vegetation sample, these counties are all in Jagang Province, and their year-to-year vegetation changes do not track those of the treated cohorts (for example, in 2003 and 2006), while the treated cohorts move closely together. Their pre-period and early post-period estimates in Panel A therefore reflect differences between these counties and the rest of the country rather than pre-trends among treated counties. Estimators that also use not-yet-treated counties as controls show no such pattern.

Panel A reports event times from nine years before through six years after first access, and Panel B from three years before through twelve years after, the same windows as the main figures. Because marketplace data begin in 2008, counties first treated in 2009 contribute no pre-treatment estimates in Panel B. Estimates are horizontally offset for readability. The vertical dashed line separates periods before and after first access, the horizontal solid line denotes zero effect, and vertical bars report 95 percent confidence intervals based on standard errors clustered at the county level.

Table E.9: Heterogeneity-Robust Estimates of the Effect of Cell Towers

	Dep. Var: NDVI, Sloping Land– Collective Farmland 100 × ln(NDVI)				Dep. Var: Marketplace
	All sloping land	Low tree cover	Medium tree cover	High tree cover	Area (m^2)
	(1)	(2)	(3)	(4)	(5)
Panel A. Two-way fixed effects					
Cumulative cell towers					
County and year fixed effects	0.130***	0.159***	0.166***	0.075	337.215***
	(0.033)	(0.044)	(0.043)	(0.057)	(93.122)
County and province × year fixed effects	0.076***	0.106***	0.114***	0.019	297.349***
	(0.024)	(0.035)	(0.034)	(0.068)	(108.895)
Sum of negative weights	-0.18	-0.18	-0.17	-0.15	-0.48
Panel B. de Chaisemartin and D'Haultfœuille					
Average total effect per cell tower					
All comparison counties	0.100	0.319**	0.184	0.136	146.257**
	(0.116)	(0.126)	(0.157)	(0.127)	(57.377)
Comparison counties in the same province	0.162	0.358	0.205	0.129	184.134
	(0.210)	(0.240)	(0.262)	(0.206)	(183.632)
Joint placebo *p*-value, all comparison counties	0.00	0.01	0.01	0.00	0.22
Joint placebo *p*-value, same province	0.40	0.58	0.41	0.78	0.40
Pre-treatment mean	17.50	11.93	21.18	18.53	9,938
Observations	2,408	2,408	2,322	1,675	2,492
N of clusters	172	172	166	121	178

NOTES: This table compares the two-way fixed effects (TWFE) estimates of the effect of one additional cell tower with the heterogeneity-robust estimator of de Chaisemartin and D'Haultfœuille (2024). All columns use the preferred Uniform measure of cumulative cell-tower exposure. In columns (1)–(4), the dependent variable is 100 × ln(NDVI) on sloping land (at least 15 degrees) minus 100 × ln(NDVI) on collective farmland (below 15 degrees), over 2002–2015. Columns (2)–(4) restrict sloping land to areas with low (below 30 percent), medium (30–70 percent), and high (at least 70 percent) tree cover, as in Table 1. In column (5), the dependent variable is total marketplace area in square meters over 2008–2021.

Panel A reports TWFE estimates with county and year fixed effects (first row) or county and province-by-year fixed effects (second row), together with the sum of the negative weights attached to the TWFE coefficient (De Chaisemartin and d'Haultfoeuille, 2020). Panel B reports the average total effect per unit of treatment from de Chaisemartin and D'Haultfœuille (2024), which compares counties whose tower stock changes with counties whose tower stock has not yet changed and normalizes the cumulative effect by the cumulative change in the number of towers, so that it is comparable to the per-tower TWFE coefficient. The first row uses all comparison counties, and the second row restricts comparisons to counties in the same province, analogous to province-by-year fixed effects. Joint placebo *p*-values test the null hypothesis that outcomes evolved similarly for switching and comparison counties before the tower stock changed, so a high *p*-value indicates that the null of no pre-trends is not rejected. For vegetation, the placebo test rejects when switchers are compared with all not-yet-switchers, driven by placebos five to seven years before the first change in tower stock. It does not reject within provinces, where point estimates are similar in magnitude but less precise. Within-province comparisons rely on few switchers at long horizons. Dynamic estimates with the continuous treatment (not reported) show the same timing as the binary event studies in Figures IV and VI.

Pre-treatment means are calculated using observations prior to the first positive tower stock and are in the units of each dependent variable. Pre-treatment means, observations, and the number of clusters refer to the TWFE specification with county and year fixed effects. Robust standard errors are clustered at the county level. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.